INTEGRATIVE PROGRAMMING

ACTIVITY COMPILATION

Activities 1 to 13

Sonajo, Christian Paolo

BSIT 3B

Subject : Integrative Programming

School Year : School Year 2026 - 2027

Date Submitted : October 26, 2026

Submitted to : _____

Activity 1 - Console Basics and Data Types

Act1_Activity202606241600 | Sonajo, Christian Paolo | BSIT 3B

Source Code

Module1.vb

```
' ============================================================
' Activity 1 - Act1_Activity202606241600
' Console basics and data types
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Module Module1

    Sub Main()
        ' program header
        Console.WriteLine("Activity202606241600")
        Console.WriteLine("Author: Sonajo, Christian Paolo")
        Console.WriteLine("Course Year Section: BSIT 3B")
        Console.WriteLine()

        ' String variable
        Dim theName As String = "Sonajo, Christian Paolo"
        Console.WriteLine("Hello " + theName)

        ' one variable for every basic data type
        Dim n1 As Integer = 0
        Dim n2 As Double = 0.0
        Dim isValid As Boolean = True
        Dim character As Char = "A"c
        Dim number1 As Integer = 12
        Dim number2 As Double = 12.12

        Console.WriteLine("n1        = " & n1.ToString())
        Console.WriteLine("n2        = " & n2.ToString())
        Console.WriteLine("isValid   = " & isValid.ToString())
        Console.WriteLine("character = " & character.ToString())
        Console.WriteLine("number1   = " & number1.ToString())
        Console.WriteLine("number2   = " & number2.ToString())
        Console.WriteLine()

        ' closing line with the name of the author
        Console.WriteLine("Prepared by: Sonajo, Christian Paolo  |  BSIT 3B")
        Console.ReadKey()
    End Sub

End Module
```

Sample Output

```
Activity202606241600
Author: Sonajo, Christian Paolo
Course Year Section: BSIT 3B

Hello Sonajo, Christian Paolo
n1        = 0
n2        = 0
isValid   = True
character = A
number1   = 12
number2   = 12.12

Prepared by: Sonajo, Christian Paolo  |  BSIT 3B
```

Activity 2 - Name Input and Celsius - Fahrenheit Converter

Act2 Celcius-Fahrenheit Converter | Sonajo, Christian Paolo | BSIT 3B

Source Code

Module1.vb

```
' ============================================================
' Activity 2 - Act2 Celcius-Fahrenheit Converter
' Name input and temperature conversion
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Module Module1

    Sub Main()
        ' program header
        Console.WriteLine("Act2 Celcius-Fahrenheit Converter")
        Console.WriteLine("Author: Sonajo, Christian Paolo")
        Console.WriteLine("Course Year Section: BSIT 3B")
        Console.WriteLine()

        Console.Write("Hello world")
        Console.ReadLine()

        Dim lastName As String = ""
        Dim firstName As String = ""

        Console.Write("Enter Lastname: ")
        lastName = Console.ReadLine()
        Console.Write("Enter Firstname: ")
        firstName = Console.ReadLine()

        Console.WriteLine("Your Lastname is " + lastName)
        Console.WriteLine("Your Firstname is " + firstName)
        Console.ReadLine()

        ' Converter 1: Celsius to Fahrenheit
        Console.WriteLine("--- Celsius to Fahrenheit Converter ---" & vbNewLine)
        Console.Write("Enter Celsius: ")
        Dim celsiusInput As Decimal = Convert.ToDecimal(Console.ReadLine())
        Dim fahrenheitResult As Decimal = (celsiusInput * 1.8) + 32
        Console.WriteLine("Result in Fahrenheit: " & fahrenheitResult)
        Console.ReadLine()

        ' Converter 2: Fahrenheit to Celsius
        Console.WriteLine("--- Fahrenheit to Celsius Converter ---" & vbNewLine)
        Console.Write("Enter Fahrenheit: ")
        Dim fahrenheitInput As Decimal = Convert.ToDecimal(Console.ReadLine())
        Dim celsiusResult As Decimal = (fahrenheitInput - 32) / 1.8
        Console.WriteLine("Result in Celsius: " & celsiusResult)

        ' closing line with the name of the author
        Console.WriteLine()
        Console.WriteLine("Activity 2 completed by: Sonajo, Christian Paolo  |  BSIT 3B")
        Console.ReadLine()

    End Sub

End Module
```

Sample Output

```
Act2 Celcius-Fahrenheit Converter
Author: Sonajo, Christian Paolo
Course Year Section: BSIT 3B

Hello world
Enter Lastname: Sonajo
Enter Firstname: Christian Paolo
Your Lastname is Sonajo
Your Firstname is Christian Paolo
--- Celsius to Fahrenheit Converter ---
Enter Celsius: 37
Result in Fahrenheit: 98.6
--- Fahrenheit to Celsius Converter ---
Enter Fahrenheit: 98.6
Result in Celsius: 37.0

Activity 2 completed by: Sonajo, Christian Paolo  |  BSIT 3B
```

Activity 3 - Sum, Difference, Product, Quotient and Average (Console with Retry)

Activity3_20260629 | Sonajo, Christian Paolo | BSIT 3B

Source Code

Module1.vb

```
Module Module1
    Sub Main()

        'Activity3_20260629'
        'Create a vb.net console app that will accept 3 numbers, and display the sum, difference, product,
    quotient, and ave of the 3 numbers'

        Try
ulit:
            Console.WriteLine("Activity3_20260629")
            Console.WriteLine("Author: Sonajo, Christian Paolo")
            Console.WriteLine("Course Year Section: BSIT 3B")

            Console.Write("Enter number 1: ")
            Dim n1 As Integer = Convert.ToInt16(Console.ReadLine())
            Console.Write("Enter number 2: ")
            Dim n2 As Integer = Convert.ToInt16(Console.ReadLine())
            Console.Write("Enter number 3: ")
            Dim n3 As Integer = Convert.ToInt16(Console.ReadLine())

            Console.Write("The sum is: ")
            Dim sum As Double = n1 + n2 + n3
            Console.WriteLine(sum)

            Console.Write("The difference is: ")
            Dim difference As Double = n1 - n2 - n3
            Console.WriteLine(difference)

            Console.Write("The product is: ")
            Dim product As Double = n1 * n2 * n3
            Console.WriteLine(product)

            Console.Write("The quotient is: ")
            Dim quotient As Double = n1 / n2 / n3
            Console.WriteLine(quotient)

            Console.Write("The average is: ")
            Dim ave As Double = sum / 3
            Console.WriteLine(ave)

        Catch ex As FormatException
            Console.WriteLine("Error: " & ex.Message)
            GoTo ulit
        End Try

        Console.ReadKey()

        'Iti nasursurwak ditoy nga taraon ket iti makuna nga "convertion method" iti agduduma nga datatypes.'
        'ken syak ket na gwapo'
    End Sub
End Module
```

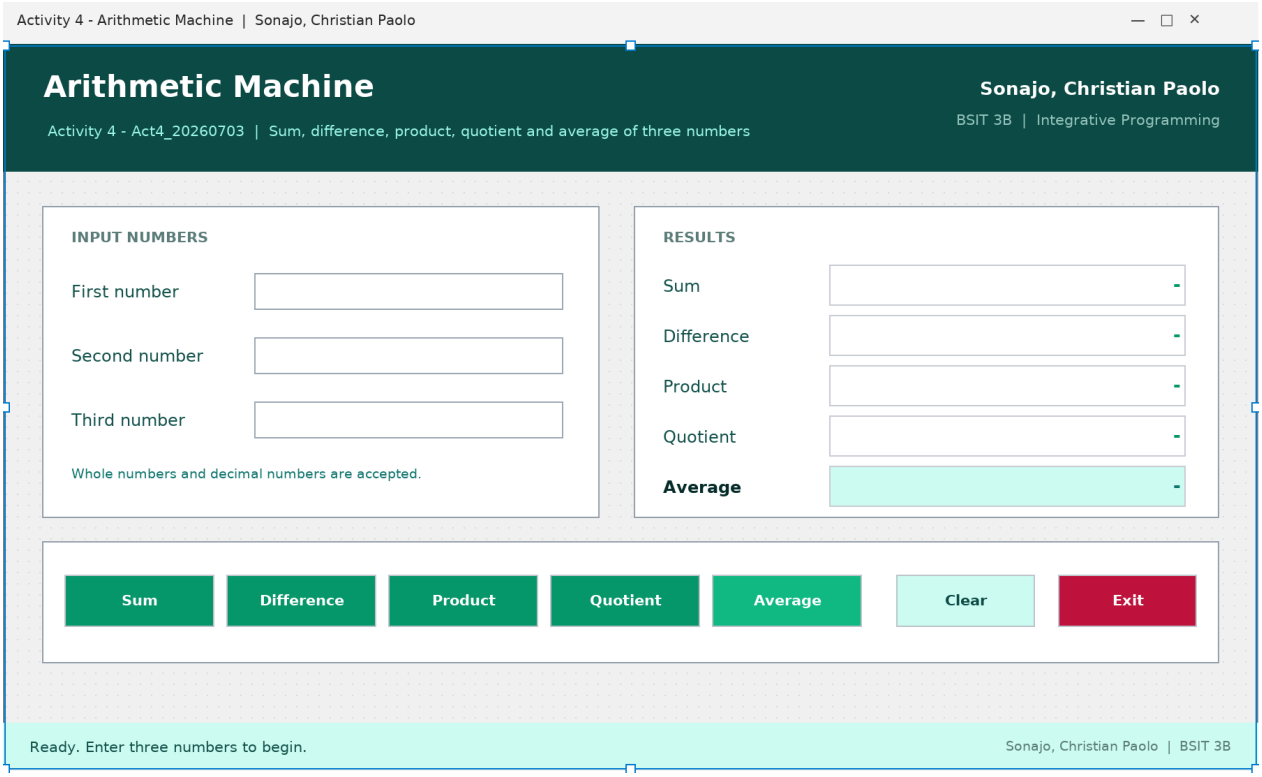
Sample Output

```
Activity3_20260629
Author: Sonajo, Christian Paolo
Course Year Section: BSIT 3B
Enter number 1: 10
Enter number 2: 5
Enter number 3: 2
The sum is: 17
The difference is: 3
The product is: 100
The quotient is: 1
The average is: 5.666666666666667
```

Activity 4 - Arithmetic Machine (Windows Form)

Activity4_20260703 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 4 - Arithmetic Machine

Source Code

Form1.vb

```
' ============================================================
' Activity 4 - Act4_20260703
' Arithmetic Machine : sum, difference, product, quotient and
' average of three numbers entered by the user.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Public Class Form1

    Private number1 As Double
    Private number2 As Double
    Private number3 As Double
    Private inputsAreValid As Boolean

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        ResetForm()
        txtN1.Focus()
    End Sub

    ' Reads the three text boxes and stores the values in the
    ' module level variables. Returns False when an entry is not a number.
    Private Function ReadInputs() As Boolean
        Try
            number1 = Convert.ToDouble(txtN1.Text.Trim())
            number2 = Convert.ToDouble(txtN2.Text.Trim())
            number3 = Convert.ToDouble(txtN3.Text.Trim())
            inputsAreValid = True
        Catch ex As Exception
            inputsAreValid = False
            ClearResults()
            lblStatus.Text = "Invalid entry. Please type numbers only (for example 12 or 12.5)."
        End Try
        Return inputsAreValid
    End Function

    Private Sub ShowResult(ByVal target As Label, ByVal value As Double, ByVal operation As String)
        target.Text = value.ToString("N4")
        lblStatus.Text = operation & " = " & value.ToString("N4")
    End Sub

    Private Sub ClearResults()
```

```
        lblSumValue.Text = "-"
        lblDiffValue.Text = "-"
        lblProdValue.Text = "-"
        lblQuotValue.Text = "-"
        lblAveValue.Text = "-"
    End Sub

    Private Sub ResetForm()
        txtN1.Clear()
        txtN2.Clear()
        txtN3.Clear()
        ClearResults()
        lblStatus.Text = "Ready. Enter three numbers to begin."
        txtN1.Focus()
    End Sub

    Private Sub btnSum_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnSum.Click
        If Not ReadInputs() Then Return
        ShowResult(lblSumValue, number1 + number2 + number3, "Sum")
    End Sub

    Private Sub btnDifference_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnDifference.Click
        If Not ReadInputs() Then Return
        ShowResult(lblDiffValue, number1 - number2 - number3, "Difference")
    End Sub

    Private Sub btnProduct_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnProduct.Click
        If Not ReadInputs() Then Return
        ShowResult(lblProdValue, number1 * number2 * number3, "Product")
    End Sub

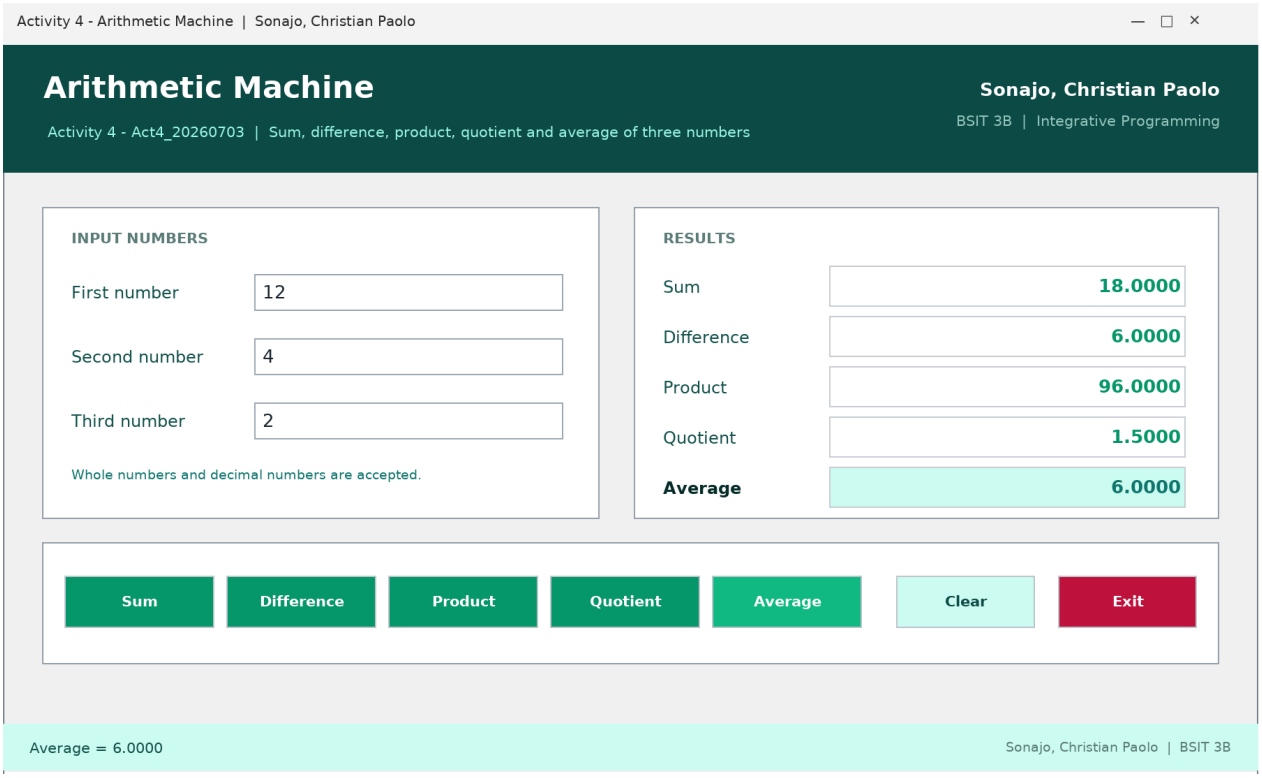
    Private Sub btnQuotient_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnQuotient.Click
        If Not ReadInputs() Then Return
        If number2 = 0 OrElse number3 = 0 Then
            lblQuotValue.Text = "undefined"
            lblStatus.Text = "Quotient is undefined because the second or third number is zero."
            Return
        End If
        ShowResult(lblQuotValue, number1 / number2 / number3, "Quotient")
    End Sub

    Private Sub btnAverage_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnAverage.Click
        If Not ReadInputs() Then Return
        ShowResult(lblAveValue, (number1 + number2 + number3) / 3, "Average")
    End Sub

    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnClear.Click
        ResetForm()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnExit.Click
        If MessageBox.Show("Close the Arithmetic Machine?", "Exit", _
                           MessageBoxButtons.YesNo, MessageBoxIcon.Question) = DialogResult.Yes Then
            Application.Exit()
        End If
    End Sub
End Class
```

Sample Output

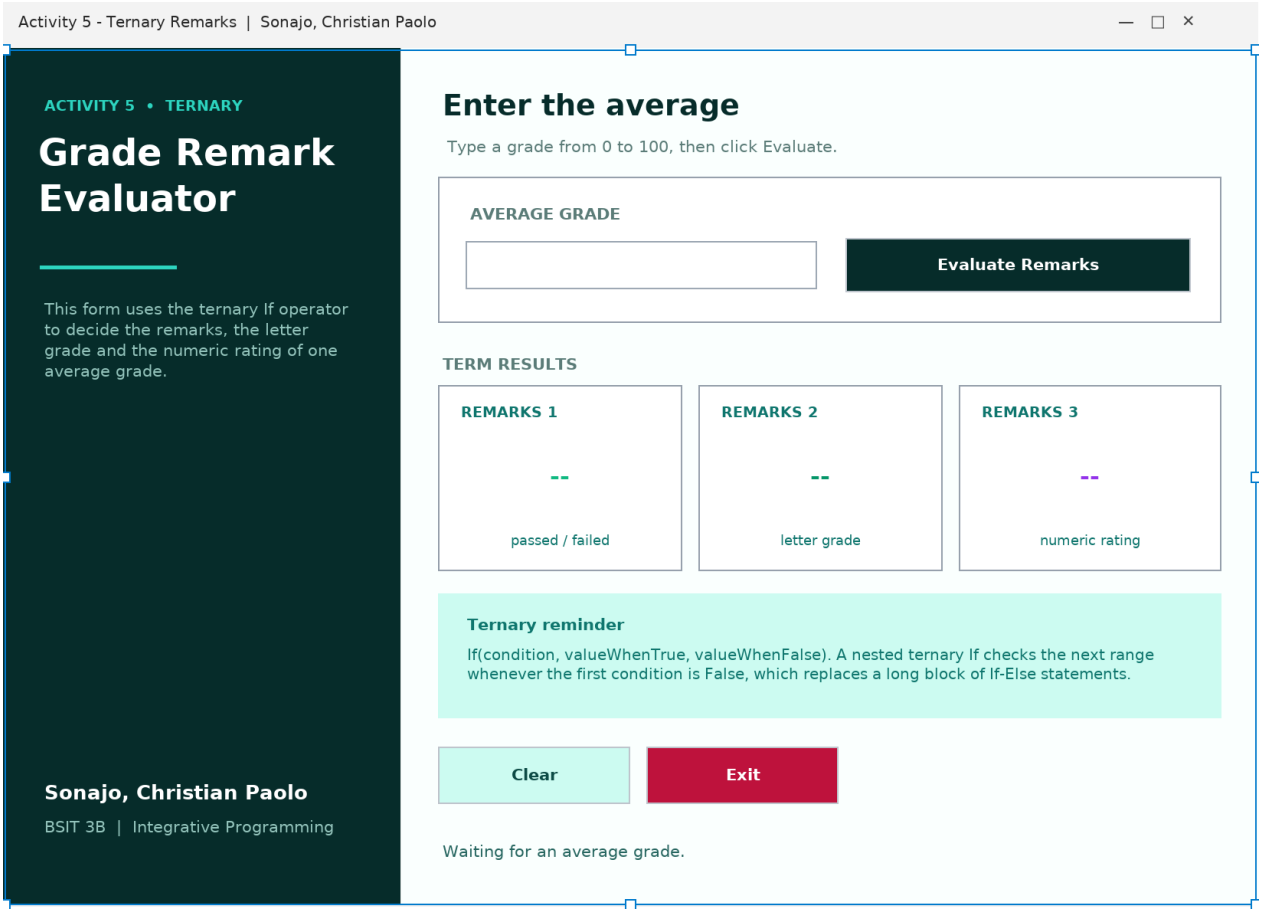


Arithmetic Machine after clicking Average

Activity 5 - Ternary Remarks (If Operator)

Act5 - Ternary2 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 5 - Ternary Remarks

Source Code

Form1.vb

```
' ============================================================
' Activity 5 - Ternary2
' Displays the remarks, the letter grade and the numeric rating
' of an average grade using nested ternary (If) operators.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Public Class Form1

    Private averageGrade As Double

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        ClearOutputs()
        lblStatus.Text = "Waiting for an average grade."
        txtAverage.Focus()
    End Sub

    Private Sub ClearOutputs()
        lblCard1Value.Text = "--"
        lblCard2Value.Text = "--"
        lblCard3Value.Text = "--"
    End Sub

    Private Sub txtAverage_KeyPress(ByVal sender As Object, ByVal e As System.Windows.Forms.KeyPressEventArgs)
    Handles txtAverage.KeyPress
        ' Enter key inside the textbox works like clicking the Evaluate button
        If e.KeyChar = Microsoft.VisualBasic.ChrW(13) Then
            btnEvaluate.PerformClick()
            e.Handled = True
        End If
    End Sub

    Private Sub btnEvaluate_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnEvaluate.Click
```

```
        Try
            averageGrade = Convert.ToDouble(txtAverage.Text.Trim())

            ' Remarks 1 : passed / conditional / failed
            lblCard1Value.Text = If(averageGrade >= 75, "Passed",
                                 If(averageGrade >= 70, "Conditional", "Failed"))

            ' Remarks 2 : letter grade
            lblCard2Value.Text = If(averageGrade >= 94, "A",
                                 If(averageGrade >= 89, "B",
                                 If(averageGrade >= 84, "C",
                                 If(averageGrade >= 80, "D",
                                 If(averageGrade >= 75, "E", "F")))))

            ' Remarks 3 : numeric rating
            lblCard3Value.Text = If(averageGrade >= 98, "1.00",
                                 If(averageGrade >= 95, "1.25",
                                 If(averageGrade >= 92, "1.50",
                                 If(averageGrade >= 89, "1.75",
                                 If(averageGrade >= 86, "2.00",
                                 If(averageGrade >= 83, "2.25",
                                 If(averageGrade >= 80, "2.50",
                                 If(averageGrade >= 77, "2.75",
                                 If(averageGrade >= 75, "3.00",
                                 If(averageGrade >= 70, "5.00", "Invalid Grade..."))))))))))

            If averageGrade > 100 OrElse averageGrade < 0 Then
                lblStatus.Text = "The grade " & averageGrade.ToString("0.##") & " is outside the 0 to 100
    range."
            Else
                lblStatus.Text = "Average " & averageGrade.ToString("0.##") & " evaluated using nested ternary
    If operators."
            End If

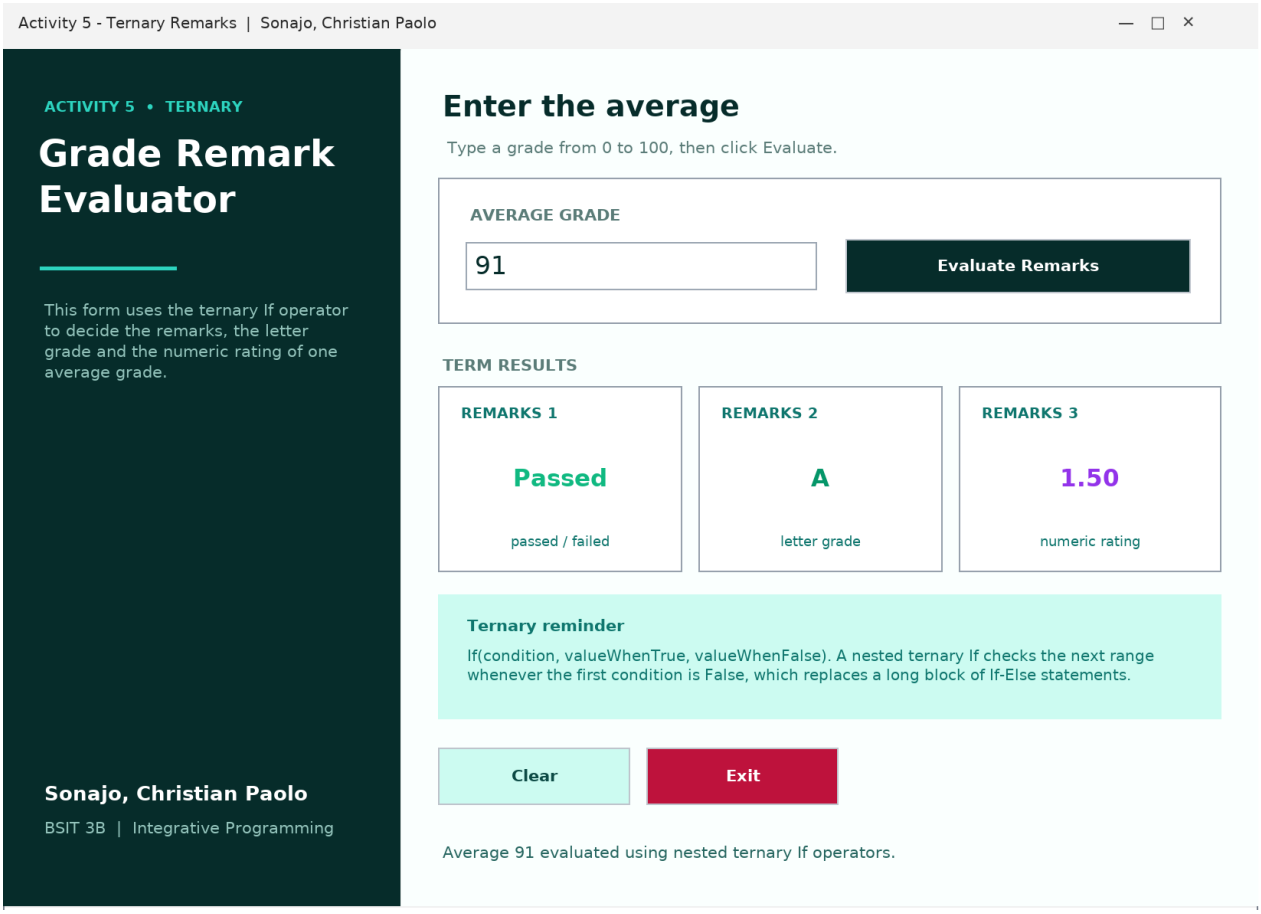
            ApplyCardColor(averageGrade)
        Catch ex As Exception
            ClearOutputs()
            lblCard1Value.Text = "Invalid"
            lblCard2Value.Text = "Invalid"
            lblCard3Value.Text = "Invalid"
            lblStatus.Text = "Please type a valid number, for example 88.5"
            txtAverage.SelectAll()
            txtAverage.Focus()
        End Try
    End Sub

    ' Small extra touch : the first card changes color with the result
    Private Sub ApplyCardColor(ByVal grade As Double)
        If grade >= 75 Then
            lblCard1Value.ForeColor = System.Drawing.Color.FromArgb(16, 185, 129)
        ElseIf grade >= 70 Then
            lblCard1Value.ForeColor = System.Drawing.Color.FromArgb(217, 119, 6)
        Else
            lblCard1Value.ForeColor = System.Drawing.Color.FromArgb(190, 18, 60)
        End If
    End Sub

    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnClear.Click
        txtAverage.Clear()
        ClearOutputs()
        lblStatus.Text = "Cleared. Type another average grade."
        txtAverage.Focus()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnExit.Click
        Application.Exit()
    End Sub
End Class
```

Sample Output

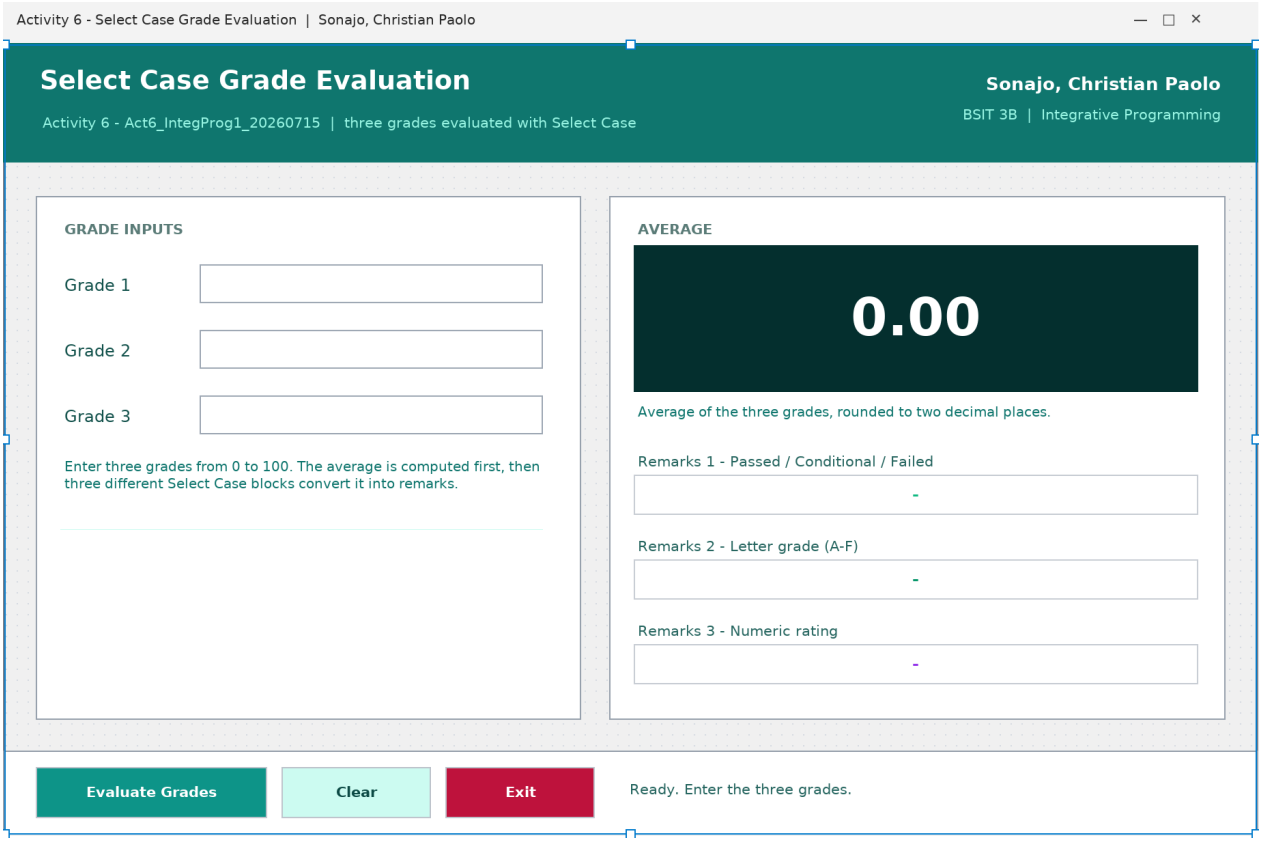


Ternary remarks for an average grade of 91

Activity 6 - Select Case Grade Evaluation

Act6_IntegProg1_20260715 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 6 - Select Case Grade Evaluation

Source Code

Form1.vb

```
' ============================================================
' Activity 6 - Act6_IntegProg1_20260715
' Reads three grades, computes the average and displays three
' different remarks using Select Case.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Public Class Form1

    Private grade1 As Double
    Private grade2 As Double
    Private grade3 As Double
    Private averageGrade As Double

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        ResetScreen()
    End Sub

    Private Sub ResetScreen()
        txtG1.Clear()
        txtG2.Clear()
        txtG3.Clear()
        lblAverage.Text = "0.00"
        lblRemarks1.Text = "-"
        lblRemarks2.Text = "-"
        lblRemarks3.Text = "-"
        lblRemarks1.ForeColor = System.Drawing.Color.FromArgb(16, 185, 129)
        pnlAveBadge.BackColor = System.Drawing.Color.FromArgb(4, 47, 46)
        lblStatus.Text = "Ready. Enter the three grades."
        txtG1.Focus()
    End Sub

    Private Function ReadGrades() As Boolean
        Try
            grade1 = Convert.ToDouble(txtG1.Text.Trim())
            grade2 = Convert.ToDouble(txtG2.Text.Trim())
            grade3 = Convert.ToDouble(txtG3.Text.Trim())
```

```
    Catch ex As Exception
        lblAverage.Text = "0.00"
        lblRemarks1.Text = "Invalid entry"
        lblRemarks2.Text = "Numbers only"
        lblRemarks3.Text = "Numbers only"
        lblStatus.Text = "Only numeric grades are accepted. Example : 88 or 91.5"
        Return False
    End Try

    If grade1 < 0 OrElse grade2 < 0 OrElse grade3 < 0 OrElse _
       grade1 > 100 OrElse grade2 > 100 OrElse grade3 > 100 Then
        lblStatus.Text = "Each grade must be from 0 to 100 only."
        lblRemarks1.Text = "Out of range"
        lblRemarks2.Text = "Out of range"
        lblRemarks3.Text = "Out of range"
        lblAverage.Text = "0.00"
        Return False
    End If

    Return True
End Function

Private Sub btnEvaluate_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
btnEvaluate.Click
    If Not ReadGrades() Then Return

    ' Average of the three grades (Double so decimals are kept)
    averageGrade = (grade1 + grade2 + grade3) / 3
    lblAverage.Text = averageGrade.ToString("0.00")

    ' Remarks 1 : passed / conditional / failed
    Select Case averageGrade
        Case Is >= 75
            lblRemarks1.Text = "Passed!"
        Case Is >= 70
            lblRemarks1.Text = "Conditional!"
        Case Else
            lblRemarks1.Text = "Failed!"
    End Select

    ' Remarks 2 : letter grade
    Select Case averageGrade
        Case Is >= 95
            lblRemarks2.Text = "A"
        Case Is >= 89
            lblRemarks2.Text = "B"
        Case Is >= 80
            lblRemarks2.Text = "C"
        Case Is >= 75
            lblRemarks2.Text = "D"
        Case Is >= 70
            lblRemarks2.Text = "E"
        Case Else
            lblRemarks2.Text = "F"
    End Select

    ' Remarks 3 : numeric rating
    Select Case averageGrade
        Case Is >= 98
            lblRemarks3.Text = "1.00"
        Case Is >= 95
            lblRemarks3.Text = "1.25"
        Case Is >= 92
            lblRemarks3.Text = "1.50"
        Case Is >= 89
            lblRemarks3.Text = "1.75"
        Case Is >= 86
            lblRemarks3.Text = "2.00"
        Case Is >= 83
            lblRemarks3.Text = "2.25"
        Case Is >= 80
            lblRemarks3.Text = "2.50"
        Case Is >= 77
            lblRemarks3.Text = "2.75"
        Case Is >= 75
            lblRemarks3.Text = "3.00"
        Case Is >= 70
            lblRemarks3.Text = "4.00"
        Case Else
            lblRemarks3.Text = "5.00"
    End Select

    ShowBadgeColor()
    lblStatus.Text = "Average " & averageGrade.ToString("0.00") & " evaluated with three Select Case
blocks."
End Sub

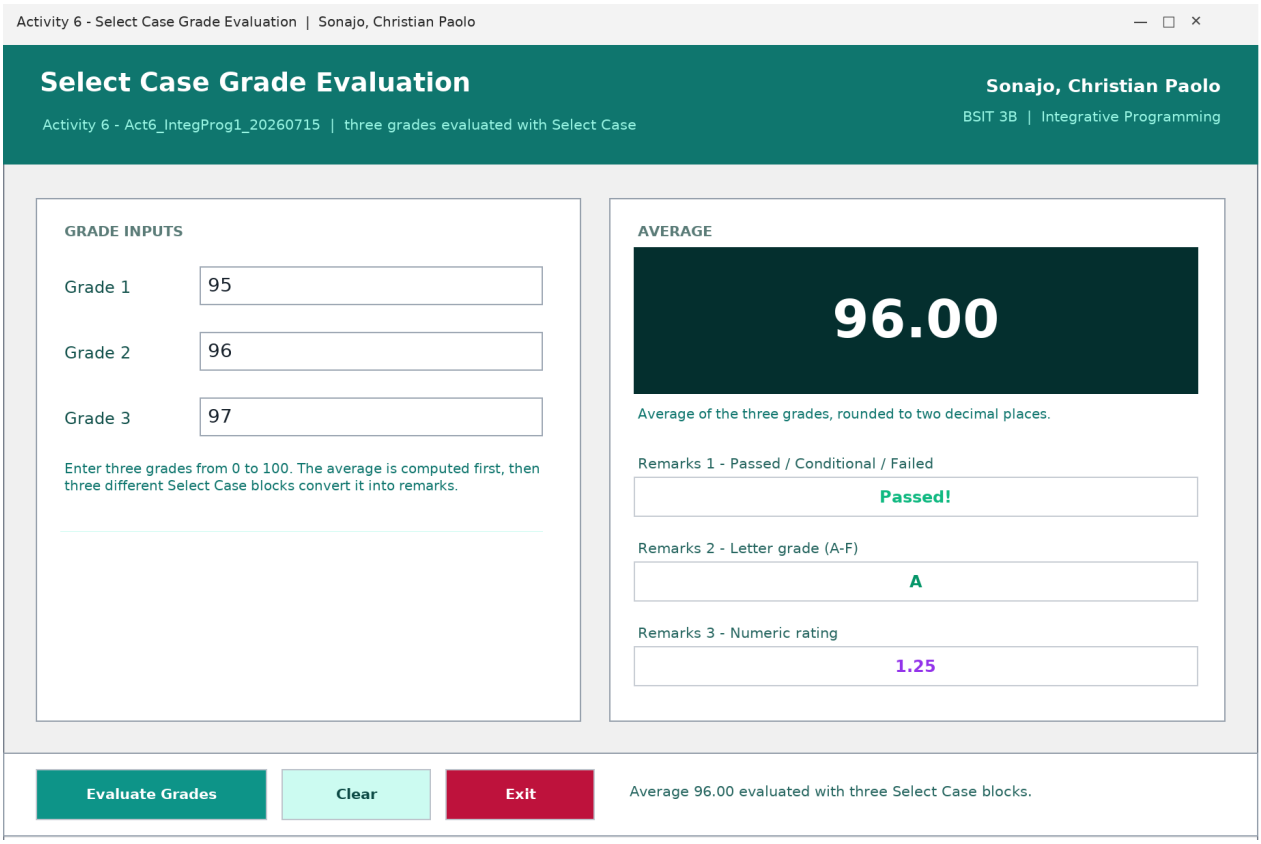
Private Sub ShowBadgeColor()
    If averageGrade >= 75 Then
        pnlAveBadge.BackColor = System.Drawing.Color.FromArgb(5, 150, 105)
        lblRemarks1.ForeColor = System.Drawing.Color.FromArgb(16, 185, 129)
    ElseIf averageGrade >= 70 Then
        pnlAveBadge.BackColor = System.Drawing.Color.FromArgb(120, 53, 15)
        lblRemarks1.ForeColor = System.Drawing.Color.FromArgb(217, 119, 6)
```

```
        Else
            pnlAveBadge.BackColor = System.Drawing.Color.FromArgb(127, 29, 29)
            lblRemarks1.ForeColor = System.Drawing.Color.FromArgb(190, 18, 60)
        End If
    End Sub

    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnClear.Click
        ResetScreen()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnExit.Click
        Application.Exit()
    End Sub
End Class
```

Sample Output

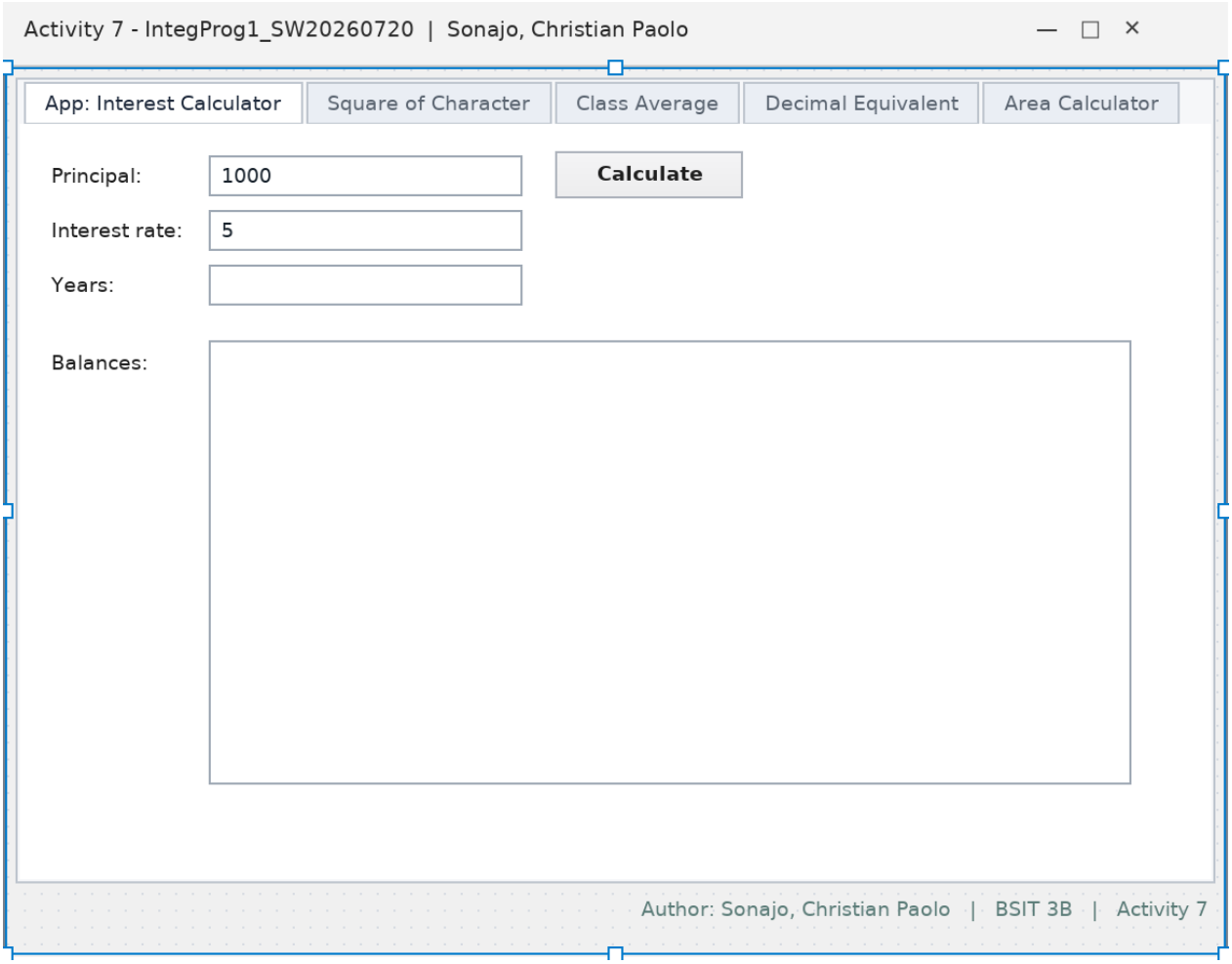


Select Case evaluation of the three grades

Activity 7 - Integrative Program 1 - Interest, Square and Class Average

Act7_IntegProg1_SW20260720 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 7 - the author label is at the bottom right

Source Code

Form1.vb

```
Public Class Form1

    Private Sub calculateButton_Click(ByVal sender As Object,
            ByVal e As EventArgs) Handles calculateButton.Click

        Dim principal As Decimal = Val(principalTextBox.Text)
        Dim rate As Decimal = Val(interestRateTextBox.Text)
        Dim amount As Decimal

        resultsListBox.Items.Add(
            "Year" & vbTab & "Amount on Deposit")

        For yearCounter As Integer = 1 To yearsNumericUpDown.Value
            amount = principal * ((1 + rate / 100) ^ yearCounter)
            resultsListBox.Items.Add(yearCounter & vbTab &
                String.Format("{0:C}", amount))
        Next
    End Sub

    Private Sub principalTextBox_TextChanged(ByVal sender As Object,
            ByVal e As EventArgs) Handles principalTextBox.TextChanged
        resultsListBox.Items.Clear()
    End Sub

    Private Sub interestRateTextBox_TextChanged(ByVal sender As Object,
            ByVal e As EventArgs) Handles interestRateTextBox.TextChanged
        resultsListBox.Items.Clear()
```

```
End Sub

Private Sub yearsNumericUpDown_ValueChanged(ByVal sender As Object,
        ByVal e As EventArgs) Handles yearsNumericUpDown.ValueChanged
    resultsListBox.Items.Clear()
End Sub

Private Sub displaySquareButton_Click(ByVal sender As Object,
        ByVal e As EventArgs) Handles displaySquareButton.Click

    outputTextBox.Clear()

    Dim fillCharacter As String = fillCharacterTextBox.Text
    Dim sideLength As Integer = sideLengthNumericUpDown.Value

    For row As Integer = 1 To sideLength
        For column As Integer = 1 To sideLength
            outputTextBox.AppendText(fillCharacter & " ")
        Next column

        outputTextBox.AppendText(vbCrLf)
    Next row
End Sub

Private Sub fillCharacterTextBox_TextChanged(ByVal sender As Object,
        ByVal e As EventArgs) Handles fillCharacterTextBox.TextChanged
    outputTextBox.Clear()
End Sub

Private Sub sideLengthNumericUpDown_ValueChanged(ByVal sender As Object,
        ByVal e As EventArgs) Handles sideLengthNumericUpDown.ValueChanged
    outputTextBox.Clear()
End Sub

Dim aCount As Integer
Dim bCount As Integer
Dim cCount As Integer
Dim dCount As Integer
Dim fCount As Integer
Dim perfectScoreCount As Integer

Private Sub submitGradeButton_Click(ByVal sender As Object,
        ByVal e As EventArgs) Handles submitGradeButton.Click

    Dim grade As Integer = Val(gradeTextBox.Text)

    gradesListBox.Items.Add(grade)
    gradeTextBox.Clear()

    Select Case grade
        Case 100
            perfectScoreCount += 1
            aCount += 1
        Case 90 To 99
            aCount += 1
        Case 80 To 89
            bCount += 1
        Case 70 To 79
            cCount += 1
        Case 60 To 69
            dCount += 1
        Case Else
            fCount += 1
    End Select

    gradeTextBox.Focus()
End Sub

Private Sub calculateAverageButton_Click(ByVal sender As Object,
        ByVal e As EventArgs) Handles calculateAverageButton.Click

    Dim total As Integer
    Dim gradeCounter As Integer
    Dim grade As Integer
    Dim average As Double

    total = 0
    gradeCounter = 0

    Do While gradeCounter < gradesListBox.Items.Count
        grade = gradesListBox.Items(gradeCounter)
        total += grade
        gradeCounter += 1
    Loop

    If gradeCounter <> 0 Then
        average = total / gradeCounter

        classAverageLabel.Text = "Total of the " & gradeCounter &
            " grade(s) is " & total & vbCrLf & "Class average is " &
            String.Format("{0:F}", average) & vbCrLf & vbCrLf

        classAverageLabel.Text &=
            "Letter grade summary:" & vbCrLf & "A: " & aCount &
```

```
                vbCrLf & "B: " & bCount & vbCrLf & "C: " & cCount &
                vbCrLf & "D: " & dCount & vbCrLf & "F: " & fCount &
                vbCrLf & "Perfect scores: " & perfectScoreCount
        Else
            classAverageLabel.Text = "No grades were entered"
        End If
    End Sub

    Private Sub clearGradesButton_Click(ByVal sender As Object,
            ByVal e As EventArgs) Handles clearGradesButton.Click

        gradesListBox.Items.Clear()
        classAverageLabel.Text = String.Empty
    End Sub

    Private Sub Form1_Load(ByVal sender As Object,
            ByVal e As EventArgs) Handles MyBase.Load

        decimalOutputTextBox.AppendText("Decimal" & vbTab & "Octal" &
            vbTab & "Hexadecimal" & vbTab & "Binary" & vbCrLf)

        For number As Integer = 1 To 255
            decimalOutputTextBox.AppendText(number & vbTab &
                Convert.ToString(number, 8) & vbTab &
                Convert.ToString(number, 16).ToUpper() & vbTab & vbTab &
                Convert.ToString(number, 2) & vbCrLf)
        Next number
    End Sub

    Private Sub decimalOutputTextBox_TextChanged(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles decimalOutputTextBox.TextChanged

    End Sub

    Private Sub tabClassAverage_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    tabClassAverage.Click

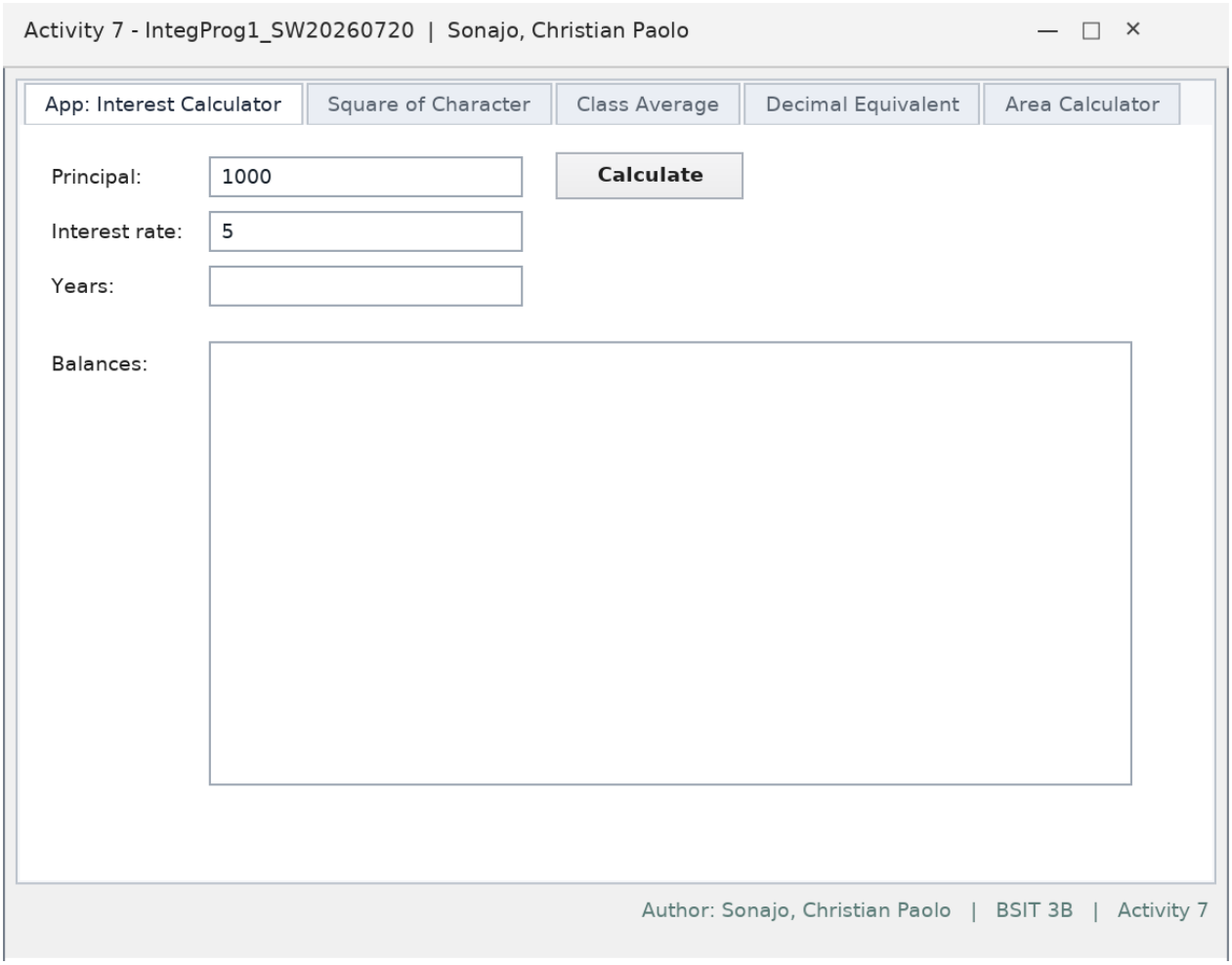
    End Sub

    Private Sub TabPage1_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    TabPage1.Click
        'square, rectangle, trapesoid, triangle, circle
    End Sub

    Private Sub TabPage2_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    TabPage2.Click

    End Sub
End Class
```

Sample Output

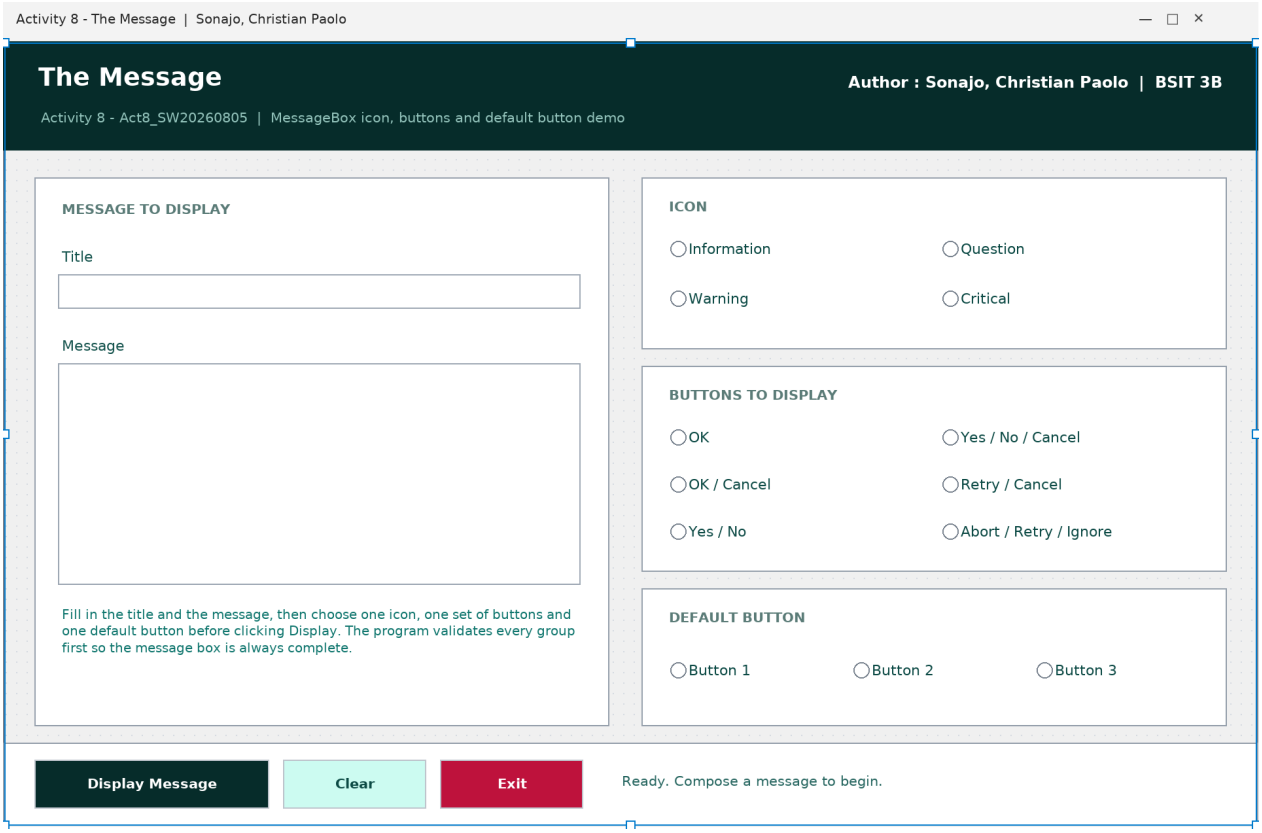


Interest, square and class average after computing

Activity 8 - The Message (MessageBox Options)

Act8_SW20260805 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 8 - The Message

Source Code

Form1.vb

```
' ============================================================
' Activity 8 - Act8_SW20260805
' The Message : composes a MessageBox with a title, a message,
' an icon, a button set and a default button.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Public Class Form1

    Private Sub Form1_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        PrepareForm()
    End Sub

    Private Sub PrepareForm()
        txtTitle.Clear()
        txtMessage.Clear()
        ClearRadioButtons(pnlIcons)
        ClearRadioButtons(pnlButtons)
        ClearRadioButtons(pnlDefault)
        lblStatus.Text = "Ready. Compose a message to begin."
        txtTitle.Focus()
    End Sub

    Private Sub ClearRadioButtons(ByVal host As Control)
        For Each child As Control In host.Controls
            If TypeOf child Is RadioButton Then
                CType(child, RadioButton).Checked = False
            End If
        Next
    End Sub

    ' ------------------------ validation ------------------------
    Private Function ValidateEntry() As Boolean
        If txtTitle.Text.Trim() = "" Then
            MessageBox.Show("Please enter a title.", "Missing Title", _
                            MessageBoxButtons.OK, MessageBoxIcon.Warning)
            txtTitle.Focus()
            Return False
```

```
    End If

    If txtMessage.Text.Trim() = "" Then
        MessageBox.Show("Please enter a message.", "Missing Message", _
                        MessageBoxButtons.OK, MessageBoxIcon.Warning)
        txtMessage.Focus()
        Return False
    End If

    If Not (rbInfo.Checked OrElse rbWarning.Checked OrElse rbQuestion.Checked OrElse rbCritical.Checked)
Then
        MessageBox.Show("Please select one icon.", "Icon Required", _
                        MessageBoxButtons.OK, MessageBoxIcon.Warning)
        Return False
    End If

    If Not (rbOk.Checked OrElse rbOkCancel.Checked OrElse rbYesNo.Checked OrElse _
            rbYesNoCancel.Checked OrElse rbRetryCancel.Checked OrElse rbAbortRetryIgnore.Checked) Then
        MessageBox.Show("Please select the buttons to display.", "Buttons Required", _
                        MessageBoxButtons.OK, MessageBoxIcon.Warning)
        Return False
    End If

    If Not (rbDefault1.Checked OrElse rbDefault2.Checked OrElse rbDefault3.Checked) Then
        MessageBox.Show("Please select the default button.", "Default Button Required", _
                        MessageBoxButtons.OK, MessageBoxIcon.Warning)
        Return False
    End If

    Return True
End Function

' --------------------- selected MessageBox parts ---------------------
Private Function ChosenIcon() As MessageBoxIcon
    If rbInfo.Checked Then Return MessageBoxIcon.Information
    If rbWarning.Checked Then Return MessageBoxIcon.Warning
    If rbQuestion.Checked Then Return MessageBoxIcon.Question
    Return MessageBoxIcon.Error
End Function

Private Function ChosenButtons() As MessageBoxButtons
    If rbOk.Checked Then Return MessageBoxButtons.OK
    If rbOkCancel.Checked Then Return MessageBoxButtons.OKCancel
    If rbYesNo.Checked Then Return MessageBoxButtons.YesNo
    If rbYesNoCancel.Checked Then Return MessageBoxButtons.YesNoCancel
    If rbRetryCancel.Checked Then Return MessageBoxButtons.RetryCancel
    Return MessageBoxButtons.AbortRetryIgnore
End Function

Private Function ChosenDefaultButton() As MessageBoxDefaultButton
    If rbDefault2.Checked Then Return MessageBoxDefaultButton.Button2
    If rbDefault3.Checked Then Return MessageBoxDefaultButton.Button3
    Return MessageBoxDefaultButton.Button1
End Function

Private Function DescribeResult(ByVal answer As DialogResult) As String
    Select Case answer
        Case DialogResult.OK
            Return "OK was clicked."
        Case DialogResult.Cancel
            Return "Cancel was clicked."
        Case DialogResult.Yes
            Return "Yes was clicked."
        Case DialogResult.No
            Return "No was clicked."
        Case DialogResult.Retry
            Return "Retry was clicked."
        Case DialogResult.Abort
            Return "Abort was clicked."
        Case DialogResult.Ignore
            Return "Ignore was clicked."
        Case Else
            Return "The message box was closed without a button answer."
    End Select
End Function

Private Sub btnDisplay_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
btnDisplay.Click
    If Not ValidateEntry() Then Return

    Try
        Dim answer As DialogResult = MessageBox.Show(txtMessage.Text, txtTitle.Text, _
                                                     ChosenButtons(), ChosenIcon(), _
                                                     ChosenDefaultButton())

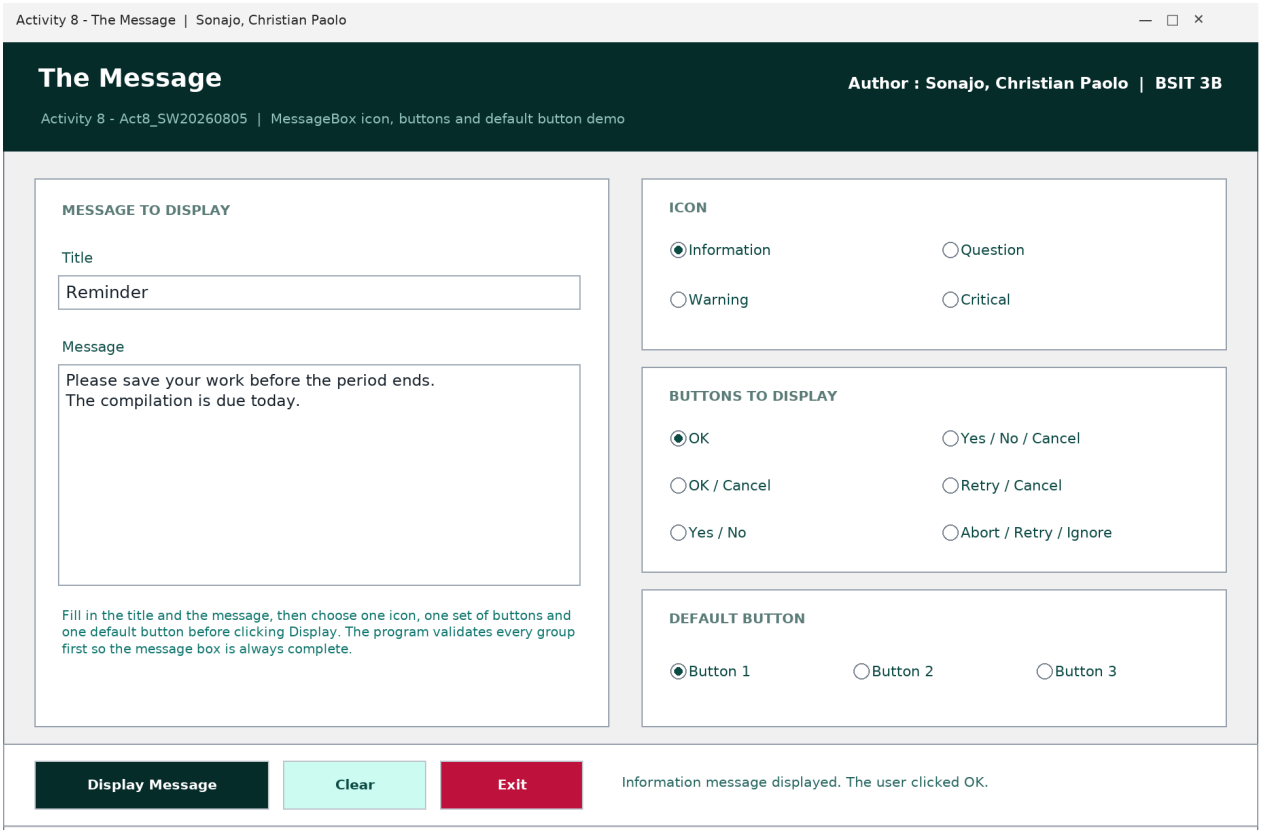
        lblStatus.Text = DescribeResult(answer)

        MessageBox.Show(DescribeResult(answer), "Result", _
                        MessageBoxButtons.OK, MessageBoxIcon.Information)
    Catch ex As Exception
        MessageBox.Show(ex.Message, "There is an error", _
                        MessageBoxButtons.OK, MessageBoxIcon.Error)
    End Try
End Sub
```

```
    Private Sub btnClear_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    btnClear.Click
        PrepareForm()
    End Sub

    Private Sub btnExit_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles btnExit.Click
        Application.Exit()
    End Sub
End Class
```

Sample Output

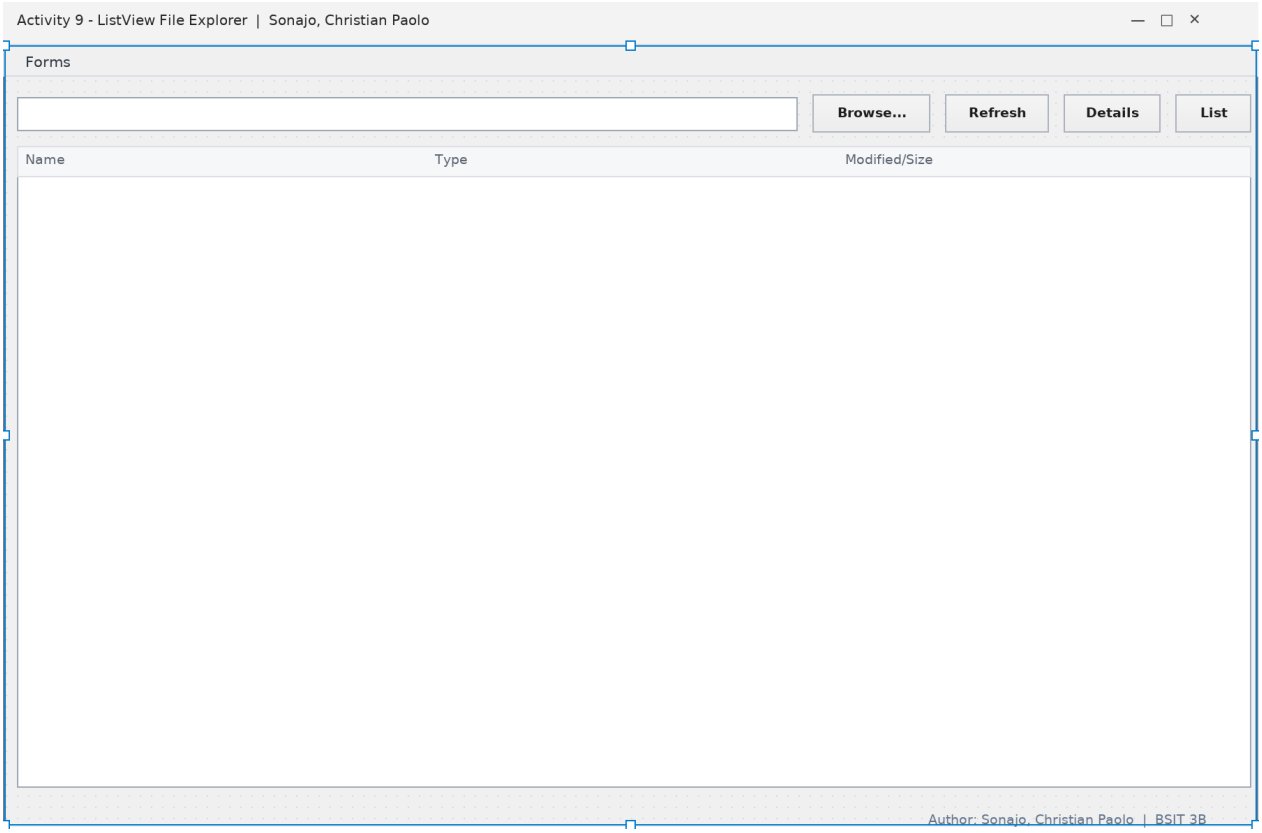


Message composed with the Information icon and the OK button

Activity 9 - ListView File Explorer

SW202609301600 | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 9 - the author label shares the status line

Source Code

ListViewSample.vb

```
Imports System
Imports System.IO
Imports System.Diagnostics
Imports System.Windows.Forms

Public Class ListViewSample
    Private currentFolder As String = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments)

    Public Sub New()
        InitializeComponent()
        AddHandler menuList.Click, AddressOf OpenListForm

        AddHandler menuExit.Click, AddressOf ExitApplication
    End Sub

    Private Sub OpenListForm(ByVal sender As Object, ByVal e As EventArgs)
        Using form As New ListViewSample()
            form.ShowDialog()
        End Using
    End Sub


    Private Sub ExitApplication(ByVal sender As Object, ByVal e As EventArgs)
        Me.Close()
    End Sub

    Private Sub Form1_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        LoadFolder(currentFolder)
    End Sub

    Private Sub LoadFolder(ByVal folder As String)
        If Not Directory.Exists(folder) Then Return
        currentFolder = folder
        txtPath.Text = folder
        lv.Items.Clear()
        Try
            For Each directoryName As String In Directory.GetDirectories(folder)
```

```
                Dim item As New ListViewItem(Path.GetFileName(directoryName))
                item.SubItems.Add("Folder")
                item.SubItems.Add(Directory.GetLastWriteTime(directoryName).ToString())
                item.Tag = directoryName
                lv.Items.Add(item)
            Next
            For Each fileName As String In Directory.GetFiles(folder)
                Dim info As New FileInfo(fileName)
                Dim item As New ListViewItem(info.Name)
                item.SubItems.Add(info.Extension)
                item.SubItems.Add(info.Length.ToString() & " bytes")
                item.Tag = fileName
                lv.Items.Add(item)
            Next
            lbl.Text = lv.Items.Count.ToString() & " item(s)"
        Catch ex As Exception
            lbl.Text = ex.Message
        End Try
    End Sub

    Private Sub browse_Click(ByVal sender As Object, ByVal e As EventArgs) Handles browse.Click
        Using dialog As New FolderBrowserDialog()
            dialog.SelectedPath = currentFolder
            dialog.Description = "Select a folder"
            If dialog.ShowDialog() = DialogResult.OK Then
                LoadFolder(dialog.SelectedPath)
            End If
        End Using
    End Sub

    Private Sub btnRefresh_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnRefresh.Click
        LoadFolder(currentFolder)
    End Sub

    Private Sub details_Click(ByVal sender As Object, ByVal e As EventArgs) Handles details.Click
        lv.View = View.Details
    End Sub

    Private Sub list_Click(ByVal sender As Object, ByVal e As EventArgs) Handles list.Click
        lv.View = View.List
    End Sub

    Private Sub lv_DoubleClick(ByVal sender As Object, ByVal e As EventArgs) Handles lv.DoubleClick
        If lv.SelectedItems.Count = 0 Then Return
        Dim selectedPath As String = CStr(lv.SelectedItems(0).Tag)
        If Directory.Exists(selectedPath) Then
            LoadFolder(selectedPath)
        ElseIf File.Exists(selectedPath) Then
            Process.Start(selectedPath)
        End If
    End Sub

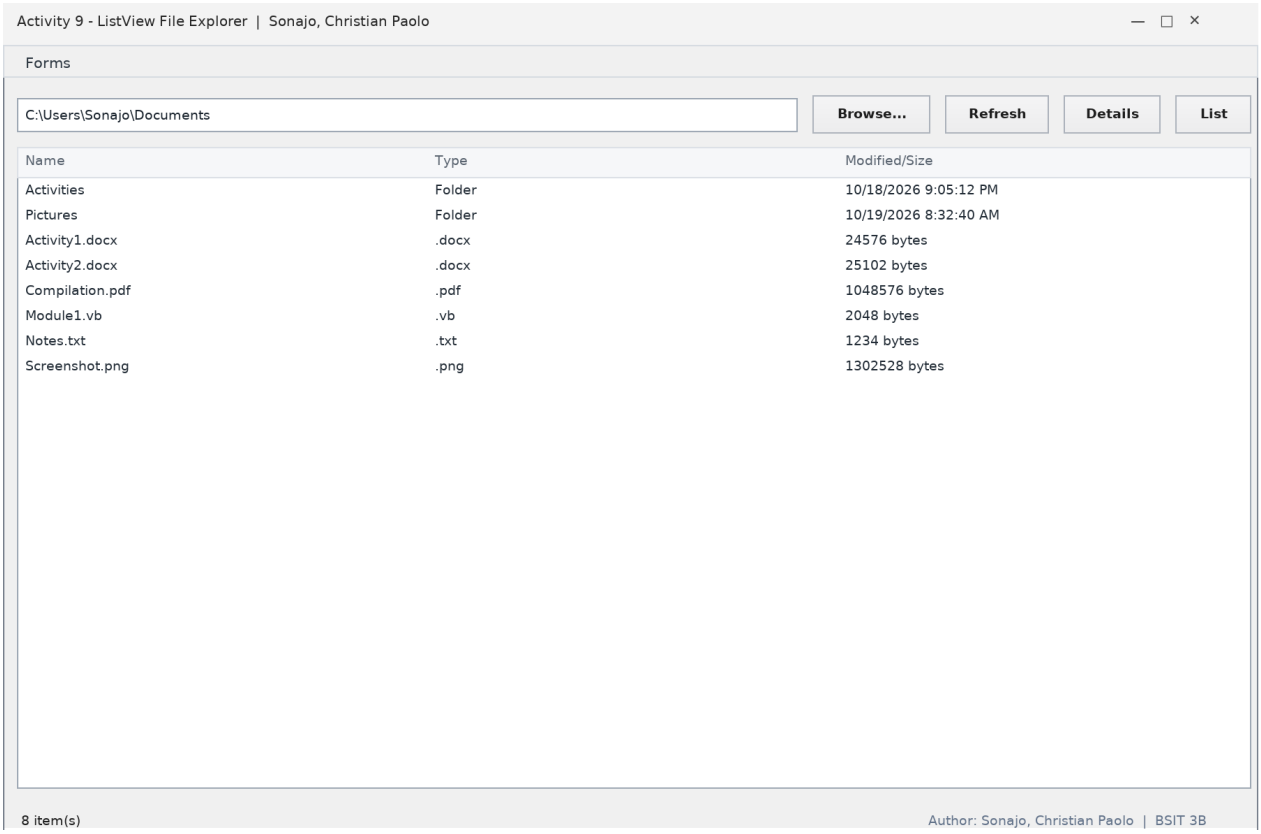
    Private Sub lv_ColumnClick(ByVal sender As Object, ByVal e As ColumnClickEventArgs) Handles lv.ColumnClick
        lv.ListViewItemSorter = New ListViewComparer(e.Column)
        lv.Sort()
    End Sub

    Private Class ListViewComparer
        Implements System.Collections.IComparer
        Private ReadOnly columnNumber As Integer

        Public Sub New(ByVal column As Integer)
            columnNumber = column
        End Sub

        Public Function Compare(ByVal first As Object, ByVal second As Object) As Integer Implements
    System.Collections.IComparer.Compare
            Dim a As ListViewItem = DirectCast(first, ListViewItem)
            Dim b As ListViewItem = DirectCast(second, ListViewItem)
            Return String.Compare(a.SubItems(columnNumber).Text, b.SubItems(columnNumber).Text, True)
        End Function
    End Class
End Class
```

Sample Output



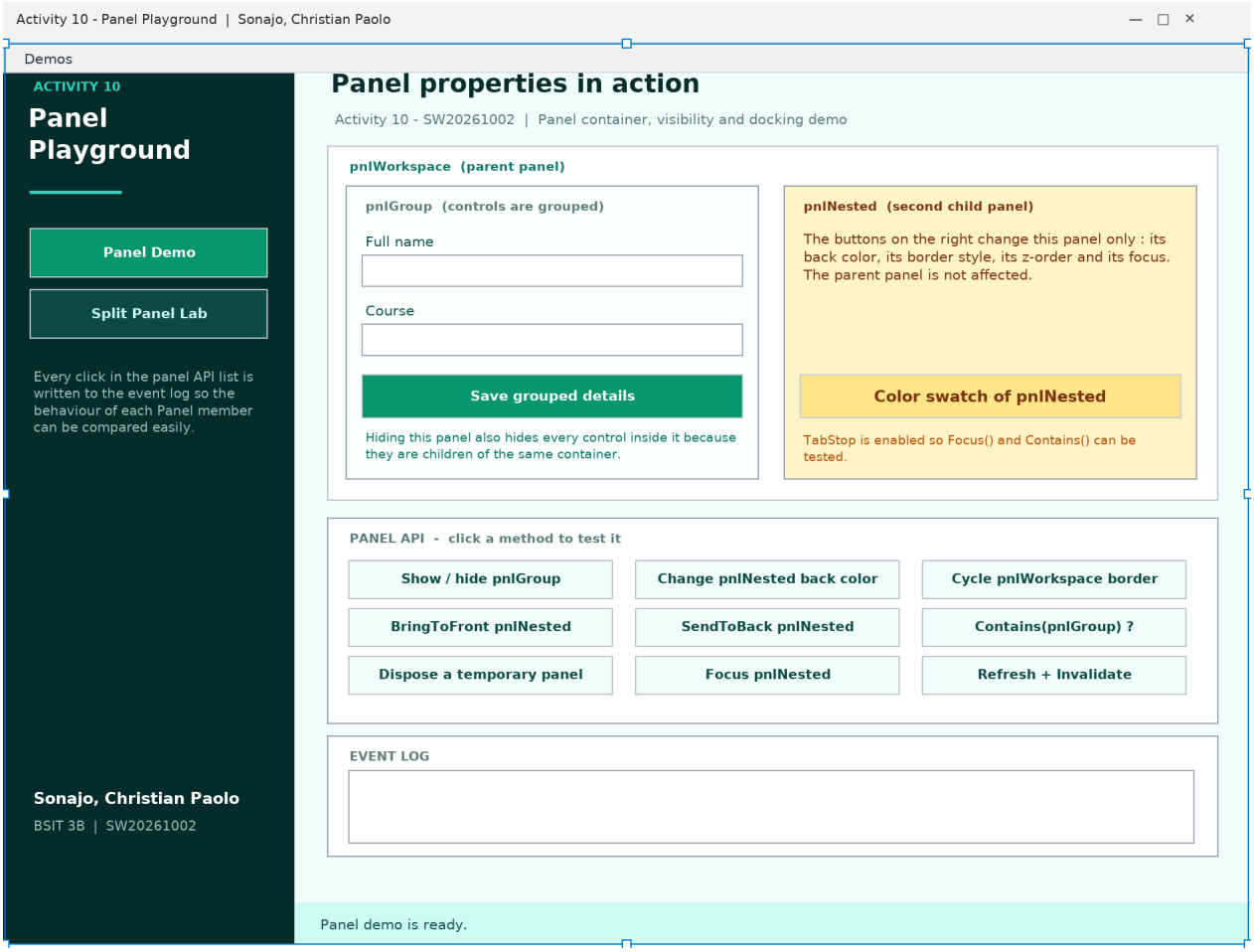
Files of a folder listed in the ListView

Activity 10 - Panel and SplitContainer Laboratory

SW20261002 - PanelSplitPanelDemo | Sonajo, Christian Paolo | BSIT 3B

Form1 - Panel Playground

Form Design



Form design - Form1

Source Code

Form1.vb

```
' ============================================================
' Activity 10 - SW20261002  (Panel Demo)
' Demonstrates the most common members of the Panel control :
' Visible, BackColor, BorderStyle, BringToFront, SendToBack,
' Contains, Focus, Refresh, Invalidate and Dispose.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Option Strict On
Option Explicit On

Public Class Form1

    Private actionCounter As Integer = 0
    Private colorIndex As Integer = 0
    Private borderIndex As Integer = 0

    Private ReadOnly palette As Color() = { _
        Color.FromArgb(254, 243, 199), _
        Color.FromArgb(204, 251, 241), _
        Color.FromArgb(220, 252, 231)}

    Private Sub Form1_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        WriteLog("Form1 loaded - Panel Playground is ready.")
        WriteLog("pnlWorkspace contains " & pnlWorkspace.Controls.Count.ToString() & " child control(s).")
        pnlNested.TabStop = True
        lblStatus.Text = "Panel demo is ready."
    End Sub
```

```
' ------------------------------------------------------------
' helper routines
' ------------------------------------------------------------
Private Sub WriteLog(ByVal message As String)
    actionCounter += 1
    lstLog.Items.Insert(0, actionCounter.ToString("00") & "  " & message)
    lblStatus.Text = message
    lblCounter.Text = actionCounter.ToString() & " action(s) performed"
End Sub

Private Sub OpenSplitLab()
    Dim lab As New Form2()
    lab.Owner = Me
    AddHandler lab.FormClosed, AddressOf SplitLabClosed
    lab.Show()
    Me.Hide()
End Sub

Private Sub SplitLabClosed(ByVal sender As Object, ByVal e As FormClosedEventArgs)
    If Not Me.IsDisposed Then
        Me.Show()
        WriteLog("Split Panel Lab was closed, Panel Playground is shown again.")
    End If
End Sub

' ------------------------------------------------------------
' navigation
' ------------------------------------------------------------
Private Sub btnNavSplit_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnNavSplit.Click
    OpenSplitLab()
End Sub

Private Sub mnuSplitLab_Click(ByVal sender As Object, ByVal e As EventArgs) Handles mnuSplitLab.Click
    OpenSplitLab()
End Sub

Private Sub btnNavPanel_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnNavPanel.Click
    WriteLog("You are already on the Panel demo.")
End Sub

Private Sub mnuExit_Click(ByVal sender As Object, ByVal e As EventArgs) Handles mnuExit.Click
    Application.Exit()
End Sub

' ------------------------------------------------------------
' grouped controls demo
' ------------------------------------------------------------
Private Sub btnSave_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnSave.Click
    Dim fullName As String = txtName.Text.Trim()
    Dim course As String = txtCourse.Text.Trim()

    If fullName = "" Then
        WriteLog("Nothing saved - the full name box is empty.")
        txtName.Focus()
        Return
    End If

    If course = "" Then
        course = "no course given"
    End If

    WriteLog("Saved from pnlGroup : " & fullName & " - " & course)
End Sub

' ------------------------------------------------------------
' Panel API demo buttons
' ------------------------------------------------------------
Private Sub btnToggleGroup_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnToggleGroup.Click
    pnlGroup.Visible = Not pnlGroup.Visible
    btnToggleGroup.Text = If(pnlGroup.Visible, "Show / hide pnlGroup", "pnlGroup is hidden - click to show")
    WriteLog("pnlGroup.Visible = " & pnlGroup.Visible.ToString())
End Sub

Private Sub btnSwapColor_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnSwapColor.Click
    colorIndex += 1
    If colorIndex > palette.Length - 1 Then colorIndex = 0

    pnlNested.BackColor = palette(colorIndex)
    lblNestedBadge.BackColor = ControlPaint.Light(palette(colorIndex))
    WriteLog("pnlNested.BackColor changed to " & ColorName(palette(colorIndex)) & ".")
End Sub

Private Function ColorName(ByVal value As Color) As String
    If value = palette(0) Then Return "amber"
    If value = palette(1) Then Return "blue"
    Return "green"
End Function

Private Sub btnCycleBorder_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnCycleBorder.Click
    borderIndex += 1
    Select Case borderIndex Mod 3
        Case 0
            pnlWorkspace.BorderStyle = BorderStyle.Fixed3D
        Case 1
```

```
                pnlWorkspace.BorderStyle = BorderStyle.FixedSingle
            Case 2
                pnlWorkspace.BorderStyle = BorderStyle.None
        End Select
        WriteLog("pnlWorkspace.BorderStyle = " & pnlWorkspace.BorderStyle.ToString())
    End Sub

    Private Sub btnBringFront_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnBringFront.Click
        pnlNested.BringToFront()
        WriteLog("pnlNested.BringToFront() - the child panel is now in front of pnlGroup.")
    End Sub

    Private Sub btnSendBack_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnSendBack.Click
        pnlNested.SendToBack()
        WriteLog("pnlNested.SendToBack() - the child panel moved behind pnlGroup.")
    End Sub

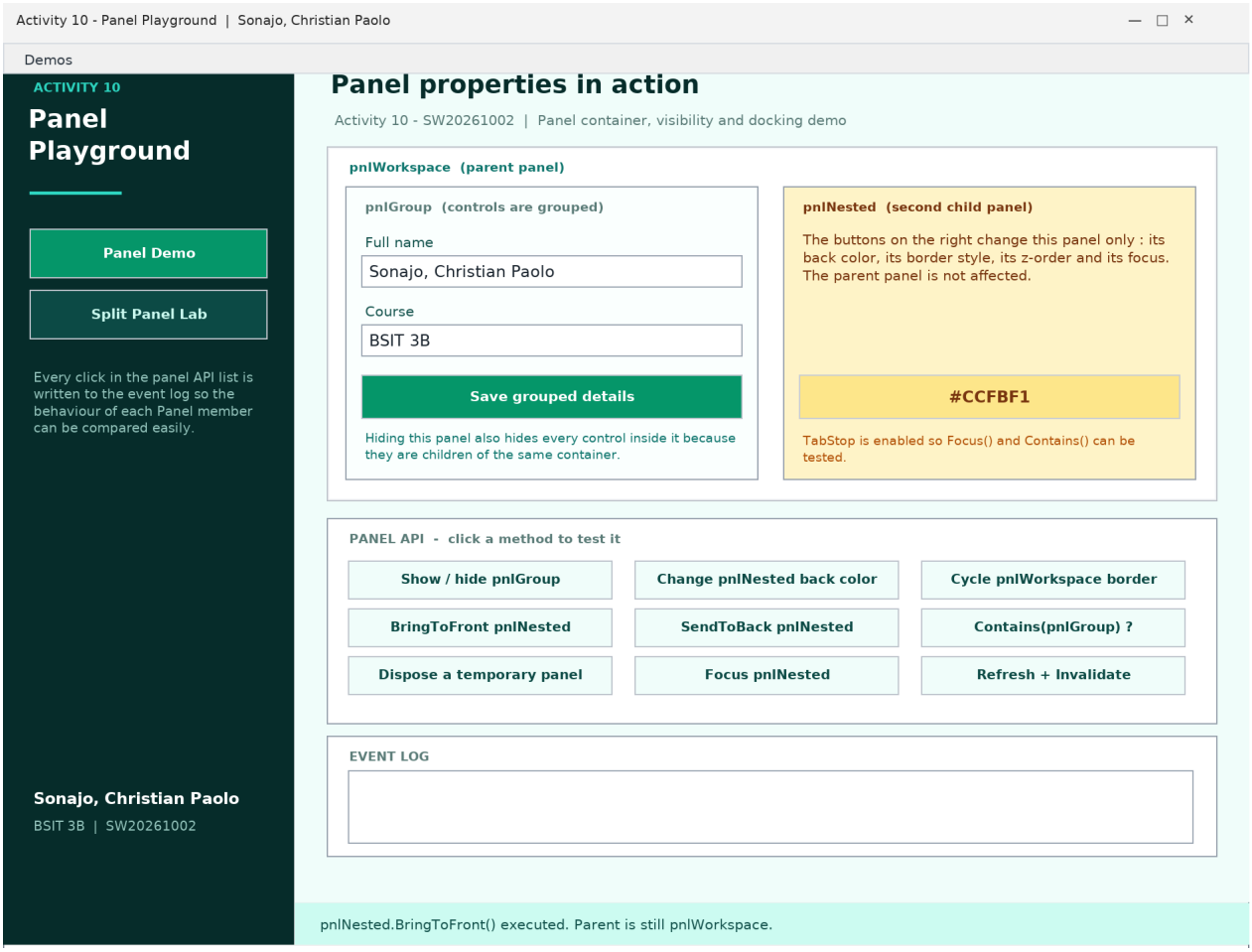
    Private Sub btnContains_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnContains.Click
        Dim hasGroup As Boolean = pnlWorkspace.Contains(pnlGroup)
        Dim hasNested As Boolean = pnlWorkspace.Contains(pnlNested)
        Dim hasOutsider As Boolean = pnlMain.Contains(pnlGroup)
        WriteLog("Contains -> pnlGroup : " & hasGroup.ToString() & _
                 ", pnlNested : " & hasNested.ToString() & _
                 ", from pnlMain : " & hasOutsider.ToString())
    End Sub

    Private Sub btnDisposeDemo_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnDisposeDemo.Click
        Dim temporary As New Panel()
        temporary.Controls.Add(New Label())
        temporary.Controls.Add(New TextBox())
        Dim childCount As Integer = temporary.Controls.Count
        temporary.Dispose()
        WriteLog("A temporary panel with " & childCount.ToString() & " child control(s) was disposed safely.")
    End Sub

    Private Sub btnFocusPanel_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnFocusPanel.Click
        Dim gotFocus As Boolean = pnlNested.Focus()
        WriteLog("pnlNested.Focus() returned " & gotFocus.ToString() & ".")
    End Sub

    Private Sub btnRedraw_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnRedraw.Click
        pnlNested.Refresh()
        pnlWorkspace.Invalidate(True)
        WriteLog("Refresh() and Invalidate(True) requested a redraw of the child panels.")
    End Sub
End Class
```

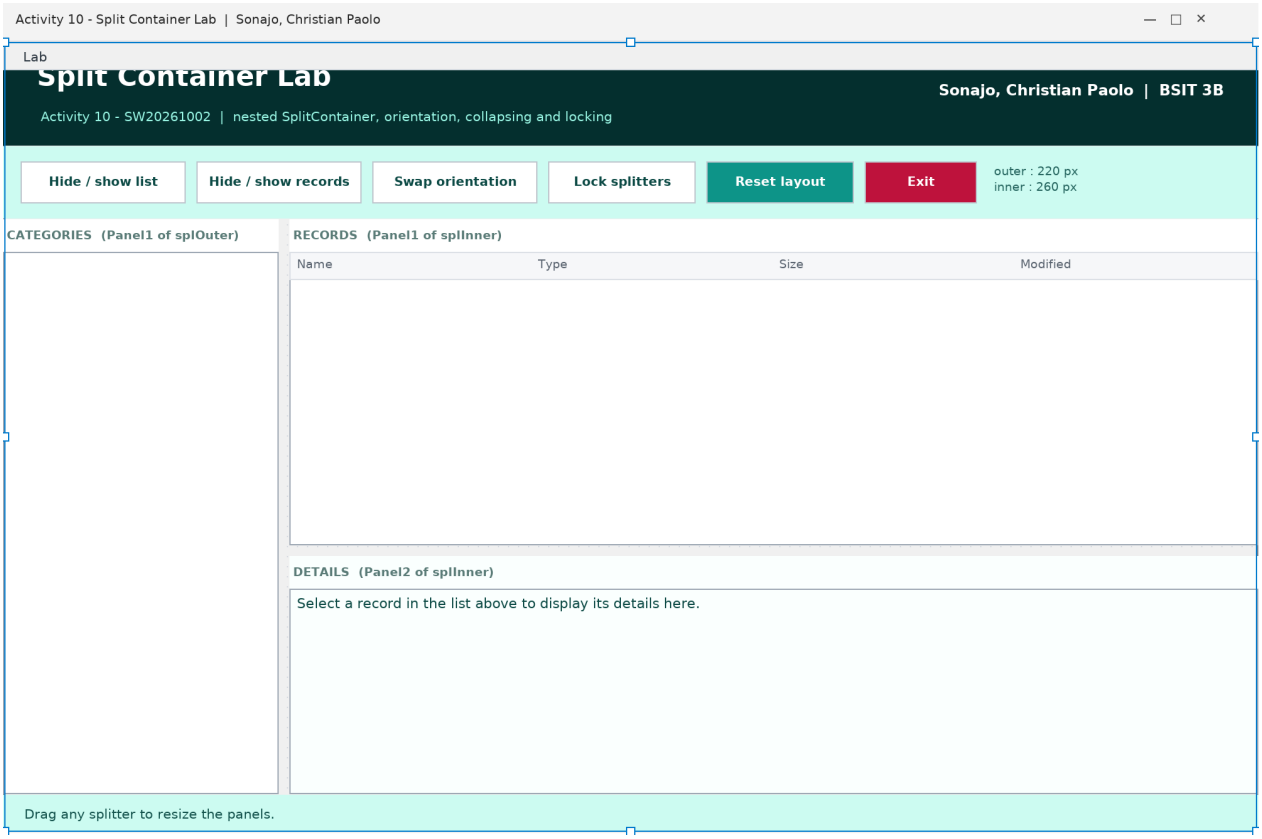
Sample Output



Panel Playground after three panel API calls

Form2 - Split Container Lab

Form Design



Form design - Form2

Source Code

Form2.vb

```
' ============================================================
' Activity 10 - SW20261002  (SplitContainer Demo)
' Nested SplitContainer practice : two panels in the outer
' splitter, two panels in the inner splitter.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Option Strict On
Option Explicit On

Public Class Form2

    ' one line of sample data
    Private Class RecordItem
        Public Name As String
        Public Category As String
        Public Status As String
        Public Detail As String

        Public Sub New(ByVal recordName As String, ByVal recordCategory As String, _
                       ByVal recordStatus As String, ByVal recordDetail As String)
            Name = recordName
            Category = recordCategory
            Status = recordStatus
            Detail = recordDetail
        End Sub
    End Class

    Private records As New List(Of RecordItem)()
    Private splittersLocked As Boolean = False

    Private Sub Form2_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        BuildSampleData()
        FillCategories()
        ApplyDefaultLayout()

        If lstCategories.Items.Count > 0 Then lstCategories.SelectedIndex = 0

        lblStatus.Text = "Drag any splitter to resize the panels."
        lblCount.Text = records.Count.ToString() & " sample record(s) loaded"
    End Sub

    ' ------------------------------------------------------------
    ' sample data
```

```
' ------------------------------------------------------------
Private Sub BuildSampleData()
    records.Add(New RecordItem("Student Masterlist", "Students", "Active", _
        "The student masterlist holds the complete list of enrolled students of BSIT 3B." & vbCrLf & _
        "Fields : student number, full name, year level, section and status."))
    records.Add(New RecordItem("Grade Sheet", "Students", "Active", _
        "The grade sheet stores the midterm and final grades of every student." & vbCrLf & _
        "Average and remarks are computed automatically."))
    records.Add(New RecordItem("Course Catalog", "Courses", "Active", _
        "The course catalog lists every subject with its code, title and unit count."))
    records.Add(New RecordItem("Class Schedule", "Courses", "Draft", _
        "The class schedule combines a course, a teacher, a day, a time and a room."))
    records.Add(New RecordItem("Enrollment Slip", "Enrollments", "Active", _
        "The enrollment slip is printed for each student after the subjects are confirmed."))
    records.Add(New RecordItem("Subject Load", "Enrollments", "Pending", _
        "The subject load shows the subjects and schedules of one student for the term."))
    records.Add(New RecordItem("Enrollment Report", "Reports", "Active", _
        "A summary of the number of enrolled students per course and per year level."))
    records.Add(New RecordItem("Grade Summary", "Reports", "Draft", _
        "A report that groups the class by remarks : passed, conditional and failed."))
    records.Add(New RecordItem("User Accounts", "Settings", "Active", _
        "User accounts control who can add, edit or delete the records of the system."))
    records.Add(New RecordItem("Backup Preferences", "Settings", "Pending", _
        "Backup preferences decide how often the database file is copied to a safe folder."))
End Sub

Private Sub FillCategories()
    lstCategories.Items.Clear()
    lstCategories.Items.Add("All records")
    lstCategories.Items.Add("Students")
    lstCategories.Items.Add("Courses")
    lstCategories.Items.Add("Enrollments")
    lstCategories.Items.Add("Reports")
    lstCategories.Items.Add("Settings")
End Sub

Private Sub lstCategories_SelectedIndexChanged(ByVal sender As Object, ByVal e As EventArgs) Handles
lstCategories.SelectedIndexChanged
    ShowRecords(CStr(lstCategories.SelectedItem))
End Sub

Private Sub ShowRecords(ByVal category As String)
    lvRecords.BeginUpdate()
    lvRecords.Items.Clear()

    For Each record As RecordItem In records
        If category = "All records" OrElse record.Category = category Then
            Dim row As New ListViewItem(record.Name)
            row.SubItems.Add(record.Category)
            row.SubItems.Add(record.Status)
            row.Tag = record
            lvRecords.Items.Add(row)
        End If
    Next

    lvRecords.EndUpdate()
    txtDetails.Text = "Select a record in the list above to display its details here."
    lblStatus.Text = If(category = "All records", "Showing every record", "Showing the " & category & "
records") & _
                     " - " & lvRecords.Items.Count.ToString() & " item(s)."
    lblCount.Text = "outer splitter : " & splOuter.SplitterDistance.ToString() & " px"
End Sub

Private Sub lvRecords_SelectedIndexChanged(ByVal sender As Object, ByVal e As EventArgs) Handles
lvRecords.SelectedIndexChanged
    If lvRecords.SelectedItems.Count = 0 Then Return

    Dim chosen As RecordItem = CType(lvRecords.SelectedItems(0).Tag, RecordItem)
    txtDetails.Text = chosen.Name & vbCrLf & _
                      "Category : " & chosen.Category & vbCrLf & _
                      "Status : " & chosen.Status & vbCrLf & vbCrLf & _
                      chosen.Detail & vbCrLf & vbCrLf & _
                      "This text is shown inside Panel2 of the inner SplitContainer."
    lblStatus.Text = "Selected : " & chosen.Name
End Sub

' ------------------------------------------------------------
' split container layout helpers
' ------------------------------------------------------------
Private Sub ApplyDefaultLayout()
    splOuter.Panel1Collapsed = False
    splOuter.Panel2Collapsed = False
    splInner.Panel1Collapsed = False
    splInner.Panel2Collapsed = False
    splInner.Orientation = Orientation.Horizontal

    SetRatioDistance(splOuter, 0.24)
    SetRatioDistance(splInner, 0.62)
    ShowSplitInfo()
End Sub

' The distance is given as a ratio of the available space so the
' splitter never goes beyond its minimum sizes.
Private Sub SetRatioDistance(ByVal splitter As SplitContainer, ByVal ratio As Double)
```

```
    Dim available As Integer
    If splitter.Orientation = Orientation.Vertical Then
        available = splitter.ClientSize.Width
    Else
        available = splitter.ClientSize.Height
    End If

    Dim lowest As Integer = splitter.Panel1MinSize
    Dim highest As Integer = available - splitter.Panel2MinSize - splitter.SplitterWidth
    If highest <= lowest Then Return

    Dim requested As Integer = CInt(available * ratio)
    If requested < lowest Then requested = lowest
    If requested > highest Then requested = highest
    splitter.SplitterDistance = requested
End Sub

Private Sub ShowSplitInfo()
    lblSplitInfo.Text = "outer : " & splOuter.SplitterDistance.ToString() & " px" & vbCrLf & _
                        "inner : " & splInner.SplitterDistance.ToString() & " px"
End Sub

Private Sub btnCollapseLeft_Click(ByVal sender As Object, ByVal e As EventArgs) Handles
btnCollapseLeft.Click
    splOuter.Panel1Collapsed = Not splOuter.Panel1Collapsed
    lblStatus.Text = "splOuter.Panel1Collapsed = " & splOuter.Panel1Collapsed.ToString()
    ShowSplitInfo()
End Sub

Private Sub btnCollapseTop_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnCollapseTop.Click
    splInner.Panel1Collapsed = Not splInner.Panel1Collapsed
    lblStatus.Text = "splInner.Panel1Collapsed = " & splInner.Panel1Collapsed.ToString()
    ShowSplitInfo()
End Sub

Private Sub btnSwapOrientation_Click(ByVal sender As Object, ByVal e As EventArgs) Handles
btnSwapOrientation.Click
    Try
        If splInner.Orientation = Orientation.Horizontal Then
            splInner.Orientation = Orientation.Vertical
            lblStatus.Text = "Orientation = Vertical, the inner panels are now left and right."
        Else
            splInner.Orientation = Orientation.Horizontal
            lblStatus.Text = "Orientation = Horizontal, the inner panels are now top and bottom."
        End If
        SetRatioDistance(splInner, 0.62)
        ShowSplitInfo()
    Catch ex As Exception
        lblStatus.Text = "The orientation could not be changed : " & ex.Message
    End Try
End Sub

Private Sub btnLock_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnLock.Click
    splittersLocked = Not splittersLocked
    splOuter.IsSplitterFixed = splittersLocked
    splInner.IsSplitterFixed = splittersLocked
    btnLock.Text = If(splittersLocked, "Unlock splitters", "Lock splitters")
    lblStatus.Text = "IsSplitterFixed = " & splittersLocked.ToString()
End Sub

Private Sub btnReset_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnReset.Click
    splittersLocked = False
    splOuter.IsSplitterFixed = False
    splInner.IsSplitterFixed = False
    btnLock.Text = "Lock splitters"
    ApplyDefaultLayout()
    lblStatus.Text = "The default layout and splitter distances were restored."
End Sub

Private Sub splOuter_SplitterMoved(ByVal sender As Object, ByVal e As SplitterEventArgs) Handles
splOuter.SplitterMoved
    ShowSplitInfo()
    lblStatus.Text = "Outer splitter moved to " & splOuter.SplitterDistance.ToString() & " pixels."
End Sub

Private Sub splInner_SplitterMoved(ByVal sender As Object, ByVal e As SplitterEventArgs) Handles
splInner.SplitterMoved
    ShowSplitInfo()
    lblStatus.Text = "Inner splitter moved to " & splInner.SplitterDistance.ToString() & " pixels."
End Sub

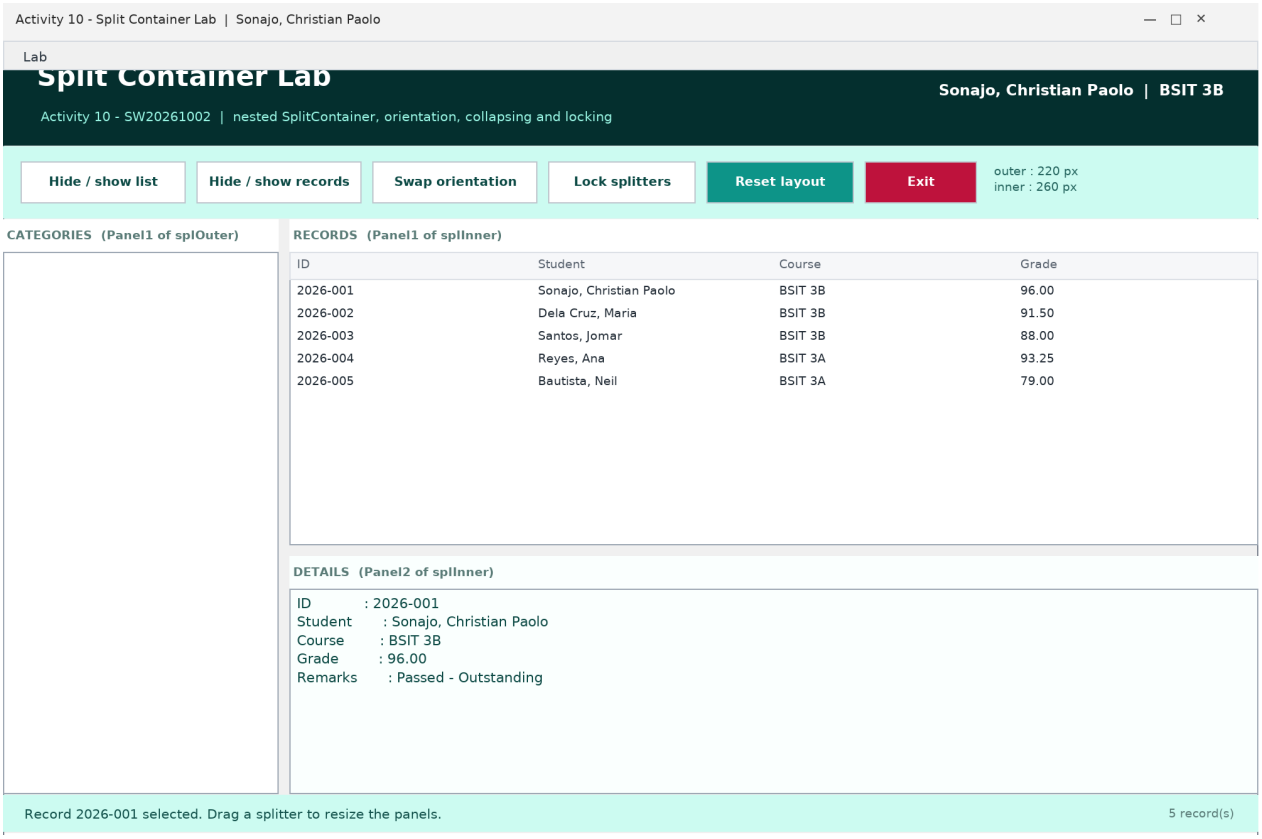
' ------------------------------------------------------------
' navigation
' ------------------------------------------------------------
Private Sub mnuPanelDemo_Click(ByVal sender As Object, ByVal e As EventArgs) Handles mnuPanelDemo.Click
    Me.Close()
End Sub

Private Sub mnuExit_Click(ByVal sender As Object, ByVal e As EventArgs) Handles mnuExit.Click
    Application.Exit()
End Sub

Private Sub btnExit_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnExit.Click
    Application.Exit()
```

```
    End Sub
End Class
```

Sample Output



Split Container Lab with a record selected

Activity 11 - Windows Forms Controls (Menu, TreeView, OpenFileDialog and Combined Explorer)

SW202610021600 | Sonajo, Christian Paolo | BSIT 3B

MenuForm

Form Design



Form design - MenuForm

Source Code

MenuForm.vb

```
Imports System
Imports System.Drawing
Imports System.Windows.Forms

Public Class MenuForm
    Inherits Form

    Private menuStrip1 As MenuStrip
    Private listMenu As ToolStripMenuItem
    Private treeMenu As ToolStripMenuItem
    Private openMenu As ToolStripMenuItem
    Private combinedMenu As ToolStripMenuItem
    Private exitMenu As ToolStripMenuItem

    Public Sub New()
        Me.Text = "Activity 11 - SW202610021600  |  Sonajo, Christian Paolo"
        Me.StartPosition = FormStartPosition.CenterScreen
        Me.Size = New Size(720, 400)
        menuStrip1 = New MenuStrip()
        treeMenu = New ToolStripMenuItem("TreeView File Explorer")
        openMenu = New ToolStripMenuItem("OpenFileDialog File Explorer")
        combinedMenu = New ToolStripMenuItem("Combined File Explorer")
        exitMenu = New ToolStripMenuItem("Exit")
        menuStrip1.Items.AddRange(New ToolStripItem() {treeMenu, openMenu, combinedMenu, exitMenu})
        menuStrip1.Dock = DockStyle.Top
        Me.Controls.Add(menuStrip1)
        Me.MainMenuStrip = menuStrip1

        AddHandler treeMenu.Click, AddressOf OpenTree
        AddHandler openMenu.Click, AddressOf OpenDialog
        AddHandler combinedMenu.Click, AddressOf OpenCombined
        AddHandler exitMenu.Click, AddressOf ExitProgram
    End Sub


    Private Sub OpenTree(ByVal sender As Object, ByVal e As EventArgs)
```

```
        Using form As New TreeViewSample()
            form.ShowDialog()
        End Using
    End Sub

    Private Sub OpenDialog(ByVal sender As Object, ByVal e As EventArgs)
        Using form As New OpenFileDialogSample()
            form.ShowDialog()
        End Using
    End Sub

    Private Sub OpenCombined(ByVal sender As Object, ByVal e As EventArgs)
        Using form As New CombinedSample()
            form.ShowDialog()
        End Using
    End Sub

    Private Sub ExitProgram(ByVal sender As Object, ByVal e As EventArgs)
        Me.Close()
    End Sub

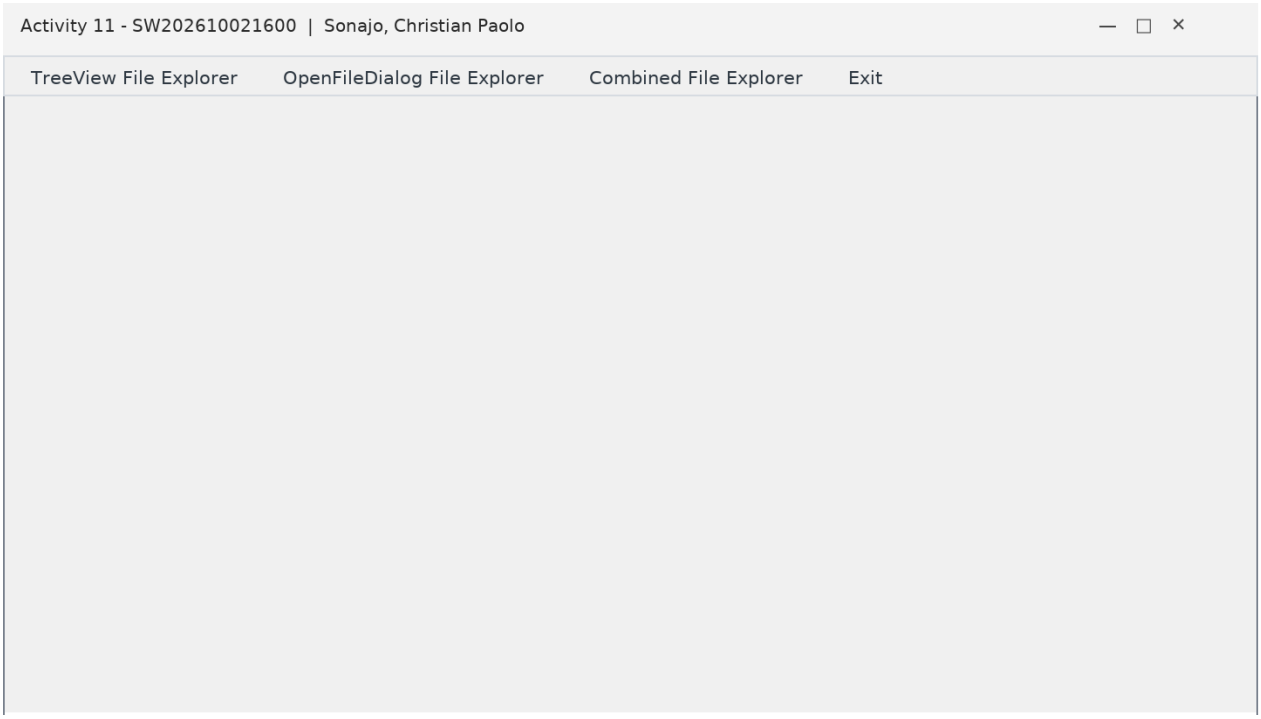
    Private Sub InitializeComponent()
        Me.SuspendLayout()
        '
        'MenuForm
        '
        Me.ClientSize = New System.Drawing.Size(729, 387)
        Me.Name = "MenuForm"
        Me.ResumeLayout(False)

    End Sub

    Private Sub MenuForm_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load

    End Sub
End Class
```

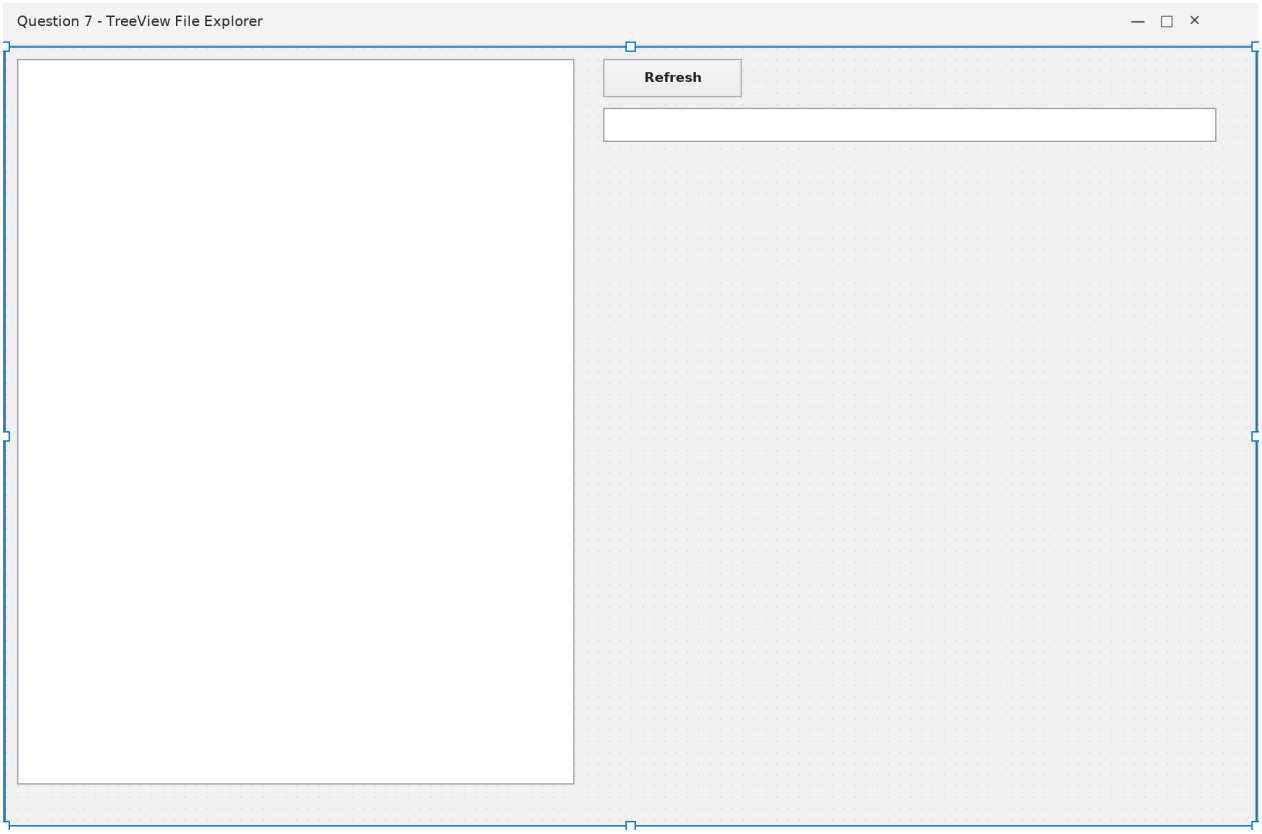
Sample Output



MenuForm - the menu that opens each explorer

TreeViewSample

Form Design



Form design - TreeViewSample

Source Code

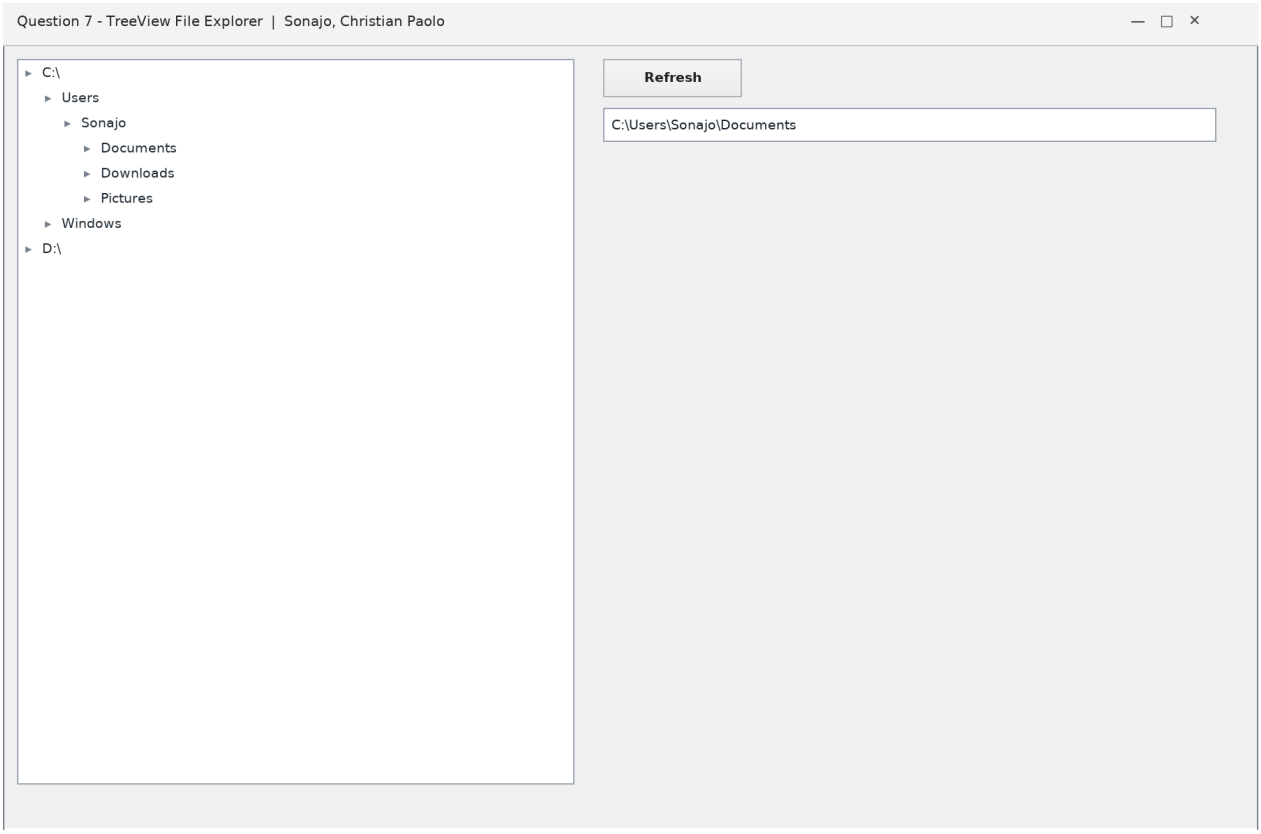
TreeViewSample.vb

```
Imports System
Imports System.IO
Imports System.Windows.Forms

Public Class TreeViewSample
    Public Sub New()
        InitializeComponent()
    End Sub
    Private Sub Form1_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        BuildTree()
    End Sub
    Private Sub BuildTree()
        tree.Nodes.Clear()
        For Each drive As DriveInfo In DriveInfo.GetDrives()
            If drive.IsReady Then
                Dim node As New TreeNode(drive.Name)
                node.Tag = drive.RootDirectory.FullName
                node.Nodes.Add("Loading...")
                tree.Nodes.Add(node)
            End If
        Next
    End Sub
    Private Sub tree_BeforeExpand(ByVal sender As Object, ByVal e As TreeViewCancelEventArgs) Handles
    tree.BeforeExpand
        Dim folder As String = CStr(e.Node.Tag)
        e.Node.Nodes.Clear()
        Try
            For Each directoryName As String In Directory.GetDirectories(folder)
                Dim child As New TreeNode(Path.GetFileName(directoryName))
                child.Tag = directoryName
                Try
                    If Directory.GetDirectories(directoryName).Length > 0 Then child.Nodes.Add("Loading...")
                Catch
                End Try
                e.Node.Nodes.Add(child)
            Next
        Catch
        End Try
    End Sub
    Private Sub tree_AfterSelect(ByVal sender As Object, ByVal e As TreeViewEventArgs) Handles tree.AfterSelect
        selectedPath.Text = CStr(e.Node.Tag)
    End Sub
    Private Sub btnRefresh_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnRefresh.Click
        BuildTree()
```

```
    End Sub
End Class
```

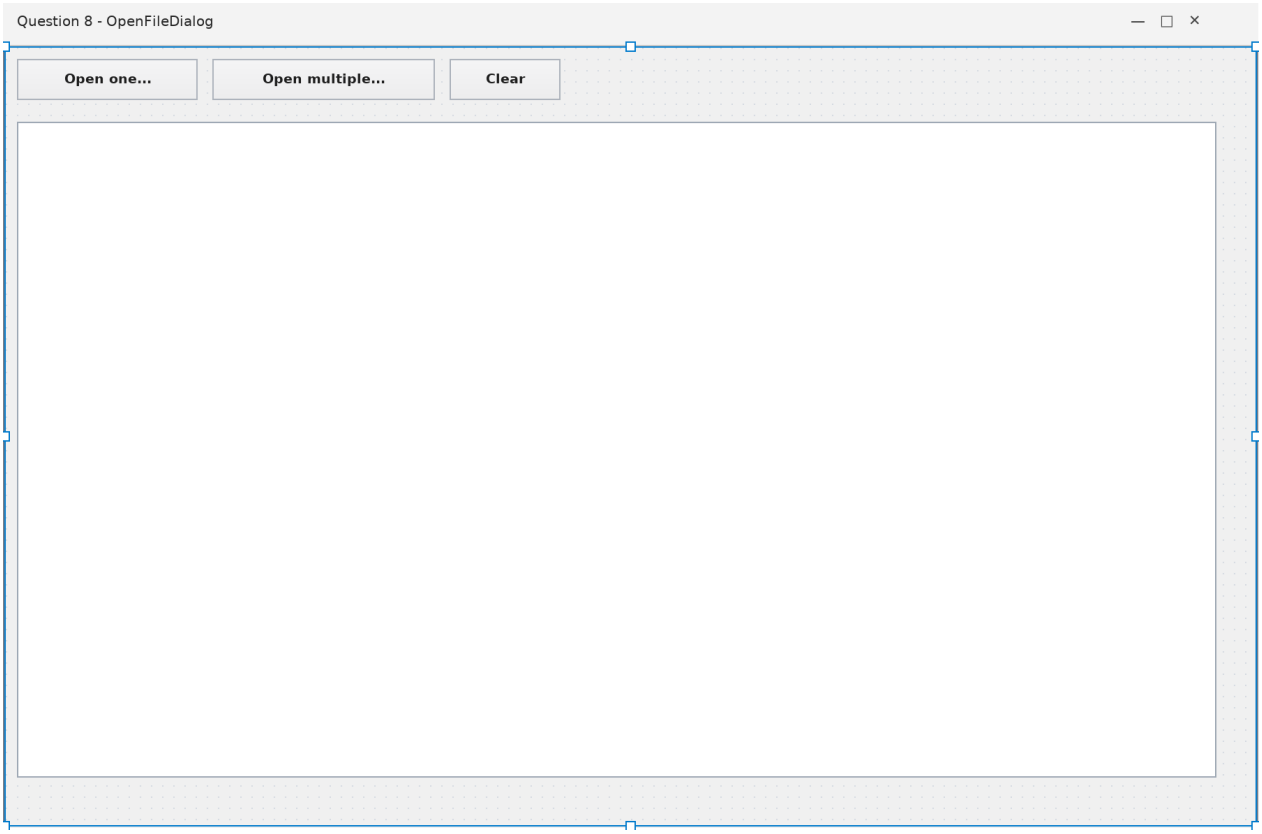
Sample Output



TreeViewSample - a node selected, its path shown

OpenFileDialogSample

Form Design



Form design - OpenFileDialogSample

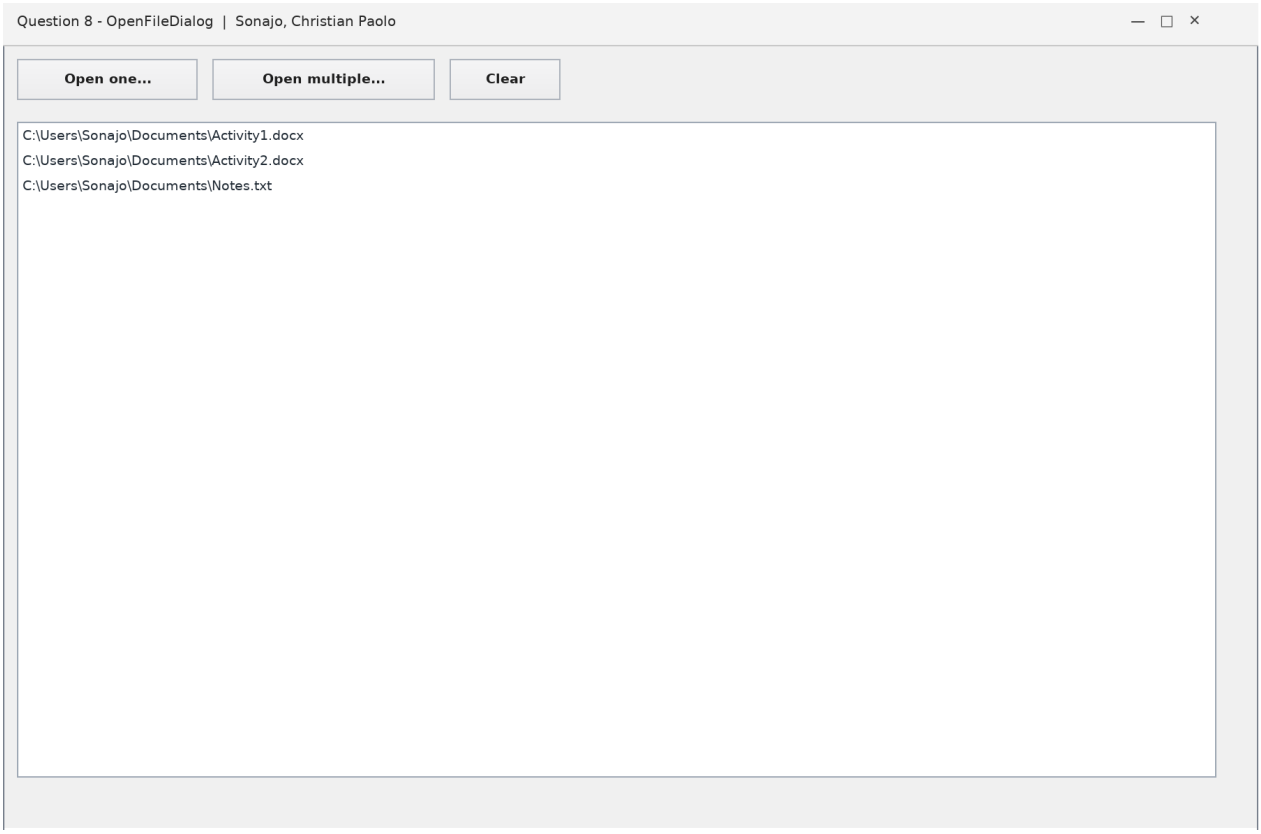
Source Code

OpenFileDialogSample.vb

```
Imports System
Imports System.Windows.Forms

Public Class OpenFileDialogSample
    Public Sub New()
        InitializeComponent()
    End Sub
    Private Sub one_Click(ByVal sender As Object, ByVal e As EventArgs) Handles one.Click
        Using dialog As New OpenFileDialog()
            Configure(dialog, False)
            If dialog.ShowDialog() = DialogResult.OK Then ShowFiles(dialog.FileNames)
        End Using
    End Sub
    Private Sub many_Click(ByVal sender As Object, ByVal e As EventArgs) Handles many.Click
        Using dialog As New OpenFileDialog()
            Configure(dialog, True)
            If dialog.ShowDialog() = DialogResult.OK Then ShowFiles(dialog.FileNames)
        End Using
    End Sub
    Private Sub Configure(ByVal dialog As OpenFileDialog, ByVal allowMultiple As Boolean)
        dialog.Title = "Select file"
        dialog.Filter = "All files (*.*)|*.*|Text files (*.txt)|*.txt|Images (*.jpg;*.png)|*.jpg;*.png"
        dialog.FilterIndex = 1
        dialog.Multiselect = allowMultiple
        dialog.CheckFileExists = True
        dialog.CheckPathExists = True
        dialog.RestoreDirectory = True
    End Sub
    Private Sub ShowFiles(ByVal files() As String)
        selected.Items.Clear()
        For Each fileName As String In files
            selected.Items.Add(fileName)
        Next
    End Sub
    Private Sub clear_Click(ByVal sender As Object, ByVal e As EventArgs) Handles clear.Click
        selected.Items.Clear()
    End Sub
End Class
```

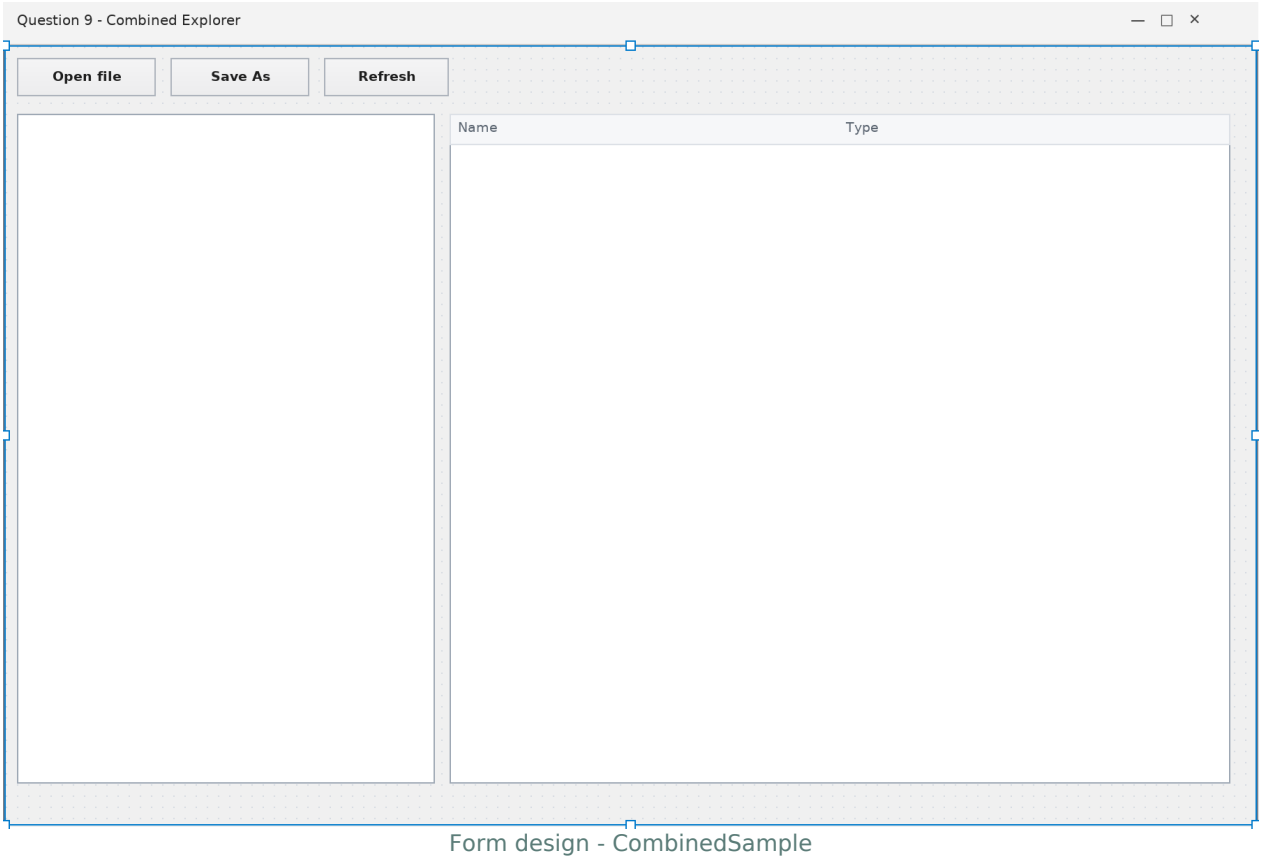
Sample Output



OpenFileDialogSample - files chosen with Open multiple...

CombinedSample

Form Design



Form design - CombinedSample

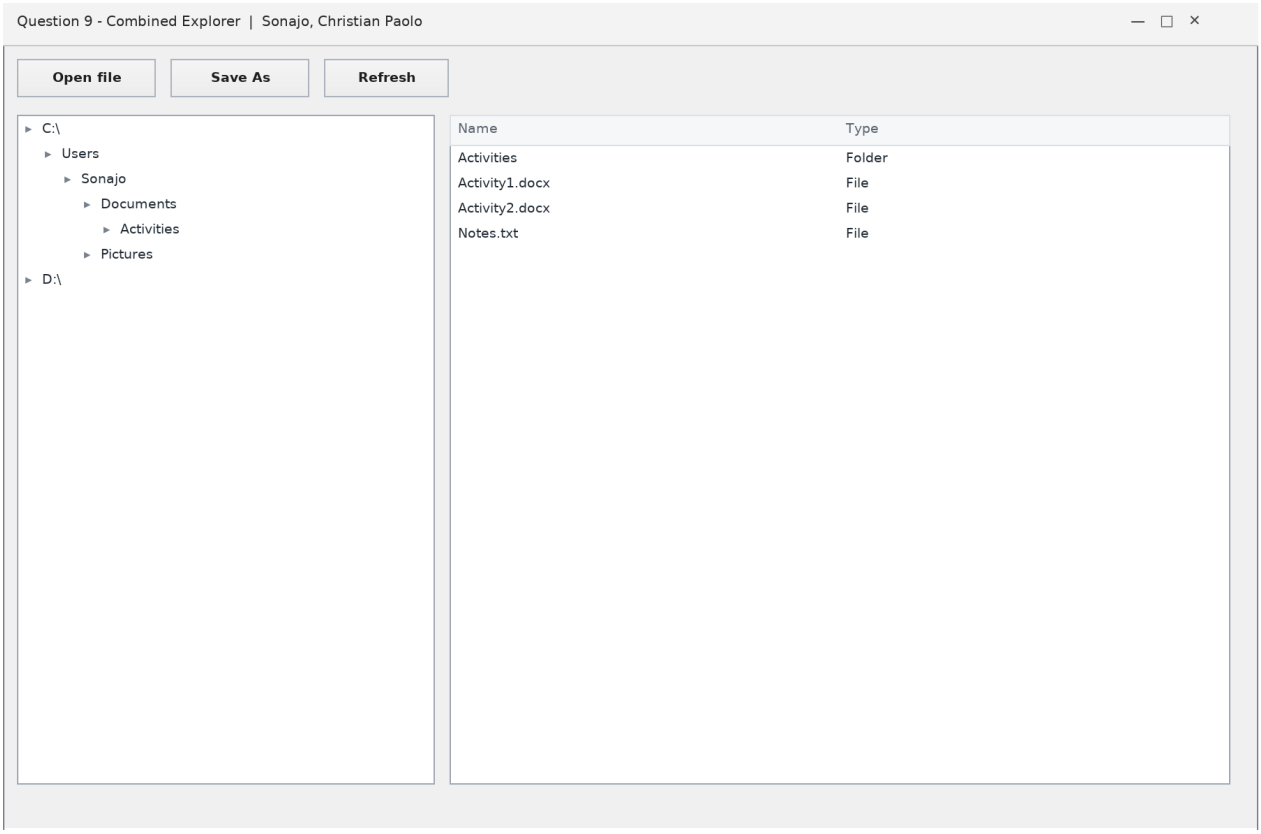
Source Code

CombinedSample.vb

```
Imports System
Imports System.IO
Imports System.Diagnostics
Imports System.Windows.Forms

Public Class CombinedSample
    Private currentFolder As String = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments)
    Public Sub New()
        InitializeComponent()
    End Sub
    Private Sub Form1_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        BuildTree()
        LoadFolder(currentFolder)
    End Sub
    Private Sub BuildTree()
        tree.Nodes.Clear()
        For Each drive As DriveInfo In DriveInfo.GetDrives()
            If drive.IsReady Then
                Dim node As New TreeNode(drive.Name)
                node.Tag = drive.RootDirectory.FullName
                node.Nodes.Add("Loading...")
                tree.Nodes.Add(node)
            End If
        Next
    End Sub
    Private Sub tree_BeforeExpand(ByVal sender As Object, ByVal e As TreeViewCancelEventArgs) Handles
    tree.BeforeExpand
        Dim folder As String = CStr(e.Node.Tag)
        e.Node.Nodes.Clear()
        Try
            For Each directoryName As String In Directory.GetDirectories(folder)
                Dim child As New TreeNode(Path.GetFileName(directoryName))
                child.Tag = directoryName
                e.Node.Nodes.Add(child)
            Next
        Catch
        End Try
    End Sub
    Private Sub tree_AfterSelect(ByVal sender As Object, ByVal e As TreeViewEventArgs) Handles tree.AfterSelect
        currentFolder = CStr(e.Node.Tag)
        LoadFolder(currentFolder)
    End Sub
    Private Sub LoadFolder(ByVal folder As String)
        If Not Directory.Exists(folder) Then Return
        lv.Items.Clear()
        Try
            For Each directoryName As String In Directory.GetDirectories(folder)
                Dim item As New ListViewItem(Path.GetFileName(directoryName))
                item.SubItems.Add("Folder")
                item.Tag = directoryName
                lv.Items.Add(item)
            Next
            For Each fileName As String In Directory.GetFiles(folder)
                Dim item As New ListViewItem(Path.GetFileName(fileName))
                item.SubItems.Add("File")
                item.Tag = fileName
                lv.Items.Add(item)
            Next
        Catch
        End Try
    End Sub
    Private Sub open_Click(ByVal sender As Object, ByVal e As EventArgs) Handles open.Click
        Using dialog As New OpenFileDialog()
            dialog.Filter = "All files (*.*)|*.*"
            dialog.CheckFileExists = True
            If dialog.ShowDialog() = DialogResult.OK Then Process.Start(dialog.FileName)
        End Using
    End Sub
    Private Sub save_Click(ByVal sender As Object, ByVal e As EventArgs) Handles save.Click
        Using dialog As New SaveFileDialog()
            dialog.Filter = "Text files (*.txt)|*.txt|All files (*.*)|*.*"
            If dialog.ShowDialog() = DialogResult.OK Then
                Using writer As New StreamWriter(dialog.FileName)
                    For Each item As ListViewItem In lv.Items
                        writer.WriteLine(item.Text & Microsoft.VisualBasic.ControlChars.Tab &
    item.SubItems(1).Text)
                    Next
                End Using
            End If
        End Using
    End Sub
    Private Sub btnRefresh_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnRefresh.Click
        BuildTree()
        LoadFolder(currentFolder)
    End Sub
End Class
```

Sample Output

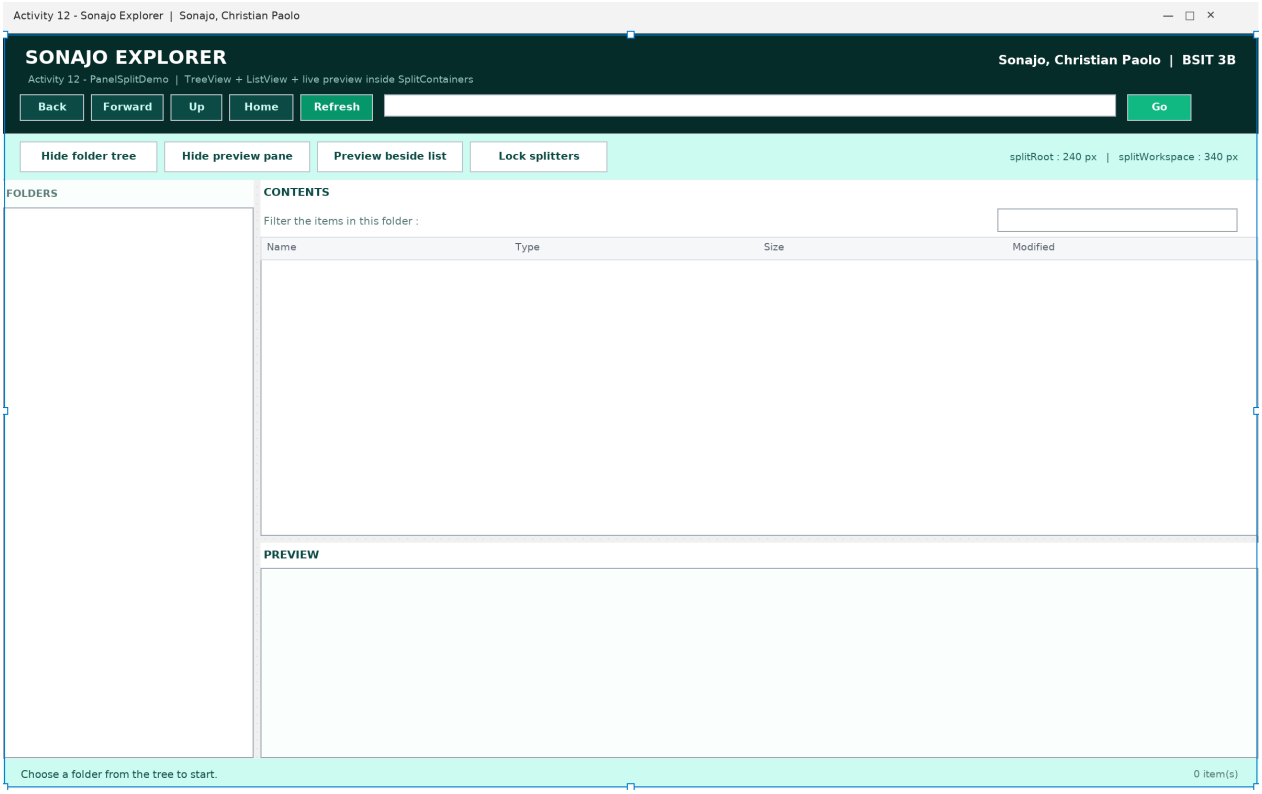


CombinedSample - TreeView on the left, ListView on the right

Activity 12 - File Explorer Clone (PanelSplitDemo)

FileExplorerClone - PanelSplitDemo | Sonajo, Christian Paolo | BSIT 3B

Form Design



Form design of Activity 12 - Sonajo Explorer

Source Code

Form1.vb

```
' ============================================================
' Activity 12 - PanelSplitDemo  (Sonajo Explorer)
' A small file explorer built with TreeView + ListView + preview
' inside nested SplitContainer controls.
' Author : Sonajo, Christian Paolo   |   BSIT 3B
' ============================================================
Option Strict On
Option Explicit On

Imports System.IO
Imports System.Text

Public Class Form1

    Private currentFolder As String = ""
    Private ReadOnly backStack As New Stack(Of String)()
    Private ReadOnly forwardStack As New Stack(Of String)()
    Private syncingTree As Boolean = False
    Private splittersLocked As Boolean = False
    Private lastSortColumn As Integer = -1
    Private sortAscending As Boolean = True

    Private ReadOnly imageTypes As String() = _
        {".jpg", ".jpeg", ".png", ".gif", ".bmp", ".ico", ".tif", ".tiff"}

    Private ReadOnly textTypes As String() = _
        {".txt", ".vb", ".cs", ".xml", ".config", ".csv", ".log", ".md", ".json", _
         ".ini", ".html", ".htm", ".css", ".js", ".sql", ".bat", ".sln", ".vbproj"}

    ' ------------------------------------------------------------
    ' start up
    ' ------------------------------------------------------------
    Private Sub Form1_Load(ByVal sender As Object, ByVal e As EventArgs) Handles MyBase.Load
        Try
            BuildFolderTree()
            ApplyDefaultLayout()
            NavigateTo(StartFolder(), False)
        Catch ex As Exception
            lblStatus.Text = "The explorer could not start : " & ex.Message
```

```
    End Try
End Sub

Private Function StartFolder() As String
    Dim documents As String = Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments)
    If Directory.Exists(documents) Then Return documents

    Dim desktopFolder As String = Environment.GetFolderPath(Environment.SpecialFolder.Desktop)
    If Directory.Exists(desktopFolder) Then Return desktopFolder

    Return Environment.CurrentDirectory
End Function

' ------------------------------------------------------------
' folder tree
' ------------------------------------------------------------
Private Sub BuildFolderTree()
    tvFolders.BeginUpdate()
    tvFolders.Nodes.Clear()

    AddRootFolder("Desktop", Environment.GetFolderPath(Environment.SpecialFolder.Desktop))
    AddRootFolder("Documents", Environment.GetFolderPath(Environment.SpecialFolder.MyDocuments))
    AddRootFolder("Pictures", Environment.GetFolderPath(Environment.SpecialFolder.MyPictures))
    AddRootFolder("Downloads",
Path.Combine(Environment.GetFolderPath(Environment.SpecialFolder.UserProfile), "Downloads"))

    For Each drive As DriveInfo In DriveInfo.GetDrives()
        If drive.IsReady Then
            AddRootFolder(drive.Name, drive.RootDirectory.FullName)
        End If
    Next

    tvFolders.EndUpdate()
    lblStatus.Text = tvFolders.Nodes.Count.ToString() & " root folder(s) listed."
End Sub

Private Sub AddRootFolder(ByVal title As String, ByVal folderPath As String)
    If folderPath = "" OrElse Not Directory.Exists(folderPath) Then Return

    Dim node As New TreeNode(title)
    node.Tag = folderPath
    node.Nodes.Add(New TreeNode("loading"))
    tvFolders.Nodes.Add(node)
End Sub

Private Sub tvFolders_BeforeExpand(ByVal sender As Object, ByVal e As TreeViewCancelEventArgs) Handles
tvFolders.BeforeExpand
    If e.Node.Tag Is Nothing Then Return
    If e.Node.Nodes.Count <> 1 OrElse e.Node.Nodes(0).Text <> "loading" Then Return
    If e.Node.Nodes(0).Tag IsNot Nothing Then Return

    Dim parentPath As String = CStr(e.Node.Tag)
    e.Node.Nodes.Clear()

    Try
        For Each childPath As String In Directory.GetDirectories(parentPath)
            Dim child As New TreeNode(Path.GetFileName(childPath))
            If child.Text = "" Then child.Text = childPath
            child.Tag = childPath
            child.Nodes.Add(New TreeNode("loading"))
            e.Node.Nodes.Add(child)
        Next
    Catch ex As Exception
        lblStatus.Text = "Some folders cannot be listed : " & ex.Message
    End Try

    If e.Node.Nodes.Count = 0 Then e.Node.Nodes.Add(New TreeNode("empty"))
End Sub

Private Sub tvFolders_AfterSelect(ByVal sender As Object, ByVal e As TreeViewEventArgs) Handles
tvFolders.AfterSelect
    If syncingTree Then Return
    If e.Node.Tag Is Nothing Then Return

    Dim folderPath As String = CStr(e.Node.Tag)
    If Directory.Exists(folderPath) Then
        NavigateTo(folderPath, True)
    End If
End Sub

' ------------------------------------------------------------
' navigation
' ------------------------------------------------------------
Private Sub NavigateTo(ByVal folderPath As String, ByVal remember As Boolean)
    If folderPath = "" OrElse Not Directory.Exists(folderPath) Then
        lblStatus.Text = "Folder not found : " & folderPath
        Return
    End If

    If remember AndAlso currentFolder <> "" AndAlso currentFolder <> folderPath Then
        backStack.Push(currentFolder)
        forwardStack.Clear()
    End If
    ShowFolder(folderPath)
```

```
End Sub

Private Sub ShowFolder(ByVal folderPath As String)
    currentFolder = folderPath
    txtPath.Text = folderPath
    RefreshFileList()
    SyncTreeSelection(folderPath)
    ShowPreview("")
    lblStatus.Text = "Now browsing : " & folderPath
End Sub

Private Sub SyncTreeSelection(ByVal folderPath As String)
    syncingTree = True
    Try
        Dim node As TreeNode = FindNode(tvFolders.Nodes, folderPath)
        If node IsNot Nothing Then
            tvFolders.SelectedNode = node
            node.EnsureVisible()
        End If
    Finally
        syncingTree = False
    End Try
End Sub

Private Function FindNode(ByVal siblings As TreeNodeCollection, ByVal folderPath As String) As TreeNode
    For Each node As TreeNode In siblings
        If node.Tag IsNot Nothing AndAlso _
           String.Equals(CStr(node.Tag), folderPath, StringComparison.OrdinalIgnoreCase) Then
            Return node
        End If

        If node.IsExpanded AndAlso node.Nodes.Count > 0 Then
            Dim found As TreeNode = FindNode(node.Nodes, folderPath)
            If found IsNot Nothing Then Return found
        End If
    Next
    Return Nothing
End Function

Private Sub btnBack_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnBack.Click
    If backStack.Count = 0 Then
        lblStatus.Text = "There is no previous folder in the history."
        Return
    End If

    forwardStack.Push(currentFolder)
    ShowFolder(backStack.Pop())
End Sub

Private Sub btnForward_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnForward.Click
    If forwardStack.Count = 0 Then
        lblStatus.Text = "There is no next folder in the history."
        Return
    End If

    backStack.Push(currentFolder)
    ShowFolder(forwardStack.Pop())
End Sub

Private Sub btnUp_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnUp.Click
    If currentFolder = "" Then Return

    Dim parent As String = Path.GetDirectoryName(currentFolder)
    If parent Is Nothing OrElse parent = "" Then
        lblStatus.Text = "This is already the top of the drive."
        Return
    End If

    NavigateTo(parent, True)
End Sub

Private Sub btnHome_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnHome.Click
    NavigateTo(StartFolder(), True)
End Sub

Private Sub btnRefresh_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnRefresh.Click
    RefreshFileList()
    lblStatus.Text = "Folder refreshed."
End Sub

Private Sub btnGo_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnGo.Click
    NavigateTo(txtPath.Text.Trim(), True)
End Sub

Private Sub txtPath_KeyDown(ByVal sender As Object, ByVal e As KeyEventArgs) Handles txtPath.KeyDown
    If e.KeyCode = Keys.Enter Then
        e.SuppressKeyPress = True
        NavigateTo(txtPath.Text.Trim(), True)
    End If
End Sub

' ------------------------------------------------------------
' file list
' ------------------------------------------------------------
```

```
Private Sub RefreshFileList()
    lvFiles.BeginUpdate()
    lvFiles.Items.Clear()

    If currentFolder = "" OrElse Not Directory.Exists(currentFolder) Then
        lvFiles.EndUpdate()
        lblCount.Text = "0 item(s)"
        Return
    End If

    Dim keyword As String = txtFilter.Text.Trim()

    Try
        For Each folderPath As String In Directory.GetDirectories(currentFolder)
            Dim folderInfo As New DirectoryInfo(folderPath)
            If IsHidden(folderInfo) Then Continue For
            If Not Matches(keyword, folderInfo.Name) Then Continue For
            AddRow(folderInfo.Name, "File folder", "", folderInfo.LastWriteTime, folderInfo.FullName)
        Next

        For Each filePath As String In Directory.GetFiles(currentFolder)
            Dim fileInfo As New FileInfo(filePath)
            If IsHidden(fileInfo) Then Continue For
            If Not Matches(keyword, fileInfo.Name) Then Continue For
            AddRow(fileInfo.Name, TypeLabel(fileInfo), SizeLabel(fileInfo.Length), _
                   fileInfo.LastWriteTime, fileInfo.FullName)
        Next
    Catch ex As Exception
        lblStatus.Text = "This folder cannot be fully listed : " & ex.Message
    Finally
        lvFiles.EndUpdate()
    End Try

    If lastSortColumn >= 0 Then ApplySort(lastSortColumn, False)

    lblCount.Text = lvFiles.Items.Count.ToString() & " item(s)   |   " & currentFolder
End Sub

Private Sub AddRow(ByVal name As String, ByVal kind As String, ByVal size As String, _
                   ByVal modified As Date, ByVal fullPath As String)
    Dim row As New ListViewItem(name)
    row.SubItems.Add(kind)
    row.SubItems.Add(size)
    row.SubItems.Add(modified.ToString("MMM d, yyyy  h:mm tt"))
    row.Tag = fullPath
    lvFiles.Items.Add(row)
End Sub

Private Function IsHidden(ByVal info As FileSystemInfo) As Boolean
    Return (info.Attributes And FileAttributes.Hidden) = FileAttributes.Hidden
End Function

Private Function Matches(ByVal keyword As String, ByVal itemName As String) As Boolean
    If keyword = "" Then Return True
    Return itemName.IndexOf(keyword, StringComparison.OrdinalIgnoreCase) >= 0
End Function

Private Function TypeLabel(ByVal info As FileInfo) As String
    If info.Extension = "" Then Return "File"
    Return info.Extension.Substring(1).ToUpperInvariant() & " file"
End Function

Private Function SizeLabel(ByVal bytes As Long) As String
    If bytes >= 1073741824 Then Return (bytes / 1073741824.0).ToString("0.00") & " GB"
    If bytes >= 1048576 Then Return (bytes / 1048576.0).ToString("0.00") & " MB"
    If bytes >= 1024 Then Return (bytes / 1024.0).ToString("0.0") & " KB"
    Return bytes.ToString() & " B"
End Function

Private Sub txtFilter_TextChanged(ByVal sender As Object, ByVal e As EventArgs) Handles
txtFilter.TextChanged
    If currentFolder = "" Then Return
    RefreshFileList()
End Sub

' ------------------------------------------------------------
' sorting
' ------------------------------------------------------------
Private Sub lvFiles_ColumnClick(ByVal sender As Object, ByVal e As ColumnClickEventArgs) Handles
lvFiles.ColumnClick
    If lastSortColumn = e.Column Then
        sortAscending = Not sortAscending
    Else
        lastSortColumn = e.Column
        sortAscending = True
    End If

    ApplySort(e.Column, sortAscending)
    lblStatus.Text = "Sorted by column " & (e.Column + 1).ToString() & If(sortAscending, " (A to Z)", " (Z
to A)")
End Sub

Private Sub ApplySort(ByVal column As Integer, ByVal ascending As Boolean)
    lvFiles.ListViewItemSorter = New ItemSorter(column, ascending)
```

```
    lvFiles.Sort()
End Sub

' Compares two ListView rows; the size column is compared as a number.
Friend Class ItemSorter
    Implements System.Collections.IComparer

    Private ReadOnly columnIndex As Integer
    Private ReadOnly ascendingOrder As Boolean

    Public Sub New(ByVal column As Integer, ByVal ascending As Boolean)
        columnIndex = column
        ascendingOrder = ascending
    End Sub

    Public Function Compare(ByVal x As Object, ByVal y As Object) As Integer Implements
System.Collections.IComparer.Compare
        Dim leftRow As ListViewItem = CType(x, ListViewItem)
        Dim rightRow As ListViewItem = CType(y, ListViewItem)

        Dim leftText As String = If(leftRow.SubItems.Count > columnIndex,
leftRow.SubItems(columnIndex).Text, "")
        Dim rightText As String = If(rightRow.SubItems.Count > columnIndex,
rightRow.SubItems(columnIndex).Text, "")

        Dim answer As Integer
        If columnIndex = 2 Then
            answer = BytesOf(leftText).CompareTo(BytesOf(rightText))
        Else
            answer = String.Compare(leftText, rightText, StringComparison.OrdinalIgnoreCase)
        End If

        If Not ascendingOrder Then answer = -answer
        Return answer
    End Function

    Private Function BytesOf(ByVal sizeText As String) As Double
        Dim numberPart As String = ""
        For Each character As Char In sizeText
            If Char.IsDigit(character) OrElse character = "."c Then
                numberPart &= character
            Else
                Exit For
            End If
        Next

        Dim value As Double = 0
        Double.TryParse(numberPart, value)

        If sizeText.Contains("GB") Then Return value * 1073741824
        If sizeText.Contains("MB") Then Return value * 1048576
        If sizeText.Contains("KB") Then Return value * 1024
        Return value
    End Function
End Class

' ------------------------------------------------------------
' preview
' ------------------------------------------------------------
Private Sub lvFiles_SelectedIndexChanged(ByVal sender As Object, ByVal e As EventArgs) Handles
lvFiles.SelectedIndexChanged
    If lvFiles.SelectedItems.Count = 0 Then
        ShowPreview("")
        Return
    End If

    ShowPreview(CStr(lvFiles.SelectedItems(0).Tag))
End Sub

Private Sub lvFiles_DoubleClick(ByVal sender As Object, ByVal e As EventArgs) Handles lvFiles.DoubleClick
    If lvFiles.SelectedItems.Count = 0 Then Return

    Dim chosenPath As String = CStr(lvFiles.SelectedItems(0).Tag)

    If Directory.Exists(chosenPath) Then
        NavigateTo(chosenPath, True)
        Return
    End If

    Try
        Process.Start(chosenPath)
    Catch ex As Exception
        lblStatus.Text = "This file cannot be opened : " & ex.Message
    End Try
End Sub

Private Sub ShowPreview(ByVal chosenPath As String)
    ReleasePicture()

    If chosenPath = "" Then
        lblPreviewTitle.Text = "PREVIEW"
        txtPreview.Visible = True
        txtPreview.Text = "Select a file or a folder to preview it here."
        Return
```

```
    End If

    Dim itemName As String = Path.GetFileName(chosenPath)
    If itemName = "" Then itemName = chosenPath
    lblPreviewTitle.Text = "PREVIEW  -  " & itemName

    If Directory.Exists(chosenPath) Then
        ShowFolderPreview(chosenPath)
        Return
    End If

    If Not File.Exists(chosenPath) Then
        txtPreview.Visible = True
        txtPreview.Text = "The selected item no longer exists."
        Return
    End If

    Dim info As New FileInfo(chosenPath)
    Dim extension As String = info.Extension.ToLowerInvariant()

    If IsOneOf(extension, imageTypes) Then
        If ShowImagePreview(chosenPath) Then Return
    End If

    txtPreview.Visible = True

    If IsOneOf(extension, textTypes) Then
        txtPreview.Text = ReadTextPreview(chosenPath, info)
    Else
        txtPreview.Text = "Name          : " & info.Name & vbCrLf & _
                          "Type          : " & TypeLabel(info) & vbCrLf & _
                          "Size          : " & SizeLabel(info.Length) & vbCrLf & _
                          "Created       : " & info.CreationTime.ToString("MMM d, yyyy  h:mm tt") & vbCrLf &
 _
                          "Last modified : " & info.LastWriteTime.ToString("MMM d, yyyy  h:mm tt") & vbCrLf
& _
                          "Attributes    : " & info.Attributes.ToString() & vbCrLf & vbCrLf & _
                          "There is no built-in preview for this file type."
    End If
End Sub

Private Function IsOneOf(ByVal value As String, ByVal choices As String()) As Boolean
    For Each choice As String In choices
        If value = choice Then Return True
    Next
    Return False
End Function

Private Function ShowImagePreview(ByVal filePath As String) As Boolean
    Try
        Using stream As New FileStream(filePath, FileMode.Open, FileAccess.Read)
            Using source As Image = Image.FromStream(stream)
                pbPreview.Image = New Bitmap(source)
            End Using
        End Using

        txtPreview.Visible = False
        pbPreview.Visible = True
        lblStatus.Text = "Image preview : " & Path.GetFileName(filePath)
        Return True
    Catch ex As Exception
        txtPreview.Visible = True
        pbPreview.Visible = False
        txtPreview.Text = "The image cannot be previewed." & vbCrLf & ex.Message
        Return True
    End Try
End Function

Private Sub ShowFolderPreview(ByVal folderPath As String)
    Dim summary As New StringBuilder()
    Dim folderCount As Integer = 0
    Dim fileCount As Integer = 0
    Dim totalSize As Long = 0

    Try
        For Each childFolder As String In Directory.GetDirectories(folderPath)
            folderCount += 1
        Next

        For Each childFile As String In Directory.GetFiles(folderPath)
            Dim childInfo As New FileInfo(childFile)
            fileCount += 1
            totalSize += childInfo.Length
        Next
    Catch ex As Exception
        summary.AppendLine("Some items cannot be counted : " & ex.Message)
    End Try

    Dim parentPath As String = Path.GetDirectoryName(folderPath)
    If parentPath Is Nothing Then parentPath = folderPath

    summary.AppendLine("Folder        : " & folderPath)
    summary.AppendLine("Parent        : " & parentPath)
    summary.AppendLine("Sub folders   : " & folderCount.ToString())
```

```
    summary.AppendLine("Files         : " & fileCount.ToString())
    summary.AppendLine("Total size    : " & SizeLabel(totalSize))
    summary.AppendLine()
    summary.AppendLine("CONTENTS (first 40 items)")
    summary.AppendLine("----------------------------------------")

    Try
        Dim shown As Integer = 0
        For Each childFolder As String In Directory.GetDirectories(folderPath)
            summary.AppendLine("[folder]  " & Path.GetFileName(childFolder))
            shown += 1
            If shown >= 40 Then Exit For
        Next

        If shown < 40 Then
            For Each childFile As String In Directory.GetFiles(folderPath)
                Dim childInfo As New FileInfo(childFile)
                summary.AppendLine("          " & childInfo.Name & "   (" & SizeLabel(childInfo.Length) &
")")
                shown += 1
                If shown >= 40 Then Exit For
            Next
        End If
    Catch ex As Exception
        summary.AppendLine(ex.Message)
    End Try

    txtPreview.Visible = True
    txtPreview.Text = summary.ToString()
End Sub

Private Function ReadTextPreview(ByVal filePath As String, ByVal info As FileInfo) As String
    Const characterLimit As Integer = 6000

    Try
        Dim buffer(characterLimit) As Char
        Dim text As New StringBuilder()

        Using reader As New StreamReader(filePath, Encoding.Default, True)
            Dim readNow As Integer = reader.Read(buffer, 0, characterLimit)
            While readNow > 0
                text.Append(New String(buffer, 0, readNow))
                readNow = reader.Read(buffer, 0, characterLimit)
            End While
        End Using

        Dim header As String = "Name : " & info.Name & "    Size : " & SizeLabel(info.Length) & vbCrLf & _
                               "Modified : " & info.LastWriteTime.ToString("MMM d, yyyy  h:mm tt") & vbCrLf
& _
                               "----------------------------------------" & vbCrLf

        Dim body As String = text.ToString()
        If body.Length >= characterLimit Then
            body &= vbCrLf & vbCrLf & "...  only the first " & characterLimit.ToString() & " characters are
shown."
        End If

        Return header & body
    Catch ex As Exception
        Return "The text of this file cannot be shown." & vbCrLf & ex.Message
    End Try
End Function

Private Sub ClearPreview(ByVal message As String)
    ReleasePicture()
    pbPreview.Visible = False
    txtPreview.Visible = True
    txtPreview.Text = message
End Sub

Private Sub ReleasePicture()
    If pbPreview.Image IsNot Nothing Then
        Dim oldPicture As Image = pbPreview.Image
        pbPreview.Image = Nothing
        oldPicture.Dispose()
    End If
    pbPreview.Visible = False
End Sub

' ------------------------------------------------------------
' split container controls
' ------------------------------------------------------------
Private Sub ApplyDefaultLayout()
    splRoot.Panel1Collapsed = False
    splRoot.Panel2Collapsed = False
    splWorkspace.Panel1Collapsed = False
    splWorkspace.Panel2Collapsed = False
    splWorkspace.Orientation = Orientation.Horizontal

    btnTreeToggle.Text = "Hide folder tree"
    btnPreviewToggle.Text = "Hide preview pane"
    btnSwapLayout.Text = "Preview beside list"

    SetRatioDistance(splRoot, 0.20)
```

```
        SetRatioDistance(splWorkspace, 0.62)
        ShowSplitInfo()
    End Sub

    Private Sub SetRatioDistance(ByVal splitter As SplitContainer, ByVal ratio As Double)
        Dim available As Integer
        If splitter.Orientation = Orientation.Vertical Then
            available = splitter.ClientSize.Width
        Else
            available = splitter.ClientSize.Height
        End If

        Dim lowest As Integer = splitter.Panel1MinSize
        Dim highest As Integer = available - splitter.Panel2MinSize - splitter.SplitterWidth
        If highest <= lowest Then Return

        Dim requested As Integer = CInt(available * ratio)
        If requested < lowest Then requested = lowest
        If requested > highest Then requested = highest
        splitter.SplitterDistance = requested
    End Sub

    Private Sub ShowSplitInfo()
        lblSplitInfo.Text = "splitRoot : " & splRoot.SplitterDistance.ToString() & " px   |   " & _
                            "splitWorkspace : " & splWorkspace.SplitterDistance.ToString() & " px"
    End Sub

    Private Sub btnTreeToggle_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnTreeToggle.Click
        splRoot.Panel1Collapsed = Not splRoot.Panel1Collapsed
        btnTreeToggle.Text = If(splRoot.Panel1Collapsed, "Show folder tree", "Hide folder tree")
        lblStatus.Text = "splRoot.Panel1Collapsed = " & splRoot.Panel1Collapsed.ToString()
        ShowSplitInfo()
    End Sub

    Private Sub btnPreviewToggle_Click(ByVal sender As Object, ByVal e As EventArgs) Handles
    btnPreviewToggle.Click
        splWorkspace.Panel2Collapsed = Not splWorkspace.Panel2Collapsed
        btnPreviewToggle.Text = If(splWorkspace.Panel2Collapsed, "Show preview pane", "Hide preview pane")
        lblStatus.Text = "splWorkspace.Panel2Collapsed = " & splWorkspace.Panel2Collapsed.ToString()
        ShowSplitInfo()
    End Sub

    Private Sub btnSwapLayout_Click(ByVal sender As Object, ByVal e As EventArgs) Handles btnSwapLayout.Click
        Try
            If splWorkspace.Orientation = Orientation.Horizontal Then
                splWorkspace.Orientation = Orientation.Vertical
                btnSwapLayout.Text = "Preview below list"
                lblStatus.Text = "The preview is now beside the file list."
            Else
                splWorkspace.Orientation = Orientation.Horizontal
                btnSwapLayout.Text = "Preview beside list"
                lblStatus.Text = "The preview is now below the file list."
            End If

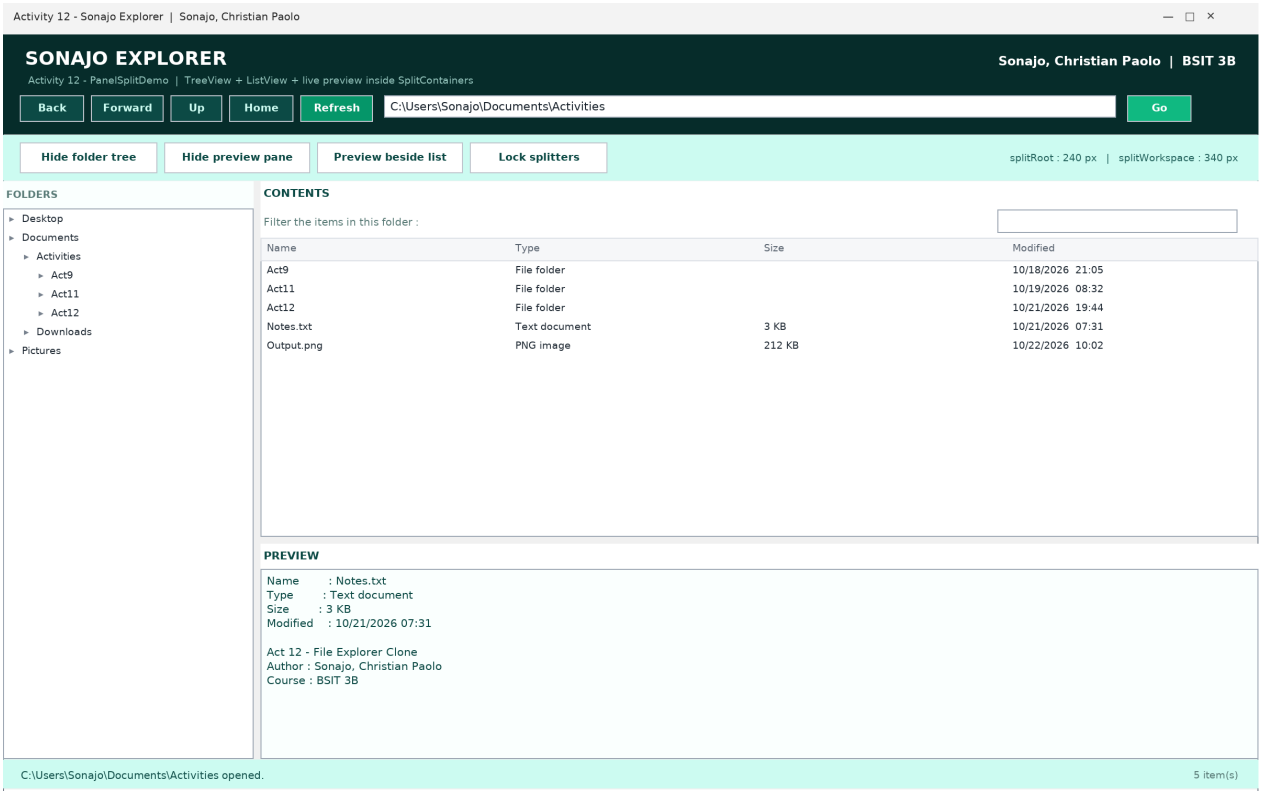
            SetRatioDistance(splWorkspace, 0.62)
            ShowSplitInfo()
        Catch ex As Exception
            lblStatus.Text = "The layout cannot be changed : " & ex.Message
        End Try
    End Sub

    Private Sub btnLockSplitters_Click(ByVal sender As Object, ByVal e As EventArgs) Handles
    btnLockSplitters.Click
        splittersLocked = Not splittersLocked
        splRoot.IsSplitterFixed = splittersLocked
        splWorkspace.IsSplitterFixed = splittersLocked
        btnLockSplitters.Text = If(splittersLocked, "Unlock splitters", "Lock splitters")
        lblStatus.Text = "IsSplitterFixed = " & splittersLocked.ToString()
    End Sub

    Private Sub splRoot_SplitterMoved(ByVal sender As Object, ByVal e As SplitterEventArgs) Handles
    splRoot.SplitterMoved
        ShowSplitInfo()
        lblStatus.Text = "Folder tree width set to " & splRoot.SplitterDistance.ToString() & " pixels."
    End Sub

    Private Sub splWorkspace_SplitterMoved(ByVal sender As Object, ByVal e As SplitterEventArgs) Handles
    splWorkspace.SplitterMoved
        ShowSplitInfo()
        lblStatus.Text = "Preview size set to " & splWorkspace.SplitterDistance.ToString() & " pixels."
    End Sub
End Class
```

Sample Output



File explorer clone showing a folder, its files and the preview

Activity 13 - Integrative Project - School Records with MDI Forms

IntegProgProject2026 | Sonajo, Christian Paolo | BSIT 3B

frmMain

Form Design



Form design - frmMain

Source Code

frmMain.vb

```
Public Class frmMain

    Private Sub LogLine(ByVal text As String)
        Try
            IO.File.AppendAllText(IO.Path.Combine(Application.StartupPath, "forms_log.txt"), _
                Date.Now.ToString("HH:mm:ss") & "  " & text & vbCrLf)
        Catch
        End Try
    End Sub

    Private Sub frmMain_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        Try
            toolstripUserInfo.Text = "UserInfo: Sonajo, Christian Paolo  |  BSIT 3B"
            UpdateDateTime()
            Timer1.Interval = 1000
            Timer1.Start()
            lblStatusBar.Text = "Choose a form from the menu."
            LogLine("main form loaded")
        Catch ex As Exception
            LogLine("main load failed: " & ex.Message)
        End Try
    End Sub

    Private Sub Timer1_Tick(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles Timer1.Tick
        Try
            UpdateDateTime()
        Catch
        End Try
    End Sub

    Private Sub UpdateDateTime()
        toolstripCurrentDate.Text = "CurrentDate: " & Date.Now.ToString("dddd, d MMMM yyyy")
        toolstripCurrentTime.Text = "Current Time: " & Date.Now.ToString("h:mm:ss tt")
    End Sub
    Private Sub SchedulesToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
```

```
    Handles SchedulesToolStripMenuItem.Click
        ShowChild(GetType(frmSchedules))
    End Sub

    Private Sub AboutToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
     AboutToolStripMenuItem.Click
        ShowChild(GetType(frmAbout))
    End Sub

    Private Sub QuotesToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles QuotesToolStripMenuItem.Click
        ShowChild(GetType(frmQuotes))
    End Sub

    Private Sub BibleVerseToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs)
    Handles BibleVerseToolStripMenuItem.Click
        ShowChild(GetType(frmBibleVerses))
    End Sub

    Private Sub ExitToolStripMenuItem_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    ExitToolStripMenuItem.Click
        If MessageBox.Show("Close the program?", "Exit", MessageBoxButtons.YesNo, MessageBoxIcon.Question) =
    DialogResult.Yes Then
            Me.Close()
        End If
    End Sub

    Private Sub ShowChild(ByVal formType As Type)
        LogLine("opening " & formType.Name)
        Try
            For Each openForm As Form In Me.MdiChildren
                If openForm.GetType() Is formType Then
                    openForm.Activate()
                    lblStatusBar.Text = "Form: " & openForm.Text
                    LogLine(formType.Name & " already open - activated")
                    Return
                End If
            Next
            Dim child As Form = CType(Activator.CreateInstance(formType), Form)
            child.MdiParent = Me
            child.Show()
            child.WindowState = FormWindowState.Maximized
            lblStatusBar.Text = "Form: " & child.Text
            LogLine(formType.Name & " shown inside main form, Visible=" & child.Visible.ToString())
        Catch ex As Exception
            LogLine(formType.Name & " FAILED: " & ex.GetType().Name & " - " & ex.Message)
            Try
                MessageBox.Show("The form could not be opened." & vbCrLf & vbCrLf & _
                    "Form: " & formType.Name & vbCrLf & vbCrLf & _
                    "Details: " & ex.Message, "Form Error", MessageBoxButtons.OK, MessageBoxIcon.Error)
            Catch
            End Try
        End Try
    End Sub

End Class
```

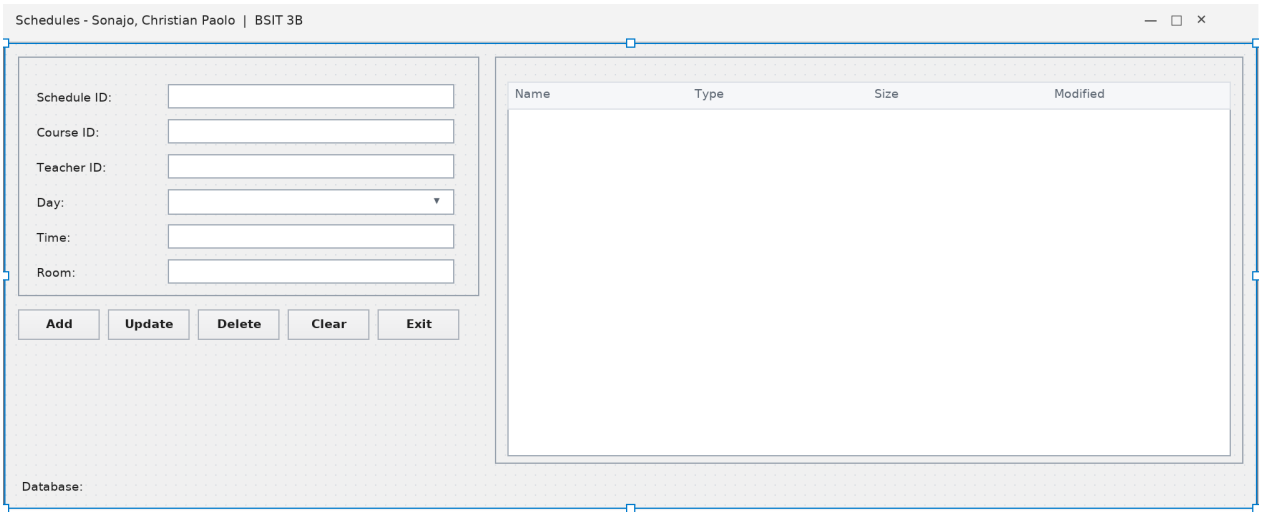
Sample Output



frmMain - the MDI parent with its menu and status strip

frmSchedules

Form Design



Form design - frmSchedules

Source Code

frmSchedules.vb

```
Public Class frmSchedules

    Private selectedId As Integer = 0
    Private isLoading As Boolean = False
    Private store As DataTable
    Private nextId As Integer = 1

    Private Sub LogStep(ByVal text As String)
        Try
            IO.File.AppendAllText(IO.Path.Combine(Application.StartupPath, "forms_log.txt"), _
                Date.Now.ToString("HH:mm:ss") & "  [frmSchedules] " & text & vbCrLf)
        Catch
        End Try
    End Sub
```

```
Private useSdf As Boolean = False

Private Const SqlCeProvider As String = "System.Data.SqlServerCe.3.5"
Private Const SqlCeEngineType As String = "System.Data.SqlServerCe.SqlCeEngine, System.Data.SqlServerCe,
Version=3.5.1.0, Culture=neutral, PublicKeyToken=89845dcd8080cc91"

Private Function XmlPath() As String
    Return IO.Path.Combine(Application.StartupPath, "projectDB_Schedules.xml")
End Function

Private Function SdfPath() As String
    Return IO.Path.Combine(Application.StartupPath, "projectDB.sdf")
End Function

Private Function StoragePath() As String
    If useSdf Then Return SdfPath()
    Return XmlPath()
End Function

Private Function NewConnection() As System.Data.Common.DbConnection
    Dim factory As System.Data.Common.DbProviderFactory =
System.Data.Common.DbProviderFactories.GetFactory(SqlCeProvider)
    Dim conn As System.Data.Common.DbConnection = factory.CreateConnection()
    conn.ConnectionString = "Data Source=" & SdfPath() & ";Persist Security Info=False;"
    Return conn
End Function

Private Sub AddParam(ByVal cmd As System.Data.Common.DbCommand, ByVal name As String, ByVal value As Object)
    Dim p As System.Data.Common.DbParameter = cmd.CreateParameter()
    p.ParameterName = name
    p.Value = value
    cmd.Parameters.Add(p)
End Sub

Private Sub frmSchedules_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
MyBase.Load
    LogStep("Load")
    Try
        SetupStore()
    Catch ex As Exception
        LogStep("SetupStore failed: " & ex.Message)
    End Try
    Try
        ClearFields()
    Catch
    End Try
    RefreshGrid()
End Sub

Private Sub SetupStore()
    store = New DataTable("Schedules")
    store.Columns.Add("ScheduleID", GetType(Integer))
    store.Columns.Add("CourseID", GetType(String))
    store.Columns.Add("TeacherID", GetType(String))
    store.Columns.Add("Day", GetType(String))
    store.Columns.Add("Time", GetType(String))
    store.Columns.Add("Room", GetType(String))
    useSdf = False
    Try
        OpenSdf()
        useSdf = True
    Catch ex As Exception
        LogStep("projectDB.sdf is not available (" & ex.Message & "), using the XML file instead")
    End Try
    If useSdf Then
        LoadFromSdf()
    Else
        LoadFromXml()
    End If
    lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & " record(s)"
End Sub

' ---------- SQL Server Compact database (bin\Debug\projectDB.sdf) ----------

Private Sub OpenSdf()
    If Not IO.File.Exists(SdfPath()) Then
        Dim engineType As Type = Type.GetType(SqlCeEngineType)
        If engineType Is Nothing Then Throw New InvalidOperationException("SQL Server Compact 3.5 is not
installed.")
        Dim engine As Object = Activator.CreateInstance(engineType, New Object() {"Data Source=" & SdfPath()
 & ";"})
        engine.CreateDatabase()
        engine.Dispose()
        LogStep("created " & SdfPath())
    End If
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Dim hasTable As Boolean = False
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME =
'Schedules'"
            hasTable = (Convert.ToInt32(cmd.ExecuteScalar()) > 0)
        End Using
        If Not hasTable Then
```

```
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "CREATE TABLE [Schedules] ([ScheduleID] INT IDENTITY(1,1) PRIMARY KEY,
[CourseID] NVARCHAR(255), [TeacherID] NVARCHAR(255), [Day] NVARCHAR(255), [Time] NVARCHAR(255), [Room]
NVARCHAR(255))"
                cmd.ExecuteNonQuery()
            End Using
            LogStep("created table Schedules")
        End If
        Dim total As Integer = 0
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM [Schedules]"
            total = Convert.ToInt32(cmd.ExecuteScalar())
        End Using
        If total = 0 Then
            SeedSampleData()
            For Each row As DataRow In store.Rows
                InsertRow(conn, row)
            Next
            store.Clear()
            nextId = 1
            LogStep("seeded sample data into projectDB.sdf")
        End If
    End Using
End Sub

Private Sub LoadFromSdf()
    store.Clear()
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT [ScheduleID], [CourseID], [TeacherID], [Day], [Time], [Room] FROM
[Schedules] ORDER BY [ScheduleID]"
            Using reader As System.Data.Common.DbDataReader = cmd.ExecuteReader()
                While reader.Read()
                    Dim row As DataRow = store.NewRow()
                    For i As Integer = 0 To store.Columns.Count - 1
                        row.Item(i) = reader.GetValue(i)
                    Next
                    store.Rows.Add(row)
                End While
            End Using
        End Using
    End Using
    store.AcceptChanges()
    LogStep("loaded rows from projectDB.sdf: " & store.Rows.Count.ToString())
End Sub

Private Function InsertRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow) As Integer
    Dim newId As Integer = 0
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "INSERT INTO [Schedules] ([CourseID], [TeacherID], [Day], [Time], [Room]) VALUES
(@p1, @p2, @p3, @p4, @p5)"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("CourseID")))
        AddParam(cmd, "@p2", Convert.ToString(row.Item("TeacherID")))
        AddParam(cmd, "@p3", Convert.ToString(row.Item("Day")))
        AddParam(cmd, "@p4", Convert.ToString(row.Item("Time")))
        AddParam(cmd, "@p5", Convert.ToString(row.Item("Room")))
        cmd.ExecuteNonQuery()
    End Using
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "SELECT @@IDENTITY"
        newId = Convert.ToInt32(cmd.ExecuteScalar())
    End Using
    Return newId
End Function

Private Sub UpdateRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow)
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "UPDATE [Schedules] SET [CourseID] = @p1, [TeacherID] = @p2, [Day] = @p3, [Time] =
 @p4, [Room] = @p5 WHERE [ScheduleID] = @id"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("CourseID")))
        AddParam(cmd, "@p2", Convert.ToString(row.Item("TeacherID")))
        AddParam(cmd, "@p3", Convert.ToString(row.Item("Day")))
        AddParam(cmd, "@p4", Convert.ToString(row.Item("Time")))
        AddParam(cmd, "@p5", Convert.ToString(row.Item("Room")))
        AddParam(cmd, "@id", Convert.ToInt32(row.Item("ScheduleID")))
        cmd.ExecuteNonQuery()
    End Using
End Sub

' ---------- XML file (only used when SQL Server Compact is not installed) ----------

Private Sub LoadFromXml()
    store.Clear()
    If IO.File.Exists(XmlPath()) Then
        Try
            store.ReadXml(XmlPath())
            LogStep("loaded rows from xml: " & store.Rows.Count.ToString())
        Catch ex As Exception
            LogStep("xml unreadable, reseeding: " & ex.Message)
            store.Clear()
        End Try
    End If
    If store.Rows.Count = 0 Then
```

```
        SeedSampleData()
        SaveXml()
        LogStep("seeded sample data")
    End If
    nextId = 1
    For Each row As DataRow In store.Rows
        Dim id As Integer = Convert.ToInt32(row.Item("ScheduleID"))
        If id >= nextId Then nextId = id + 1
    Next
End Sub

Private Sub SaveXml()
    Try
        store.AcceptChanges()
        store.WriteXml(XmlPath())
    Catch ex As Exception
        LogStep("WriteXml failed: " & ex.Message)
    End Try
End Sub

' ---------- Create / Update / Delete ----------

Private Function SaveNewRecord(ByVal row As DataRow) As Integer
    Dim newId As Integer = 0
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            newId = InsertRow(conn, row)
        End Using
    Else
        newId = nextId
        nextId += 1
    End If
    row.Item("ScheduleID") = newId
    store.Rows.Add(row)
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
    Return newId
End Function

Private Sub SaveChangedRecord(ByVal row As DataRow)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            UpdateRow(conn, row)
        End Using
    End If
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub

Private Sub DeleteRecord(ByVal id As Integer)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "DELETE FROM [Schedules] WHERE [ScheduleID] = @id"
                AddParam(cmd, "@id", id)
                cmd.ExecuteNonQuery()
            End Using
        End Using
    End If
    For i As Integer = store.Rows.Count - 1 To 0 Step -1
        If Convert.ToInt32(store.Rows(i).Item("ScheduleID")) = id Then
            store.Rows.RemoveAt(i)
        End If
    Next
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub

Private Sub SeedSampleData()
    Dim rows As String() = New String() { _
        "IT301|T-005|Monday|07:30|Room 205", _
        "IT302|T-006|Monday|11:00|Lab 3", _
        "CS201|T-007|Tuesday|09:30|Room 206", _
        "IT303|T-005|Wednesday|08:00|Lab 4", _
        "CS202|T-006|Thursday|13:30|Room 207", _
        "IT304|T-007|Friday|10:00|Room 208", _
        "CS203|T-005|Saturday|09:00|Lab 5"
        }
    For Each line As String In rows
        Dim parts As String() = line.Split("|"c)
        If parts.Length = 5 Then
            Dim row As DataRow = store.NewRow()
            row.Item("ScheduleID") = nextId
            row.Item("CourseID") = parts(0)
            row.Item("TeacherID") = parts(1)
            row.Item("Day") = parts(2)
            row.Item("Time") = parts(3)
            row.Item("Room") = parts(4)
            store.Rows.Add(row)
            nextId += 1
        End If
```

```
    Next
End Sub

Private Sub RefreshGrid()
    Try
        If store Is Nothing Then Return
        Dim view As New DataTable("Schedules")
        view = store.Copy()
        dataGridView1.DataSource = view
        FormatGrid()
        lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & "
record(s)"
    Catch ex As Exception
        LogStep("RefreshGrid failed: " & ex.Message)
    End Try
End Sub

Private Sub FormatGrid()
    If dataGridView1.Columns.Count = 0 Then Return
    With dataGridView1
        .Columns("ScheduleID").HeaderText = "ID"
        .Columns("CourseID").HeaderText = "Course ID"
        .Columns("TeacherID").HeaderText = "Teacher ID"
        .Columns("Day").HeaderText = "Day"
        .Columns("Time").HeaderText = "Time"
        .Columns("Room").HeaderText = "Room"
    End With
End Sub

Private Sub addButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
addButton.Click
    Try
        If Not IsValidInput() Then Return
        Dim row As DataRow = store.NewRow()
        row.Item("ScheduleID") = nextId
        row.Item("CourseID") = txtCourseID.Text.Trim()
        row.Item("TeacherID") = txtTeacherID.Text.Trim()
        row.Item("Day") = cboDay.Text.Trim()
        row.Item("Time") = dtpTime.Value.ToString("HH:mm")
        row.Item("Room") = txtRoom.Text.Trim()
        Dim newId As Integer = SaveNewRecord(row)
        MessageBox.Show("The record was saved." & vbCrLf & vbCrLf & "New ID: " & newId.ToString(), _
            "Record Added", MessageBoxButtons.OK, MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Add failed: " & ex.Message)
        MessageBox.Show("The record could not be saved." & vbCrLf & vbCrLf & ex.Message, _
            "Save Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub updateButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
updateButton.Click
    Try
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If Not IsValidInput() Then Return
        Dim found As DataRow = Nothing
        For Each row As DataRow In store.Rows
            If Convert.ToInt32(row.Item("ScheduleID")) = selectedId Then
                found = row
                Exit For
            End If
        Next
        If found Is Nothing Then
            MessageBox.Show("That record no longer exists.", "Record Not Found", _
                MessageBoxButtons.OK, MessageBoxIcon.Warning)
            RefreshGrid()
            Return
        End If
        found.Item("CourseID") = txtCourseID.Text.Trim()
        found.Item("TeacherID") = txtTeacherID.Text.Trim()
        found.Item("Day") = cboDay.Text.Trim()
        found.Item("Time") = dtpTime.Value.ToString("HH:mm")
        found.Item("Room") = txtRoom.Text.Trim()
        SaveChangedRecord(found)
        MessageBox.Show("The record was updated.", "Record Updated", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Update failed: " & ex.Message)
        MessageBox.Show("The record could not be updated." & vbCrLf & vbCrLf & ex.Message, _
            "Update Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub deleteButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
deleteButton.Click
    Try
```

```
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If MessageBox.Show("Delete this record?" & vbCrLf & vbCrLf & "ID: " & selectedId.ToString(), _
                "Confirm Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Question) <> DialogResult.Yes Then
            Return
        End If
        DeleteRecord(selectedId)
        MessageBox.Show("The record was deleted.", "Record Deleted", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Delete failed: " & ex.Message)
        MessageBox.Show("The record could not be deleted." & vbCrLf & vbCrLf & ex.Message, _
            "Delete Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub clearButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
clearButton.Click
    ClearFields()
End Sub

Private Sub exitButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
exitButton.Click
    Me.Close()
End Sub

Private Sub dataGridView1_SelectionChanged(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles dataGridView1.SelectionChanged
    If isLoading Then Return
    If dataGridView1.CurrentRow Is Nothing Then Return
    FillFromRow()
End Sub

Private Sub FillFromRow()
    Dim row As DataGridViewRow = dataGridView1.CurrentRow
    If row Is Nothing Then Return
    If row.Cells("ScheduleID").Value Is Nothing Then Return
    If row.Cells("ScheduleID").Value Is DBNull.Value Then Return
    isLoading = True
    Try
        selectedId = Convert.ToInt32(row.Cells("ScheduleID").Value)
        txtScheduleID.Text = selectedId.ToString()
        txtCourseID.Text = Convert.ToString(row.Cells("CourseID").Value)
        txtTeacherID.Text = Convert.ToString(row.Cells("TeacherID").Value)
        cboDay.Text = Convert.ToString(row.Cells("Day").Value)
        Dim TimeText As String = Convert.ToString(row.Cells("Time").Value)
        Dim TimeParsed As Date
        If Date.TryParse(DateTime.Today.ToShortDateString() & " " & TimeText, TimeParsed) Then
            dtpTime.Value = TimeParsed
        End If
        txtRoom.Text = Convert.ToString(row.Cells("Room").Value)
    Catch ex As Exception
        LogStep("FillFromRow failed: " & ex.Message)
    Finally
        isLoading = False
    End Try
End Sub

Private Sub ClearFields()
    isLoading = True
    Try
        dataGridView1.CurrentCell = Nothing
    Catch
    End Try
    selectedId = 0
    txtScheduleID.Clear()
    txtCourseID.Clear()
    txtTeacherID.Clear()
    cboDay.SelectedIndex = -1
    dtpTime.Value = DateTime.Today.AddHours(8)
    txtRoom.Clear()
    txtCourseID.Focus()
    isLoading = False
End Sub

Private Function IsValidInput() As Boolean
    If txtCourseID.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Course ID.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtCourseID.Focus()
        Return False
    End If
    If txtTeacherID.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Teacher ID.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtTeacherID.Focus()
        Return False
    End If
    If cboDay.SelectedIndex < 0 Then
```

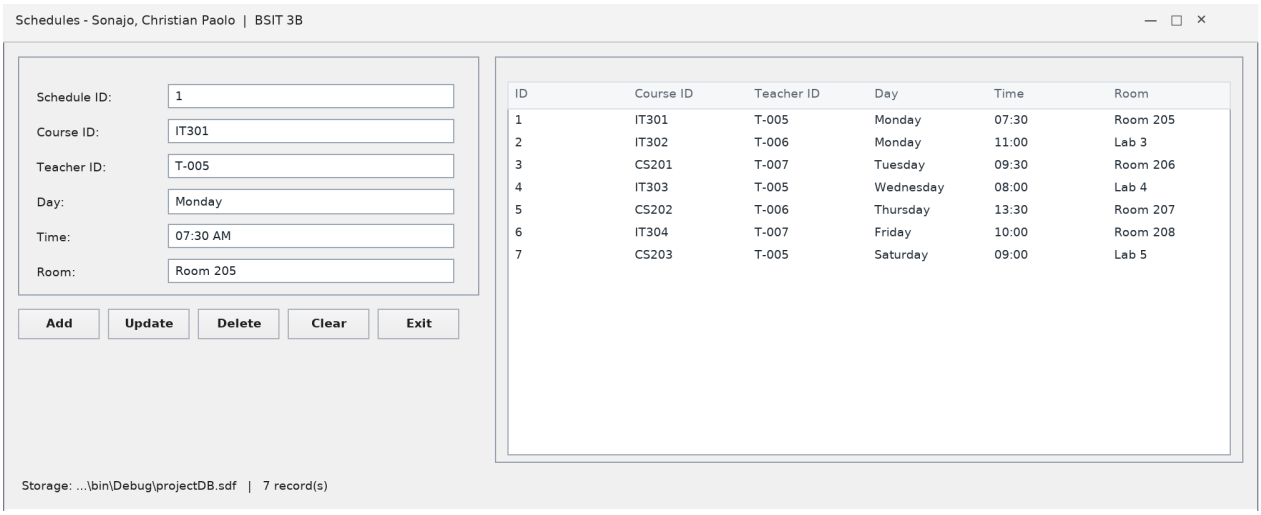
```
            MessageBox.Show("Please choose the Day.", "Missing Information", MessageBoxButtons.OK,
    MessageBoxIcon.Exclamation)
            cboDay.Focus()
            Return False
        End If
        If txtRoom.Text.Trim() = "" Then
            MessageBox.Show("Please enter the Room.", "Missing Information", MessageBoxButtons.OK,
    MessageBoxIcon.Exclamation)
            txtRoom.Focus()
            Return False
        End If
        Return True
    End Function


    Private Function IntValue(ByVal text As String) As Integer
        Dim value As Integer = 0
        If Integer.TryParse(text.Trim(), value) Then Return value
        Return 0
    End Function

    Private Function DateValue(ByVal text As String) As Date
        Dim parts As String() = text.Split("-"c)
        If parts.Length = 3 Then
            Dim y As Integer = 0
            Dim m As Integer = 0
            Dim d As Integer = 0
            If Integer.TryParse(parts(0), y) AndAlso Integer.TryParse(parts(1), m) AndAlso
    Integer.TryParse(parts(2), d) Then
                If y >= 1 AndAlso m >= 1 AndAlso m <= 12 AndAlso d >= 1 AndAlso d <= 31 Then
                    Return New Date(y, m, d)
                End If
            End If
        End If
        Return Date.Today
    End Function

End Class
```

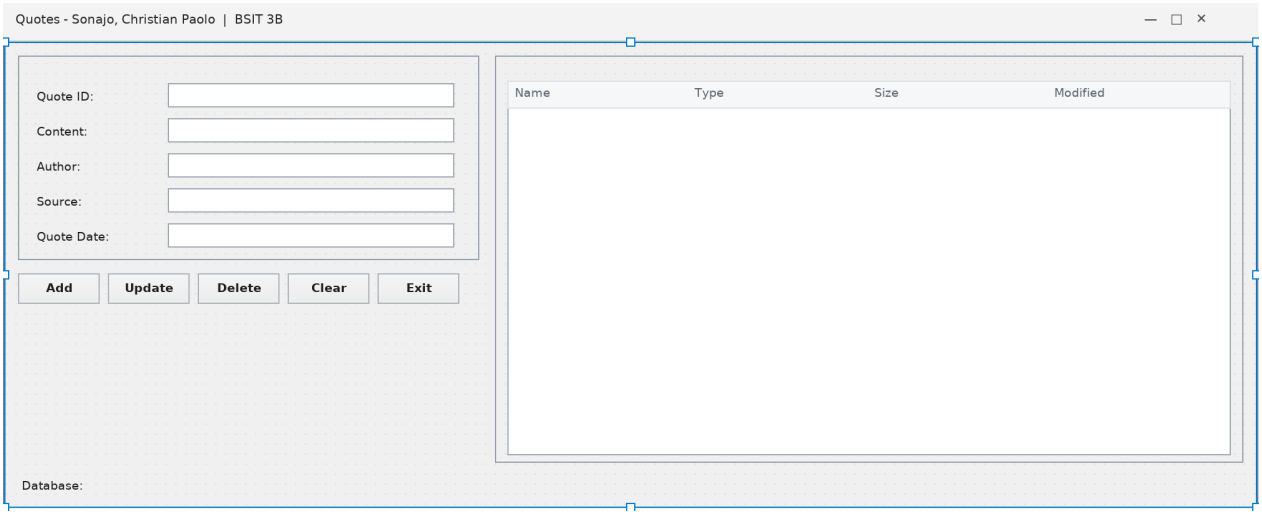
Sample Output



frmSchedules - first record selected in the grid

frmQuotes

Form Design



Form design - frmQuotes

Source Code

frmQuotes.vb

```
Public Class frmQuotes

    Private selectedId As Integer = 0
    Private isLoading As Boolean = False
    Private store As DataTable
    Private nextId As Integer = 1

    Private Sub LogStep(ByVal text As String)
        Try
            IO.File.AppendAllText(IO.Path.Combine(Application.StartupPath, "forms_log.txt"), _
                Date.Now.ToString("HH:mm:ss") & "  [frmQuotes] " & text & vbCrLf)
        Catch
        End Try
    End Sub

    Private useSdf As Boolean = False

    Private Const SqlCeProvider As String = "System.Data.SqlServerCe.3.5"
    Private Const SqlCeEngineType As String = "System.Data.SqlServerCe.SqlCeEngine, System.Data.SqlServerCe,
    Version=3.5.1.0, Culture=neutral, PublicKeyToken=89845dcd8080cc91"

    Private Function XmlPath() As String
        Return IO.Path.Combine(Application.StartupPath, "projectDB_Quotes.xml")
    End Function

    Private Function SdfPath() As String
        Return IO.Path.Combine(Application.StartupPath, "projectDB.sdf")
    End Function

    Private Function StoragePath() As String
        If useSdf Then Return SdfPath()
        Return XmlPath()
    End Function

    Private Function NewConnection() As System.Data.Common.DbConnection
        Dim factory As System.Data.Common.DbProviderFactory =
    System.Data.Common.DbProviderFactories.GetFactory(SqlCeProvider)
        Dim conn As System.Data.Common.DbConnection = factory.CreateConnection()
        conn.ConnectionString = "Data Source=" & SdfPath() & ";Persist Security Info=False;"
        Return conn
    End Function

    Private Sub AddParam(ByVal cmd As System.Data.Common.DbCommand, ByVal name As String, ByVal value As Object)
        Dim p As System.Data.Common.DbParameter = cmd.CreateParameter()
        p.ParameterName = name
        p.Value = value
        cmd.Parameters.Add(p)
    End Sub

    Private Sub frmQuotes_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        LogStep("Load")
        Try
            SetupStore()
        Catch ex As Exception
            LogStep("SetupStore failed: " & ex.Message)
        End Try
        Try
            ClearFields()
        Catch
```

```
    End Try
    RefreshGrid()
End Sub

Private Sub SetupStore()
    store = New DataTable("Quotes")
    store.Columns.Add("QuoteID", GetType(Integer))
    store.Columns.Add("Content", GetType(String))
    store.Columns.Add("Author", GetType(String))
    store.Columns.Add("Source", GetType(String))
    store.Columns.Add("QuoteDate", GetType(Date))
    useSdf = False
    Try
        OpenSdf()
        useSdf = True
    Catch ex As Exception
        LogStep("projectDB.sdf is not available (" & ex.Message & "), using the XML file instead")
    End Try
    If useSdf Then
        LoadFromSdf()
    Else
        LoadFromXml()
    End If
    lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & " record(s)"
End Sub

' ---------- SQL Server Compact database (bin\Debug\projectDB.sdf) ----------

Private Sub OpenSdf()
    If Not IO.File.Exists(SdfPath()) Then
        Dim engineType As Type = Type.GetType(SqlCeEngineType)
        If engineType Is Nothing Then Throw New InvalidOperationException("SQL Server Compact 3.5 is not
installed.")
        Dim engine As Object = Activator.CreateInstance(engineType, New Object() {"Data Source=" & SdfPath()
 & ";"})
        engine.CreateDatabase()
        engine.Dispose()
        LogStep("created " & SdfPath())
    End If
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Dim hasTable As Boolean = False
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME = 'Quotes'"
            hasTable = (Convert.ToInt32(cmd.ExecuteScalar()) > 0)
        End Using
        If Not hasTable Then
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "CREATE TABLE [Quotes] ([QuoteID] INT IDENTITY(1,1) PRIMARY KEY, [Content]
 NVARCHAR(4000), [Author] NVARCHAR(255), [Source] NVARCHAR(255), [QuoteDate] DATETIME)"
                cmd.ExecuteNonQuery()
            End Using
            LogStep("created table Quotes")
        End If
        Dim total As Integer = 0
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM [Quotes]"
            total = Convert.ToInt32(cmd.ExecuteScalar())
        End Using
        If total = 0 Then
            SeedSampleData()
            For Each row As DataRow In store.Rows
                InsertRow(conn, row)
            Next
            store.Clear()
            nextId = 1
            LogStep("seeded sample data into projectDB.sdf")
        End If
    End Using
End Sub

Private Sub LoadFromSdf()
    store.Clear()
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT [QuoteID], [Content], [Author], [Source], [QuoteDate] FROM [Quotes]
ORDER BY [QuoteID]"
            Using reader As System.Data.Common.DbDataReader = cmd.ExecuteReader()
                While reader.Read()
                    Dim row As DataRow = store.NewRow()
                    For i As Integer = 0 To store.Columns.Count - 1
                        row.Item(i) = reader.GetValue(i)
                    Next
                    store.Rows.Add(row)
                End While
            End Using
        End Using
    End Using
    store.AcceptChanges()
    LogStep("loaded rows from projectDB.sdf: " & store.Rows.Count.ToString())
End Sub

Private Function InsertRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow) As Integer
```

```
    Dim newId As Integer = 0
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "INSERT INTO [Quotes] ([Content], [Author], [Source], [QuoteDate]) VALUES (@p1,
@p2, @p3, @p4)"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("Content")))
        AddParam(cmd, "@p2", Convert.ToString(row.Item("Author")))
        AddParam(cmd, "@p3", Convert.ToString(row.Item("Source")))
        AddParam(cmd, "@p4", Convert.ToDateTime(row.Item("QuoteDate")))
        cmd.ExecuteNonQuery()
    End Using
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "SELECT @@IDENTITY"
        newId = Convert.ToInt32(cmd.ExecuteScalar())
    End Using
    Return newId
End Function

Private Sub UpdateRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow)
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "UPDATE [Quotes] SET [Content] = @p1, [Author] = @p2, [Source] = @p3, [QuoteDate]
= @p4 WHERE [QuoteID] = @id"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("Content")))
        AddParam(cmd, "@p2", Convert.ToString(row.Item("Author")))
        AddParam(cmd, "@p3", Convert.ToString(row.Item("Source")))
        AddParam(cmd, "@p4", Convert.ToDateTime(row.Item("QuoteDate")))
        AddParam(cmd, "@id", Convert.ToInt32(row.Item("QuoteID")))
        cmd.ExecuteNonQuery()
    End Using
End Sub

' ---------- XML file (only used when SQL Server Compact is not installed) ----------

Private Sub LoadFromXml()
    store.Clear()
    If IO.File.Exists(XmlPath()) Then
        Try
            store.ReadXml(XmlPath())
            LogStep("loaded rows from xml: " & store.Rows.Count.ToString())
        Catch ex As Exception
            LogStep("xml unreadable, reseeding: " & ex.Message)
            store.Clear()
        End Try
    End If
    If store.Rows.Count = 0 Then
        SeedSampleData()
        SaveXml()
        LogStep("seeded sample data")
    End If
    nextId = 1
    For Each row As DataRow In store.Rows
        Dim id As Integer = Convert.ToInt32(row.Item("QuoteID"))
        If id >= nextId Then nextId = id + 1
    Next
End Sub

Private Sub SaveXml()
    Try
        store.AcceptChanges()
        store.WriteXml(XmlPath())
    Catch ex As Exception
        LogStep("WriteXml failed: " & ex.Message)
    End Try
End Sub

' ---------- Create / Update / Delete ----------

Private Function SaveNewRecord(ByVal row As DataRow) As Integer
    Dim newId As Integer = 0
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            newId = InsertRow(conn, row)
        End Using
    Else
        newId = nextId
        nextId += 1
    End If
    row.Item("QuoteID") = newId
    store.Rows.Add(row)
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
    Return newId
End Function

Private Sub SaveChangedRecord(ByVal row As DataRow)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            UpdateRow(conn, row)
        End Using
    End If
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub
```

```
Private Sub DeleteRecord(ByVal id As Integer)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "DELETE FROM [Quotes] WHERE [QuoteID] = @id"
                AddParam(cmd, "@id", id)
                cmd.ExecuteNonQuery()
            End Using
        End Using
    End If
    For i As Integer = store.Rows.Count - 1 To 0 Step -1
        If Convert.ToInt32(store.Rows(i).Item("QuoteID")) = id Then
            store.Rows.RemoveAt(i)
        End If
    Next
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub

Private Sub SeedSampleData()
    Dim rows As String() = New String() { _
        "Simplicity is the soul of efficiency.|Austin Freeman|The Social Welfare of the Blind|1912-01-01", _
        "First, solve the problem. Then, write the code.|John Johnson|Tech talk|1980-05-20", _
        "Experience is the name everyone gives to their mistakes.|Oscar Wilde|Lady Windermere
Fan|1892-02-20", _
        "Code is like humor. When you have to explain it, it is bad.|Cory House|Blog post|2015-03-11", _
        "Make it work, make it right, make it fast.|Kent Beck|Extreme Programming|1999-10-05", _
        "The best error message is the one that never shows up.|Thomas Fuchs|Conference talk|2011-07-19"
        }
    For Each line As String In rows
        Dim parts As String() = line.Split("|"c)
        If parts.Length = 4 Then
            Dim row As DataRow = store.NewRow()
            row.Item("QuoteID") = nextId
            row.Item("Content") = parts(0)
            row.Item("Author") = parts(1)
            row.Item("Source") = parts(2)
            row.Item("QuoteDate") = DateValue(parts(3))
            store.Rows.Add(row)
            nextId += 1
        End If
    Next
End Sub

Private Sub RefreshGrid()
    Try
        If store Is Nothing Then Return
        Dim view As New DataTable("Quotes")
        view = store.Copy()
        dataGridView1.DataSource = view
        FormatGrid()
        lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & "
record(s)"
    Catch ex As Exception
        LogStep("RefreshGrid failed: " & ex.Message)
    End Try
End Sub

Private Sub FormatGrid()
    If dataGridView1.Columns.Count = 0 Then Return
    With dataGridView1
        .Columns("QuoteID").HeaderText = "ID"
        .Columns("Content").HeaderText = "Content"
        .Columns("Author").HeaderText = "Author"
        .Columns("Source").HeaderText = "Source"
        .Columns("QuoteDate").HeaderText = "Quote Date"
    End With
End Sub

Private Sub addButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
addButton.Click
    Try
        If Not IsValidInput() Then Return
        Dim row As DataRow = store.NewRow()
        row.Item("QuoteID") = nextId
        row.Item("Content") = txtContent.Text.Trim()
        row.Item("Author") = txtAuthor.Text.Trim()
        row.Item("Source") = txtSource.Text.Trim()
        row.Item("QuoteDate") = dtpQuoteDate.Value.Date
        Dim newId As Integer = SaveNewRecord(row)
        MessageBox.Show("The record was saved." & vbCrLf & vbCrLf & "New ID: " & newId.ToString(), _
            "Record Added", MessageBoxButtons.OK, MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Add failed: " & ex.Message)
        MessageBox.Show("The record could not be saved." & vbCrLf & vbCrLf & ex.Message, _
            "Save Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub updateButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
updateButton.Click
```

```
    Try
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If Not IsValidInput() Then Return
        Dim found As DataRow = Nothing
        For Each row As DataRow In store.Rows
            If Convert.ToInt32(row.Item("QuoteID")) = selectedId Then
                found = row
                Exit For
            End If
        Next
        If found Is Nothing Then
            MessageBox.Show("That record no longer exists.", "Record Not Found", _
                MessageBoxButtons.OK, MessageBoxIcon.Warning)
            RefreshGrid()
            Return
        End If
        found.Item("Content") = txtContent.Text.Trim()
        found.Item("Author") = txtAuthor.Text.Trim()
        found.Item("Source") = txtSource.Text.Trim()
        found.Item("QuoteDate") = dtpQuoteDate.Value.Date
        SaveChangedRecord(found)
        MessageBox.Show("The record was updated.", "Record Updated", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Update failed: " & ex.Message)
        MessageBox.Show("The record could not be updated." & vbCrLf & vbCrLf & ex.Message, _
            "Update Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub deleteButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
deleteButton.Click
    Try
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If MessageBox.Show("Delete this record?" & vbCrLf & vbCrLf & "ID: " & selectedId.ToString(), _
                "Confirm Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Question) <> DialogResult.Yes Then
            Return
        End If
        DeleteRecord(selectedId)
        MessageBox.Show("The record was deleted.", "Record Deleted", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Delete failed: " & ex.Message)
        MessageBox.Show("The record could not be deleted." & vbCrLf & vbCrLf & ex.Message, _
            "Delete Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub clearButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
clearButton.Click
    ClearFields()
End Sub

Private Sub exitButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
exitButton.Click
    Me.Close()
End Sub

Private Sub dataGridView1_SelectionChanged(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles dataGridView1.SelectionChanged
    If isLoading Then Return
    If dataGridView1.CurrentRow Is Nothing Then Return
    FillFromRow()
End Sub

Private Sub FillFromRow()
    Dim row As DataGridViewRow = dataGridView1.CurrentRow
    If row Is Nothing Then Return
    If row.Cells("QuoteID").Value Is Nothing Then Return
    If row.Cells("QuoteID").Value Is DBNull.Value Then Return
    isLoading = True
    Try
        selectedId = Convert.ToInt32(row.Cells("QuoteID").Value)
        txtQuoteID.Text = selectedId.ToString()
        txtContent.Text = Convert.ToString(row.Cells("Content").Value)
        txtAuthor.Text = Convert.ToString(row.Cells("Author").Value)
        txtSource.Text = Convert.ToString(row.Cells("Source").Value)
        Dim QuoteDateDate As Date
        If Date.TryParse(Convert.ToString(row.Cells("QuoteDate").Value), QuoteDateDate) Then
            dtpQuoteDate.Value = QuoteDateDate
        End If
    Catch ex As Exception
```

```
            LogStep("FillFromRow failed: " & ex.Message)
        Finally
            isLoading = False
        End Try
    End Sub

    Private Sub ClearFields()
        isLoading = True
        Try
            dataGridView1.CurrentCell = Nothing
        Catch
        End Try
        selectedId = 0
        txtQuoteID.Clear()
        txtContent.Clear()
        txtAuthor.Clear()
        txtSource.Clear()
        dtpQuoteDate.Value = Date.Today
        txtContent.Focus()
        isLoading = False
    End Sub

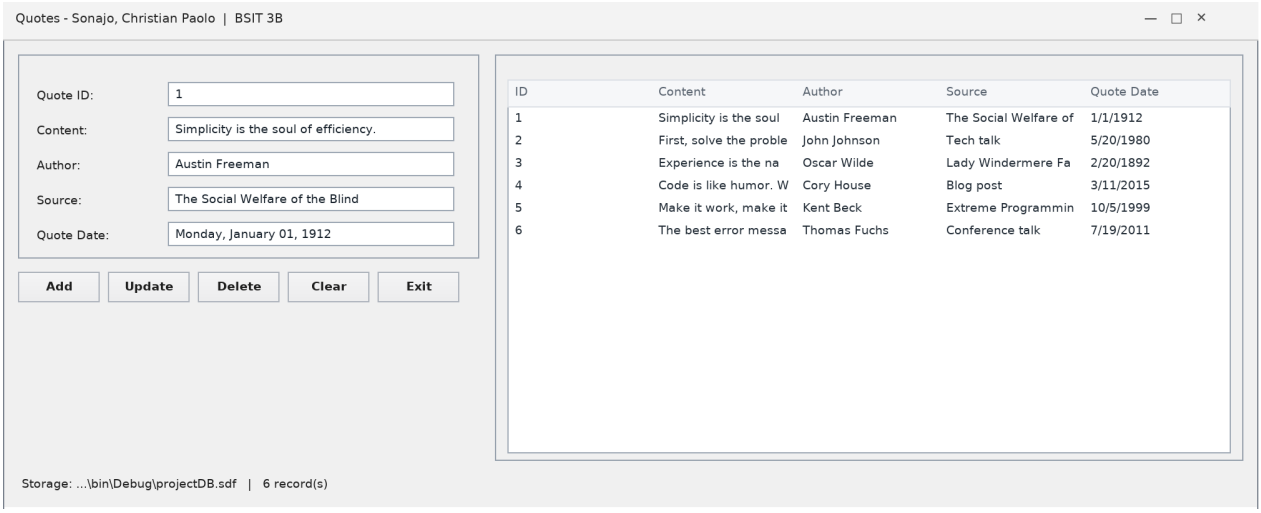
    Private Function IsValidInput() As Boolean
        If txtContent.Text.Trim() = "" Then
            MessageBox.Show("Please enter the Content.", "Missing Information", MessageBoxButtons.OK,
    MessageBoxIcon.Exclamation)
            txtContent.Focus()
            Return False
        End If
        If txtAuthor.Text.Trim() = "" Then
            MessageBox.Show("Please enter the Author.", "Missing Information", MessageBoxButtons.OK,
    MessageBoxIcon.Exclamation)
            txtAuthor.Focus()
            Return False
        End If
        Return True
    End Function


    Private Function IntValue(ByVal text As String) As Integer
        Dim value As Integer = 0
        If Integer.TryParse(text.Trim(), value) Then Return value
        Return 0
    End Function

    Private Function DateValue(ByVal text As String) As Date
        Dim parts As String() = text.Split("-"c)
        If parts.Length = 3 Then
            Dim y As Integer = 0
            Dim m As Integer = 0
            Dim d As Integer = 0
            If Integer.TryParse(parts(0), y) AndAlso Integer.TryParse(parts(1), m) AndAlso
    Integer.TryParse(parts(2), d) Then
                If y >= 1 AndAlso m >= 1 AndAlso m <= 12 AndAlso d >= 1 AndAlso d <= 31 Then
                    Return New Date(y, m, d)
                End If
            End If
        End If
        Return Date.Today
    End Function

End Class
```

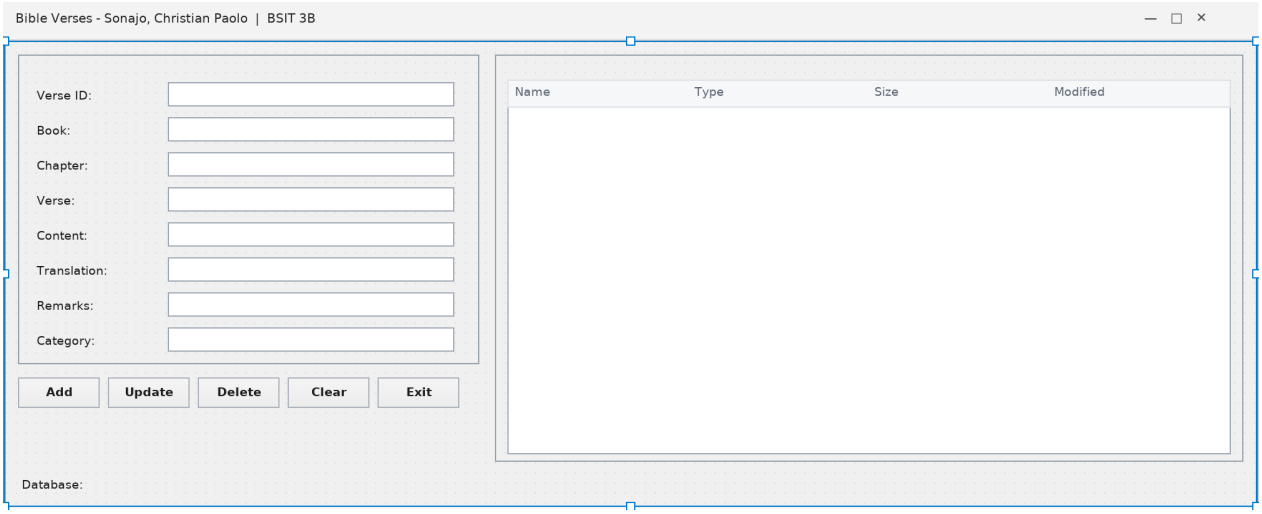
Sample Output



frmQuotes - first record selected in the grid

frmBibleVerses

Form Design



Form design - frmBibleVerses

Source Code

frmBibleVerses.vb

```
Public Class frmBibleVerses

    Private selectedId As Integer = 0
    Private isLoading As Boolean = False
    Private store As DataTable
    Private nextId As Integer = 1

    Private Sub LogStep(ByVal text As String)
        Try
            IO.File.AppendAllText(IO.Path.Combine(Application.StartupPath, "forms_log.txt"), _
                Date.Now.ToString("HH:mm:ss") & "  [frmBibleVerses] " & text & vbCrLf)
        Catch
        End Try
    End Sub

    Private useSdf As Boolean = False

    Private Const SqlCeProvider As String = "System.Data.SqlServerCe.3.5"
    Private Const SqlCeEngineType As String = "System.Data.SqlServerCe.SqlCeEngine, System.Data.SqlServerCe,
    Version=3.5.1.0, Culture=neutral, PublicKeyToken=89845dcd8080cc91"

    Private Function XmlPath() As String
        Return IO.Path.Combine(Application.StartupPath, "projectDB_BibleVerses.xml")
    End Function

    Private Function SdfPath() As String
        Return IO.Path.Combine(Application.StartupPath, "projectDB.sdf")
    End Function

    Private Function StoragePath() As String
        If useSdf Then Return SdfPath()
        Return XmlPath()
    End Function

    Private Function NewConnection() As System.Data.Common.DbConnection
        Dim factory As System.Data.Common.DbProviderFactory =
    System.Data.Common.DbProviderFactories.GetFactory(SqlCeProvider)
        Dim conn As System.Data.Common.DbConnection = factory.CreateConnection()
        conn.ConnectionString = "Data Source=" & SdfPath() & ";Persist Security Info=False;"
        Return conn
    End Function

    Private Sub AddParam(ByVal cmd As System.Data.Common.DbCommand, ByVal name As String, ByVal value As Object)
        Dim p As System.Data.Common.DbParameter = cmd.CreateParameter()
        p.ParameterName = name
        p.Value = value
        cmd.Parameters.Add(p)
    End Sub

    Private Sub frmBibleVerses_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
    MyBase.Load
        LogStep("Load")
        Try
            SetupStore()
        Catch ex As Exception
```

```
        LogStep("SetupStore failed: " & ex.Message)
    End Try
    Try
        ClearFields()
    Catch
    End Try
    RefreshGrid()
End Sub

Private Sub SetupStore()
    store = New DataTable("BibleVerses")
    store.Columns.Add("VerseID", GetType(Integer))
    store.Columns.Add("Book", GetType(String))
    store.Columns.Add("Chapter", GetType(Integer))
    store.Columns.Add("Verse", GetType(Integer))
    store.Columns.Add("Content", GetType(String))
    store.Columns.Add("Translation", GetType(String))
    store.Columns.Add("Remarks", GetType(String))
    store.Columns.Add("Category", GetType(String))
    useSdf = False
    Try
        OpenSdf()
        useSdf = True
    Catch ex As Exception
        LogStep("projectDB.sdf is not available (" & ex.Message & "), using the XML file instead")
    End Try
    If useSdf Then
        LoadFromSdf()
    Else
        LoadFromXml()
    End If
    lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & " record(s)"
End Sub

' ---------- SQL Server Compact database (bin\Debug\projectDB.sdf) ----------

Private Sub OpenSdf()
    If Not IO.File.Exists(SdfPath()) Then
        Dim engineType As Type = Type.GetType(SqlCeEngineType)
        If engineType Is Nothing Then Throw New InvalidOperationException("SQL Server Compact 3.5 is not
installed.")
        Dim engine As Object = Activator.CreateInstance(engineType, New Object() {"Data Source=" & SdfPath()
 & ";"})
        engine.CreateDatabase()
        engine.Dispose()
        LogStep("created " & SdfPath())
    End If
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Dim hasTable As Boolean = False
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM INFORMATION_SCHEMA.TABLES WHERE TABLE_NAME =
'BibleVerses'"
            hasTable = (Convert.ToInt32(cmd.ExecuteScalar()) > 0)
        End Using
        If Not hasTable Then
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "CREATE TABLE [BibleVerses] ([VerseID] INT IDENTITY(1,1) PRIMARY KEY,
[Book] NVARCHAR(255), [Chapter] INT, [Verse] INT, [Content] NVARCHAR(4000), [Translation] NVARCHAR(255),
[Remarks] NVARCHAR(4000), [Category] NVARCHAR(255))"
                cmd.ExecuteNonQuery()
            End Using
            LogStep("created table BibleVerses")
        End If
        Dim total As Integer = 0
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT COUNT(*) FROM [BibleVerses]"
            total = Convert.ToInt32(cmd.ExecuteScalar())
        End Using
        If total = 0 Then
            SeedSampleData()
            For Each row As DataRow In store.Rows
                InsertRow(conn, row)
            Next
            store.Clear()
            nextId = 1
            LogStep("seeded sample data into projectDB.sdf")
        End If
    End Using
End Sub

Private Sub LoadFromSdf()
    store.Clear()
    Using conn As System.Data.Common.DbConnection = NewConnection()
        conn.Open()
        Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
            cmd.CommandText = "SELECT [VerseID], [Book], [Chapter], [Verse], [Content], [Translation],
[Remarks], [Category] FROM [BibleVerses] ORDER BY [VerseID]"
            Using reader As System.Data.Common.DbDataReader = cmd.ExecuteReader()
                While reader.Read()
                    Dim row As DataRow = store.NewRow()
                    For i As Integer = 0 To store.Columns.Count - 1
                        row.Item(i) = reader.GetValue(i)
                    Next
```

```
                    store.Rows.Add(row)
                End While
            End Using
        End Using
    End Using
    store.AcceptChanges()
    LogStep("loaded rows from projectDB.sdf: " & store.Rows.Count.ToString())
End Sub

Private Function InsertRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow) As Integer
    Dim newId As Integer = 0
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "INSERT INTO [BibleVerses] ([Book], [Chapter], [Verse], [Content], [Translation],
[Remarks], [Category]) VALUES (@p1, @p2, @p3, @p4, @p5, @p6, @p7)"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("Book")))
        AddParam(cmd, "@p2", IntValue(Convert.ToString(row.Item("Chapter"))))
        AddParam(cmd, "@p3", IntValue(Convert.ToString(row.Item("Verse"))))
        AddParam(cmd, "@p4", Convert.ToString(row.Item("Content")))
        AddParam(cmd, "@p5", Convert.ToString(row.Item("Translation")))
        AddParam(cmd, "@p6", Convert.ToString(row.Item("Remarks")))
        AddParam(cmd, "@p7", Convert.ToString(row.Item("Category")))
        cmd.ExecuteNonQuery()
    End Using
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "SELECT @@IDENTITY"
        newId = Convert.ToInt32(cmd.ExecuteScalar())
    End Using
    Return newId
End Function

Private Sub UpdateRow(ByVal conn As System.Data.Common.DbConnection, ByVal row As DataRow)
    Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
        cmd.CommandText = "UPDATE [BibleVerses] SET [Book] = @p1, [Chapter] = @p2, [Verse] = @p3, [Content]
= @p4, [Translation] = @p5, [Remarks] = @p6, [Category] = @p7 WHERE [VerseID] = @id"
        AddParam(cmd, "@p1", Convert.ToString(row.Item("Book")))
        AddParam(cmd, "@p2", IntValue(Convert.ToString(row.Item("Chapter"))))
        AddParam(cmd, "@p3", IntValue(Convert.ToString(row.Item("Verse"))))
        AddParam(cmd, "@p4", Convert.ToString(row.Item("Content")))
        AddParam(cmd, "@p5", Convert.ToString(row.Item("Translation")))
        AddParam(cmd, "@p6", Convert.ToString(row.Item("Remarks")))
        AddParam(cmd, "@p7", Convert.ToString(row.Item("Category")))
        AddParam(cmd, "@id", Convert.ToInt32(row.Item("VerseID")))
        cmd.ExecuteNonQuery()
    End Using
End Sub

' ---------- XML file (only used when SQL Server Compact is not installed) ----------

Private Sub LoadFromXml()
    store.Clear()
    If IO.File.Exists(XmlPath()) Then
        Try
            store.ReadXml(XmlPath())
            LogStep("loaded rows from xml: " & store.Rows.Count.ToString())
        Catch ex As Exception
            LogStep("xml unreadable, reseeding: " & ex.Message)
            store.Clear()
        End Try
    End If
    If store.Rows.Count = 0 Then
        SeedSampleData()
        SaveXml()
        LogStep("seeded sample data")
    End If
    nextId = 1
    For Each row As DataRow In store.Rows
        Dim id As Integer = Convert.ToInt32(row.Item("VerseID"))
        If id >= nextId Then nextId = id + 1
    Next
End Sub

Private Sub SaveXml()
    Try
        store.AcceptChanges()
        store.WriteXml(XmlPath())
    Catch ex As Exception
        LogStep("WriteXml failed: " & ex.Message)
    End Try
End Sub

' ---------- Create / Update / Delete ----------

Private Function SaveNewRecord(ByVal row As DataRow) As Integer
    Dim newId As Integer = 0
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            newId = InsertRow(conn, row)
        End Using
    Else
        newId = nextId
        nextId += 1
    End If
    row.Item("VerseID") = newId
```

```
    store.Rows.Add(row)
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
    Return newId
End Function

Private Sub SaveChangedRecord(ByVal row As DataRow)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            UpdateRow(conn, row)
        End Using
    End If
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub

Private Sub DeleteRecord(ByVal id As Integer)
    If useSdf Then
        Using conn As System.Data.Common.DbConnection = NewConnection()
            conn.Open()
            Using cmd As System.Data.Common.DbCommand = conn.CreateCommand()
                cmd.CommandText = "DELETE FROM [BibleVerses] WHERE [VerseID] = @id"
                AddParam(cmd, "@id", id)
                cmd.ExecuteNonQuery()
            End Using
        End Using
    End If
    For i As Integer = store.Rows.Count - 1 To 0 Step -1
        If Convert.ToInt32(store.Rows(i).Item("VerseID")) = id Then
            store.Rows.RemoveAt(i)
        End If
    Next
    store.AcceptChanges()
    If Not useSdf Then SaveXml()
End Sub

Private Sub SeedSampleData()
    Dim rows As String() = New String() { _
        "Genesis|1|1|In the beginning God created the heavens and the earth.|NIV|Creation|Old Testament", _
        "Exodus|20|3|You shall have no other gods before me.|NIV|Commandments|Old Testament", _
        "Joshua|1|9|Be strong and courageous. Do not be afraid.|NIV|Courage|History", _
        "Isaiah|41|10|So do not fear, for I am with you.|NIV|Comfort|Prophecy", _
        "Matthew|5|9|Blessed are the peacemakers.|NIV|Peace|Gospel", _
        "James|1|22|Do not merely listen to the word, and so deceive yourselves. Do what it
says.|NIV|Action|Epistle"
        }
    For Each line As String In rows
        Dim parts As String() = line.Split("|"c)
        If parts.Length = 7 Then
            Dim row As DataRow = store.NewRow()
            row.Item("VerseID") = nextId
            row.Item("Book") = parts(0)
            row.Item("Chapter") = IntValue(parts(1))
            row.Item("Verse") = IntValue(parts(2))
            row.Item("Content") = parts(3)
            row.Item("Translation") = parts(4)
            row.Item("Remarks") = parts(5)
            row.Item("Category") = parts(6)
            store.Rows.Add(row)
            nextId += 1
        End If
    Next
End Sub

Private Sub RefreshGrid()
    Try
        If store Is Nothing Then Return
        Dim view As New DataTable("BibleVerses")
        view = store.Copy()
        dataGridView1.DataSource = view
        FormatGrid()
        lblDatabase.Text = "Storage: " & StoragePath() & "   |   " & store.Rows.Count.ToString() & "
record(s)"
    Catch ex As Exception
        LogStep("RefreshGrid failed: " & ex.Message)
    End Try
End Sub

Private Sub FormatGrid()
    If dataGridView1.Columns.Count = 0 Then Return
    With dataGridView1
        .Columns("VerseID").HeaderText = "ID"
        .Columns("Book").HeaderText = "Book"
        .Columns("Chapter").HeaderText = "Chapter"
        .Columns("Verse").HeaderText = "Verse"
        .Columns("Content").HeaderText = "Content"
        .Columns("Translation").HeaderText = "Translation"
        .Columns("Remarks").HeaderText = "Remarks"
        .Columns("Category").HeaderText = "Category"
    End With
End Sub

Private Sub addButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
```

```
addButton.Click
    Try
        If Not IsValidInput() Then Return
        Dim row As DataRow = store.NewRow()
        row.Item("VerseID") = nextId
        row.Item("Book") = txtBook.Text.Trim()
        row.Item("Chapter") = IntValue(txtChapter.Text)
        row.Item("Verse") = IntValue(txtVerse.Text)
        row.Item("Content") = txtContent.Text.Trim()
        row.Item("Translation") = txtTranslation.Text.Trim()
        row.Item("Remarks") = txtRemarks.Text.Trim()
        row.Item("Category") = txtCategory.Text.Trim()
        Dim newId As Integer = SaveNewRecord(row)
        MessageBox.Show("The record was saved." & vbCrLf & vbCrLf & "New ID: " & newId.ToString(), _
            "Record Added", MessageBoxButtons.OK, MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Add failed: " & ex.Message)
        MessageBox.Show("The record could not be saved." & vbCrLf & vbCrLf & ex.Message, _
            "Save Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub updateButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
updateButton.Click
    Try
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If Not IsValidInput() Then Return
        Dim found As DataRow = Nothing
        For Each row As DataRow In store.Rows
            If Convert.ToInt32(row.Item("VerseID")) = selectedId Then
                found = row
                Exit For
            End If
        Next
        If found Is Nothing Then
            MessageBox.Show("That record no longer exists.", "Record Not Found", _
                MessageBoxButtons.OK, MessageBoxIcon.Warning)
            RefreshGrid()
            Return
        End If
        found.Item("Book") = txtBook.Text.Trim()
        found.Item("Chapter") = IntValue(txtChapter.Text)
        found.Item("Verse") = IntValue(txtVerse.Text)
        found.Item("Content") = txtContent.Text.Trim()
        found.Item("Translation") = txtTranslation.Text.Trim()
        found.Item("Remarks") = txtRemarks.Text.Trim()
        found.Item("Category") = txtCategory.Text.Trim()
        SaveChangedRecord(found)
        MessageBox.Show("The record was updated.", "Record Updated", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Update failed: " & ex.Message)
        MessageBox.Show("The record could not be updated." & vbCrLf & vbCrLf & ex.Message, _
            "Update Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub deleteButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
deleteButton.Click
    Try
        If selectedId = 0 Then
            MessageBox.Show("Select a record in the table first.", "No Record Selected", _
                MessageBoxButtons.OK, MessageBoxIcon.Information)
            Return
        End If
        If MessageBox.Show("Delete this record?" & vbCrLf & vbCrLf & "ID: " & selectedId.ToString(), _
                "Confirm Delete", MessageBoxButtons.YesNo, MessageBoxIcon.Question) <> DialogResult.Yes Then
            Return
        End If
        DeleteRecord(selectedId)
        MessageBox.Show("The record was deleted.", "Record Deleted", MessageBoxButtons.OK,
MessageBoxIcon.Information)
        ClearFields()
        RefreshGrid()
    Catch ex As Exception
        LogStep("Delete failed: " & ex.Message)
        MessageBox.Show("The record could not be deleted." & vbCrLf & vbCrLf & ex.Message, _
            "Delete Error", MessageBoxButtons.OK, MessageBoxIcon.Warning)
    End Try
End Sub

Private Sub clearButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
clearButton.Click
    ClearFields()
End Sub
Private Sub exitButton_Click(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles
```

```
exitButton.Click
    Me.Close()
End Sub

Private Sub dataGridView1_SelectionChanged(ByVal sender As System.Object, ByVal e As System.EventArgs)
Handles dataGridView1.SelectionChanged
    If isLoading Then Return
    If dataGridView1.CurrentRow Is Nothing Then Return
    FillFromRow()
End Sub

Private Sub FillFromRow()
    Dim row As DataGridViewRow = dataGridView1.CurrentRow
    If row Is Nothing Then Return
    If row.Cells("VerseID").Value Is Nothing Then Return
    If row.Cells("VerseID").Value Is DBNull.Value Then Return
    isLoading = True
    Try
        selectedId = Convert.ToInt32(row.Cells("VerseID").Value)
        txtVerseID.Text = selectedId.ToString()
        txtBook.Text = Convert.ToString(row.Cells("Book").Value)
        txtChapter.Text = Convert.ToString(row.Cells("Chapter").Value)
        txtVerse.Text = Convert.ToString(row.Cells("Verse").Value)
        txtContent.Text = Convert.ToString(row.Cells("Content").Value)
        txtTranslation.Text = Convert.ToString(row.Cells("Translation").Value)
        txtRemarks.Text = Convert.ToString(row.Cells("Remarks").Value)
        txtCategory.Text = Convert.ToString(row.Cells("Category").Value)
    Catch ex As Exception
        LogStep("FillFromRow failed: " & ex.Message)
    Finally
        isLoading = False
    End Try
End Sub

Private Sub ClearFields()
    isLoading = True
    Try
        dataGridView1.CurrentCell = Nothing
    Catch
    End Try
    selectedId = 0
    txtVerseID.Clear()
    txtBook.Clear()
    txtChapter.Clear()
    txtVerse.Clear()
    txtContent.Clear()
    txtTranslation.Clear()
    txtRemarks.Clear()
    txtCategory.Clear()
    txtBook.Focus()
    isLoading = False
End Sub

Private Function IsValidInput() As Boolean
    If txtBook.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Book.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtBook.Focus()
        Return False
    End If
    If txtChapter.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Chapter.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtChapter.Focus()
        Return False
    End If
    Dim ChapterNum As Integer = 0
    If txtChapter.Text.Trim() <> "" AndAlso Not Integer.TryParse(txtChapter.Text.Trim(), ChapterNum) Then
        MessageBox.Show("The Chapter must be a whole number.", "Invalid Number", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtChapter.Focus()
        Return False
    End If
    If txtVerse.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Verse.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtVerse.Focus()
        Return False
    End If
    Dim VerseNum As Integer = 0
    If txtVerse.Text.Trim() <> "" AndAlso Not Integer.TryParse(txtVerse.Text.Trim(), VerseNum) Then
        MessageBox.Show("The Verse must be a whole number.", "Invalid Number", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtVerse.Focus()
        Return False
    End If
    If txtContent.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Content.", "Missing Information", MessageBoxButtons.OK,
MessageBoxIcon.Exclamation)
        txtContent.Focus()
        Return False
    End If
    If txtTranslation.Text.Trim() = "" Then
        MessageBox.Show("Please enter the Translation.", "Missing Information", MessageBoxButtons.OK,
```

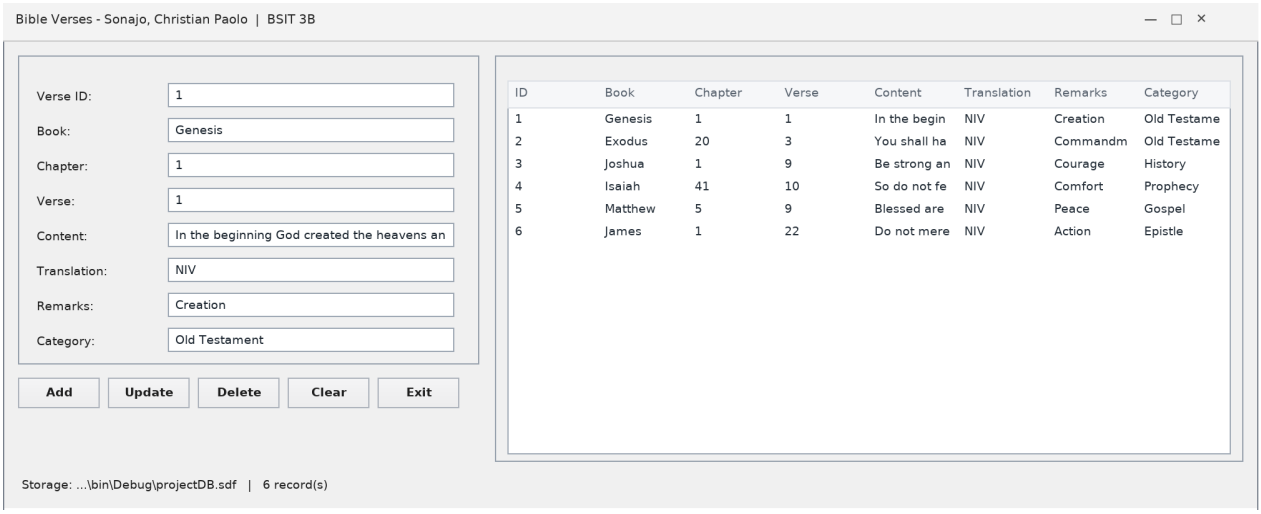
```
    MessageBoxIcon.Exclamation)
            txtTranslation.Focus()
            Return False
        End If
        Return True
    End Function


    Private Function IntValue(ByVal text As String) As Integer
        Dim value As Integer = 0
        If Integer.TryParse(text.Trim(), value) Then Return value
        Return 0
    End Function

    Private Function DateValue(ByVal text As String) As Date
        Dim parts As String() = text.Split("-"c)
        If parts.Length = 3 Then
            Dim y As Integer = 0
            Dim m As Integer = 0
            Dim d As Integer = 0
            If Integer.TryParse(parts(0), y) AndAlso Integer.TryParse(parts(1), m) AndAlso
    Integer.TryParse(parts(2), d) Then
                If y >= 1 AndAlso m >= 1 AndAlso m <= 12 AndAlso d >= 1 AndAlso d <= 31 Then
                    Return New Date(y, m, d)
                End If
            End If
        End If
        Return Date.Today
    End Function

End Class
```

Sample Output



frmBibleVerses - first record selected in the grid

frmAbout

Form Design



Form design - frmAbout

Source Code

frmAbout.vb

```
Public Class frmAbout

    Private Sub LogStep(ByVal text As String)
        Try
            IO.File.AppendAllText(IO.Path.Combine(Application.StartupPath, "forms_log.txt"), _
                Date.Now.ToString("HH:mm:ss") & "  [frmAbout] " & text & vbCrLf)
        Catch
        End Try
    End Sub

    Private Const PhotoData As String = _

    "/9j/4AAQSkZJRgABAQAAAQABAAD/2wBDAAUDBAQEAwUEBAQFBQUGBwwIBwcHBw8LCwkMEQ8SEhEPERETFhwXExQaFRERGCEYGh0dHx8fExc
    iJCI ...
' [very long data line omitted in this document]

    Private Sub frmAbout_Load(ByVal sender As System.Object, ByVal e As System.EventArgs) Handles MyBase.Load
        LogStep("Load start")
        Try
            Dim bytes As Byte() = Convert.FromBase64String(PhotoData)
            Using stream As New IO.MemoryStream(bytes)
                Using photo As Image = Image.FromStream(stream)
                    PictureBox1.Image = New Bitmap(photo)
                End Using
            End Using
        Catch ex As Exception
        End Try
    LogStep("Load end")
    End Sub

End Class
```

Sample Output



frmAbout - the About form (the programmer's photo goes in the picture box)